Capitulo 1

Introduction

A database-management system (DBMS) is a collection of interrelated data and a set of programs to access those data. The collection of data, usually referred to as the database, contains information relevant to an enterprise.

Database systems are designed to manage large bodies of information. Management of data involves both defining structures for storage of information and providing mechanisms for the manipulation of information.

Database-system application

The earliest database systems arose in the 1960s in response to the computerized management of commercial data.

Database systems are used to manage collections of data that:

- ➔ are highly valuable
- ➔ are relatively large
- ➔ are accessed by multiple users and applications, often at the same time.

Modern database systems exploit commonalities in the structure of data to gain efficiency but also allow for weakly structured data and for data whose formats are highly variable.

Capitulo 3

Review Terms

- Data-definition language
- Data-manipulation language
- Database schema
- Database instance
- Relation schema
- Relation instance
- Primary key
- Foreign key
  - Referencing relation
  - Referenced relation
- Null value
- Query language
- SQL query structure
  - **select** clause
  - **from** clause
  - **where** clause
- Multiset relational algebra
- **as** clause
- **order by** clause
- Table alias
- Correlation name (correlation variable, tuple variable)
- Set operations
  - **union**
  - **intersect**
  - **except**
- Aggregate functions
  - **avg, min, max, sum, count**
  - **group by**
  - **having**
- Nested subqueries
- Set comparisons
  - {<, <=, >, >=} { **some, all** }
  - **exists**
  - **unique**
- **lateral** clause
- **with** clause
- Scalar subquery
- Database modification
  - Delete
  - Insert
  - Update

SQL

Data-definition language (DDL): The SQL DDL provides commands for defining relation schemas, deleting relations, and modifying relation schemas.
Data-manipulation language (DML): The SQL DML provides the ability to query information from the database and to insert tuples into, delete tuples from, and modify tuples in the database.

We define an SQL relation by using the **create table** command. The following command creates a relation department in the database:

create table department
(dept name **varchar** (20),
building **varchar** (15),
budget **numeric** (12,2),
primary key (dept name));

The general form of the create table command is:

create table r
(A1 D1 ,
A2 D2 ,
...,
An Dn ,
⟨integrity-constraint1 ⟩,
…,
⟨integrity-constraintk ⟩);

where r is the name of the relation, each Ai is the name of an attribute in the schema of relation r, and Di is the domain of attribute Ai ; that is, Di specifies the type of attribute Ai

SQL supports a number of different integrity constraints:

- **primary key** (Aj1 , Aj2 , … , Ajm): The primary-key specification says that attributes Aj1 , Aj2 , … , Ajm form the primary key for the relation. The primary-key attributes are required to be nonnull and unique.
- **foreign key** (Ak1 , Ak2 , … , Akn) references s: The foreign key specification says that the values of attributes (Ak1 , Ak2 , … , Akn) for any tuple in the relation must correspond to values of the primary key attributes of some tuple in relation s. The definition of the course table has a declaration “foreign key (dept name) references department”.
- **not null:** The not null constraint on an attribute specifies that the null value is not allowed for that attribute;

To remove a relation from an SQL database, we use the

drop table

command. This one is more drastic than

delete from r;

The latter retains relation r, but deletes all tuples in r. The former deletes not only all tuples of r, but also the schema for r.

We use the alter table command to add attributes to an existing relation.The form of the alter table command is **alter table r add A D;** where r is the name of an existing relation, A is the name of the attribute to be added, and D is the type of the added attribute.

We can drop attributes from a relation by the command **alter table r drop A;**

Operaciones

El select de toda la vida:

```
select dept name
from instructor;
```

Select que elimina los duplicados:

```
select distinct dept name
from instructor;
```

SQL allows us to use the keyword **all** to specify explicitly that duplicates are not removed:

```
select all dept name
from instructor;
```

The select clause may also contain arithmetic expressions involving the operators +, −, ∗, and / operating on constants or attributes of tuples. For example, the query:

```
select ID, name, dept name, salary * 1.1
from instructor;
```

returns a relation that is the same as the instructor relation, except that the attribute salary is multiplied by 1.1.

The **where** clause allows us to select only those rows in the result relation of the from clause that satisfy a specified predicate.

```
select name
from instructor
where dept name = 'Comp. Sci.' and salary > 70000;
```

- The **select** clause is used to list the attributes desired in the result of a query.
- The **from** clause is a list of the relations to be accessed in the evaluation of the query.

• The **where** clause is a predicate involving attributes of the relation in the from clause.

A typical SQL query has the form:

```
select A1 , A2 , … , An
from r1 , r2 , … , rm
where P;
```

Each Ai represents an attribute, and each ri a relation. P is a predicate. If the where clause is omitted, the predicate P is true.

Hence, SQL provides a way of renaming the attributes of a result relation.

```
select name as instructor name, course id
from instructor, teaches
where instructor.ID= teaches.ID;
```

The **as** clause is particularly useful in renaming relations. One reason to rename a relation is to replace a long relation name with a shortened version that is more convenient to use elsewhere in the query.

```
select T.name, S.course id
from instructor as T, teaches as S
where T.ID= S.ID;
```

SQL also permits a variety of functions on character strings, such as:

- ➔ concatenating (using " // ")
- ➔ converting strings to uppercase (using the function **upper**(s))
- ➔ converting strings to lowercase (using the function **lower**(s))
- ➔ removing spaces at the end of the string (using **trim**(s))

Pattern matching can be performed on strings using the operator like. We describe patterns by using two special characters:

- ➔ **Percent (%)**: The **%** character matches any substring.
- ➔ **Underscore (_)**: The **_** character matches any character

Patterns are case sensitive; that is, uppercase characters do not match lowercase characters, or vice versa.

- 'Intro%' matches any string beginning with “Intro”.
- '%Comp%' matches any string containing “Comp” as a substring, for example, 'Intro. to Computer Science', and 'Computational Biology'.
- '_ _ _' matches any string of exactly three characters.
- '_ _ _%' matches any string of at least three characters.

SQL expresses patterns by using the like comparison operator.
The escape character is used immediately before a special pattern character to indicate that the special pattern character is to be treated like a normal character.

like 'ab\%cd%' **escape** '\' matches all strings beginning with “ab%cd”.
like 'ab\\cd%' **escape** '\' matches all strings beginning with “ab\cd”.

The asterisk symbol “ * ” can be used in the select clause to denote “all attributes.”

```
select instructor.*
from instructor, teaches
where instructor.ID= teaches.ID
order by name;
```

The **order by** clause causes the tuples in the result of a query to appear in sorted order. To specify the sort order, we may specify **desc** for descending order or **asc** for ascending order.

Between:

```
select name
from instructor
where salary between 90000 and 100000;
```

instead of:

```
select name
from instructor
where salary <= 100000 and salary >= 90000;
```

Operaciones con conjuntos

Union: (**select** …) **union** (**select** …);
If we want to retain all duplicates, we must write union all in place of union.

Intersección: (**select** …) **intersect** (**select** …);
If we want to retain all duplicates, we must write intersect all in place of intersect.

Resta: (**select** …) **except** (**select** …);
Once again… If we want to retain duplicates, we must write except all in place of except

null y *unknown*

SQL uses the special keyword **null** in a predicate to test for a null value.

```
select name
from instructor
where salary is null;
```

SQL allows us to test whether the result of a comparison is unknown, rather than true or false, by using the clauses **is unknown** and **is not unknown**.

```
select name
from instructor
where salary > 10000 is unknown;
```

Funciones de agregación

Aggregate functions are functions that take a collection of values as input and return a single value. SQL offers five standard built-in aggregate functions:

- Average: **avg** → **select avg** (salary)
- Minimum: **min**
- Maximum: **max**
- Total: **sum**
- Count: **count** → **select count** (**distinct** ID)

Tuples with the same value on all attributes in the **group by** clause are placed in one group.

When an SQL query uses grouping, it is important to ensure that…

The only attributes that appear in the select statement without being aggregated are those that are present in the group by clause.

It is useful to state a condition that applies to groups rather than to tuples. For example, we might be interested in only those departments where the average salary of the instructors is more than $42,000. To express such a query, we use the **having** clause.

```
select dept_name, avg (salary) as avg_salary
from instructor
group by dept_name
having avg_salary > 42000;
```

Any attribute that is present in the **having** clause without being aggregated must appear in the group by clause, otherwise the query is erroneous.

Subqueries anidadas

select … **and** course id **in** (**select** .. **and** year= 2018);

select … **and** course id **not in** (**select** .. **and** year= 2018); SIMILAR AL **except**

❀ The in and not in operators can also be used on enumerated sets.

…
where name **not in** ('Mozart', 'Einstein');

❀ The phrase "greater than at least one" is represented in SQL by **> some**.

select name
from instructor
where salary **> some** (**select** salary
from instructor
where dept name = 'Biology');

❀ The construct **> all** corresponds to the phrase "greater than all".

❀ The **exists/ not exists** construct returns the value true/false if the argument subquery is nonempty.

…
where exists (**select** * …);

❀ The **unique/not unique** construct returns the value true/false if the argument subquery contains no duplicate tuples.

…
where unique(**select** * …);

❀ The **with** clause provides a way of defining a temporary relation whose definition is available only to the query in which the with clause occurs.

with max_budget (value) **as**
(**select max**(budget)
from department)
select budget
from department, max_budget
where department.budget = max budget.value;

❀ Scalar subqueries

-

Modificación de la bbdd

❀ A **delete** request is expressed in much the same way as a query.

```
delete from r
where P;
```

where P represents a predicate and r represents a relation.

The delete statement first finds all tuples t in r for which P(t) is true, and then deletes them from r

Note that a delete command operates on only one relation.

❀ The simplest **insert** statement is a request to insert one tuple

```
insert into course (course id, title, dept name, credits)
values ('CS-437', 'Database Systems', 'Comp. Sci.', 4);
```

More generally, we might want to insert tuples on the basis of the result of a query:

```
insert into instructor
        select … from … where …
```

❀ The **update** statement can be used to change a value in a tuple without changing all values in the tuple. We can choose the tuples to be updated by using a query.

```
update instructor
set salary = salary * 1.05
where salary < 70000;
```

SQL provides a **case** construct that we can use to perform both updates with a single update statement, avoiding the problem with the order of updates.

```
update instructor
set salary = case
                when salary <= 100000 then salary * 1.05
                …
                when predn then resultn
                else salary * 1.03
                end
```

Capitulo 4

Review Terms

- Join types
  - Natural join
  - Inner join with **using** and **on**
  - Left, right and full outer join
  - Outer join with **using** and **on**
- View definition
  - Materialized views
  - View maintenance
  - View update
- Transactions
  - Commit work
  - Rollback work
  - Atomic transaction
- Constraints
  - Integrity constraints
  - Domain constraints
  - Unique constraint
  - Check clause
  - Referential integrity
    - Cascading deletes
    - Cascading updates
  - Assertions
- Data types
  - Date and time types
  - Default values
  - Large objects
    - clob
    - blob
  - User-defined types
  - distinct types
  - Domains
  - Type conversions
- Catalogs
- Schemas
- Indices
- Privileges
  - Types of privileges
    - **select**
    - **insert**
    - **update**
  - Granting of privileges
  - Revoking of privileges
  - Privilege to grant privileges
  - Grant option
- Roles
- Authorization on views
- Execute authorization
- Invoker privileges
- Row-level authorization
- Virtual private database (VPD)

Join expressions

❀ **Natural join**: It considers only those pairs of tuples with the same value on those attributes that appear in the schemas of both relations.

student **natural join** takes

considers only those pairs of tuples where both the tuple from student and the tuple from takes have the same value on the common attribute, ID.

```
select A1, A2, …, An
from r1 natural join r2 natural join ... natural join rm
where P;
```

❀ **Join conditions:** The **on** condition allows a general predicate over the relations being joined.

```
select *
from student join takes on student.ID = takes.ID;
```

❀ **Outer joins:** It works in a manner similar to the join operations we have already studied, but it preserves those tuples that would be lost in a join by creating tuples in the result containing null values.

- ➔ **left outer join:** preserves tuples only in the relation named before
- ➔ **right outer join:** preserves tuples only in the relation named after
- ➔ **full outer join:** preserves tuples in both relations.

¿Cómo funciona el outer join?

Primero se hace un inner join y después se agregan las tuplas correspondientes del lado que debe estar y se completa con valores null.

```
select *
from student left outer join takes on student.ID = takes.ID;
```

on and **where** behave differently for outer join. The reason for this is that outer join adds null-padded tuples only for those tuples that do not contribute to the result of the corresponding "inner" join. The **on** condition is part of the outer join specification, but a **where** clause is not.

```
select *
from student left outer join takes on true
where student.ID = takes.ID;
```

La primera query incluye la tupla: (70557, Snow, Physics, 0, null, null, null, null, null, null) , mientras que la segunda no.

❀ **Inner joins:** Operations that do not preserve nonmatched tuples (intersección para los panas)

❀ **Join types and conditions:** Normal **joins** are called **inner joins** in SQL. The keyword inner is, however, optional. The default join type, when the join clause is used without the outer prefix, is the inner join.

```
select *
from student inner join takes using (ID);
```

Vistas

The **with** clause allows us to assign a name to a subquery for use as often as desired, but in one particular query only. On the other hand, **views** present a way to extend this concept beyond a single query.

❀ **Definición:** Con el comando create view → **create view** v **as** <query expression>;

When we define a view, the database system stores the definition of the view itself, rather than the result of evaluation of the query expression that defines the view. Wherever a view relation appears in a query, it is replaced by the stored query expression. Thus, whenever we evaluate the query, the view relation is recomputed.

Views, once created, remain available until explicitly dropped.

```
create view departments_total_salary(dept_name, total_salary) as
select dept name, sum (salary)
from instructor
group by dept name;
```

❀ **Materialized views:** Certain database systems allow view relations to be stored, but they make sure that, if the actual relations used in the view definition change, the view is kept up-to-date. The process of keeping the materialized view up-to-date is called **materialized view maintenance.** Applications that use a view frequently or demand fast response may benefit if the view is materialized.

❀ **Update of a view:** An SQL view is said to be **updatable** (i.e., inserts, updates, or deletes can be applied on the view) if the following conditions are all satisfied by the query defining the view:

- ➔ The **from** clause has only one database relation.
- ➔ The **select** clause contains only attribute names of the relation and does not have any expressions, aggregates, or distinct specification.
- ➔ Any attribute not listed in the **select** clause can be set to null; that is, it does not have a not null constraint and is not part of a primary key.
- ➔ The query does not have a **group by** or **having** clause.

Views can be defined with a **with check option** clause at the end of the view definition.

Transactions

A transaction consists of a sequence of query and/or update statements. The SQL standard specifies that a transaction begins implicitly when an SQL statement is executed.

❀ **Commit:** commits the current transaction; that is, it makes the updates performed by the transaction become permanent in the database.

❀ **Rollback:** causes the current transaction to be rolled back; that is, it undoes all the updates performed by the SQL statements in the transaction.

Transaction rollback is useful if some error condition is detected during execution of a transaction. Once a transaction has executed **commit**, its effects can no longer be undone by **rollback**. The database provides an abstraction of a transaction as being **atomic**, that is, indivisible. How to turn off automatic commit depends on the specific SQL implementation, although many databases support the command set autocommit off.

However, several other databases, such as MySQL and PostgreSQL, support a **begin** statement which starts a transaction containing all subsequent SQL statements, but do not support the **end** statement; instead, the transaction must be ended by either a **commit** or a **rollback** command

Integrity constraints

Integrity constraints ensure that changes made to the database by authorized users do not result in a loss of data consistency. Thus, integrity constraints guard against accidental damage to the database.

* We study another form of integrity constraint, called **functional dependencies**, that is used primarily in the process of schema design.

Integrity constraints are usually identified as part of the database schema design process and declared as part of the create table command used to create relations. However, integrity constraints can also be added to an existing relation by using the command alter table-name add constraint, where constraint can be any constraint on the relation.

❀ **Constraints on a single relation:** In addition to the primary-key constraint, there are a number of other ones that can be included in the create table command. Such as: **not null, unique, check(**<predicate>**)**.

> ❀ **Not null constraint:** The not null constraint prohibits the insertion of a null value for the attribute, and is an example of a domain constraint. Any database modification that would cause a null to be inserted in an attribute declared to be not null generates an error diagnostic.

❀ **Unique constraint:** The unique specification says that attributes Aj 1 , Aj 2 , …, Aj m form a superkey; that is, no two tuples in the relation can be equal on all the listed attributes.

❀ **Check clause:** The clause check(P) specifies a predicate P that must be satisfied by every tuple in a relation. A common use of the check clause is to ensure that attribute values satisfy specified conditions, in effect creating a powerful type system. For instance, a clause **check(budget > 0)** in the create table command for relation *department* would ensure that the value of **budget** is nonnegative.

```
create table section
        (..,
        time_slot_id varchar (4),
        primary key (course_id, sec_id, semester, year),
        check (semester in ('Fall', 'Winter', 'Spring', 'Summer')));
```

A check clause is satisfied if it is not false, so clauses that evaluate to **unknown** are not violations. If null values are not desired, a separate not null constraint must be specified

❀ **Referential integrity:** We wish to ensure that a value that appears in one relation for a given set of attributes also appears for a certain set of attributes in another relation. **Foreign keys** are a form of a referential integrity constraint where the referenced attributes form a primary key of the referenced relation.

Foreign keys can be specified as part of the SQL create table statement by using the foreign key clause:

```
foreign key (dept_name) references department
```

By default, in SQL a foreign key references the primary-key attributes of the referenced table. The specified list of attributes must, however, be declared as a superkey of the referenced relation, using either a primary key constraint or a unique constraint.

Note that the foreign key must reference a compatible set of attributes, that is, the number of attributes must be the same and the data types of corresponding attributes must be compatible.

```
dept_name varchar(20) references department
```

However, a **foreign key clause** can specify that if a delete or update action on the referenced relation violates the constraint, then, instead of rejecting the action, the system must take steps to change the tuple in the referencing relation to restore the constraint.

```
create table course
        ( … foreign key (dept_name) references department
        on delete cascade
        on update cascade, … );
```

❀ **Names to constraints:** To name a constraint, we precede the constraint with the keyword constraint and the name we wish to assign it.

```
salary numeric(8,2), constraint minsalary check (salary > 29000)
```

If we decide we no longer want this constraint:

```
alter table instructor drop constraint minsalary;
```

❀ **Constraint violation during a transaction:** Transactions may consist of several steps, and integrity constraints may be violated temporarily after one step, but a later step may remove the violation. To handle such situations, the SQL standard allows a clause **initially deferred** to be added to a constraint specification; the constraint would then be checked at the end of a transaction and not at intermediate steps. A constraint can alternatively be specified as **deferrable**, which means it is checked immediately.

```
set constraints constraint_list deferred
```

❀ **Complex check conditions:** The predicate in the check clause can be an arbitrary predicate that can include a subquery.

```
check (time_slot_id in (select time_slot_id from time_slot))
```

Complex check conditions can be useful when we want to ensure the integrity of data, but they may be costly to test.

❀ **Assertions:** Is a predicate expressing a condition that we wish the database always to satisfy. Consider the following constraints, which can be expressed using assertions.

- For each tuple in the *student* relation, the value of the attribute *tot_cred* must equal the sum of credits of courses that the student has completed successfully.
- An instructor cannot teach in two different classrooms in a semester in the same time slot.

```
create assertion credits_earned_constraint check
(not exists (select ID
                from student
                where tot_cred <> (select coalesce(sum(credits), 0)
                                        from takes natural join course
                                        where student.ID= takes.ID
                                        and grade is not null and grade<> 'F' )))

create assertion  <assertion-name> check <predicate>;
```

SQL data types and schemas

In addition to the basic data types, SQL standard supports data types such as dates and times:

➔ **date**: A calendar date containing a (four-digit) year, month, and day of the month.

- ➔ **time**: The time of day, in hours, minutes, and seconds. A variant, **time**(p), can be used to specify the number of fractional digits for seconds (the default being 0). It is also possible to store time-zone information along with the time by specifying time with timezone.
- ➔ **timestamp**: A combination of date and time. A variant, timestamp(p), can be used to specify the number of fractional digits for seconds (the default here being 6). Time-zone information is also stored if **with timezone** is specified.

Date and time values can be specified like this:

```
date '2018-04-25'
time '09:30:00'
timestamp '2018-04-25 10:29:01.45'
```

❀ **Type conversion and formatting functions:** We can use an expression of the form:
cast (e **as** t) to convert an expression e to the type t.

Database systems offer a variety of formatting functions, and details vary among the leading systems. MySQL offers a **format** function. Oracle and PostgreSQL offer a set of functions, **to_char**, **to_number**, and **to_date**. SQL Server offers a **convert** function. We can choose how null values are output in a query result using the **coalesce** function.

```
select ID, coalesce(salary, 0) as salary
from instructor
```

❀ **Default values:** SQL allows a **default** value to be specified for an attribute.

```
create table student
(..., tot_cred numeric(3,0) default 0,
primary key (ID));
```

❀ **Large object types:** SQL provides **large-object data types** for character data (**clob**) and binary data (**blob**). The letters “lob” in these data types stand for “Large OBject.” For example, we may declare attributes:

```
book_review clob(10KB)
image blob(10MB)
movie blob(2GB)
```

For result tuples containing large objects (multiple megabytes to gigabytes), it is inefficient or impractical to retrieve an entire large object into memory. Instead, an application would usually use an SQL query to retrieve a “locator” for a large object and then use the locator to manipulate the object from the host language in which the application itself is written.

❀ **User defined types:** The first form is called **distinct types**. The other form, called **structured data types**, allows the creation of complex data types with nested record structures, arrays, and multisets.

The create type clause can be used to define new types.

create type *Dollars* **as numeric**(12,2) **final**;
create type *Pounds* **as numeric**(12,2) **final**;

SQL provides **drop type** and **alter type** clauses to drop or modify types that have been created earlier.

Differences between domains and types:

- ➔ Domains can have constraints, such as not null, specified on them, and can have default values defined for variables of the domain type, whereas user-defined types cannot have constraints or default values specified on them.
- ➔ Domains are not strongly typed.

❀ **Generating unique key values:** Database systems offer automatic management of unique key-value generation. Since this feature works only for numeric key-value data types, we change the type of ID to number, and write:

ID **number**(5) **generated always as identity;**

When the **always** option is used, any **insert** statement must avoid specifying a value for the automatically generated key. In **PostgreSQL**, we can define the type of ID as **serial**, which tells PostgreSQL to automatically generate identifiers; in MySQL we use **auto_increment** in place of **generated always as identity**, while in SQL Server we can use just **identity**.

❀ **Create table extensions:** SQL provides a **create table like** extension to support this task: **create table** *temp_instructor* **like** *instructor*; The above statement creates a new table temp instructor that has the same schema as instructor.

❀ **Schemas, catalogs and environments:** Consider how files are named in a file system. In database systems users had to coordinate to make sure they did not try to use the same name for different relations. Database systems provide a three-level hierarchy for naming relations. The top level of the hierarchy consists of **catalogs**, each of which can contain **schemas**.

The default catalog and schema are part of an SQL **environment** that is set up for each connection. The environment additionally contains the user identifier. We can create and drop schemas by means of **create schema** and **drop schema** statements.

Index definition

An **index** on an attribute of a relation is a data structure that allows the database system to find those tuples in the relation that have a specified value for that attribute efficiently, without scanning through all the tuples of the relation. Indices form part of the physical schema of the database, as opposed to its logical schema.

Indices are important for efficient processing of transactions. We create an index with the create index command, which takes the form:

create index <index-name> **on** <relation-name> (<attribute-list>);

If we wish to declare that the search key is a candidate key, we add the attribute **unique** to the index definition.

Authorization

We may assign a user several forms of authorizations on parts of the database.

- ➔ Authorization to read data.
- ➔ Authorization to insert new data.
- ➔ Authorization to update data.
- ➔ Authorization to delete data.

Each of these types of authorizations is called a **privilege**. When a user submits a query or an update, the SQL implementation first checks if the query or update is authorized. In addition to authorizations on data, users may also be granted authorizations on the database schema, allowing them, for example, to create, modify, or drop relations. A user who has some form of authorization may be allowed to pass on (**grant**) this authorization to other users, or to withdraw (**revoke**) an authorization that was granted earlier. The database administrator may authorize new users, restructure the database, and so on; (**superuser)**.

❀ **Granting and revoking of privileges:** The SQL standard includes the **privileges** select, insert, update, and delete (**all privileges** for all the allowable privileges).
The grant statement is used to confer authorization. The basic form of this statement is:

grant <privilege list>
on <relation name or view name>
to <user/role list>;

For update you can specify the attribute → **grant update** (budget) **on…**

❀ **Roles:** A set of roles is created in the database. Authorizations can be granted to roles, in exactly the same fashion as they are granted to individual users. Each database user is granted a set of roles (which may be empty) that she is authorized to perform. Roles can be created in SQL as follows:

create role instructor;

Roles can be granted to users, as well as to other roles:

create role *dean*;
grant *instructor* **to** *dean*;
grant *dean* **to** Satoshi;

❀ **Authorization on views:** A user who creates a view does not necessarily receive all privileges on that view. For example, a user who creates a view cannot be given update authorization on a view without having update authorization on the relations used to define the view. The **execute** privilege can be granted on a function or procedure, enabling a user to execute the function or procedure.

❀ **Authorization on schemas:** SQL includes a **references** privilege that permits a user to declare foreign keys when creating relations:

grant references (dept name) **on** department **to** Mariano**;**

Initially, it may appear that there is no reason ever to prevent users from creating foreign keys referencing another relation. However, recall that foreign-key constraints restrict deletion and update operations on the referenced relation. The **references privilege** on *department* is also required to create a **check** constraint on a relation r if the constraint has a subquery referencing department.

❀ **Transfer of privileges:** If we wish to grant a privilege and to allow the recipient to pass the privilege on to other users, we append the **with grant option** clause to the appropriate grant command. The passing of a specific authorization from one user to another can be represented by an **authorization graph**.

❀ **Revoking of privileges:** Revocation of a privilege from a user/role may cause other users/roles also to lose that privilege (**cascading revocation**). The revoke statement may specify restrict in order to prevent cascading revocation:

revoke select on department **from** Amit, Satoshi **restrict**;

Capitulo 5

Advanced SQL

A database programmer must have access to a general-purpose programming language for at least two reasons:

- ➔ Not all queries can be expressed in SQL, since SQL does not provide the full expressive power of a general-purpose language.
- ➔ Nondeclarative actions—such as printing a report, interacting with a user, or sending the results of a query to a graphical user interface—cannot be done from within SQL.

There are two approaches to accessing SQL from a general-purpose programming language:

- ➔ **Dynamic SQL:** Dynamic SQL allows the program to construct an SQL query as a character string at runtime, submit the query, and then retrieve the result into program variables a tuple at a time.
- ➔ **Embedded SQL:** under embedded SQL, the SQL statements are identified at compile time using a preprocessor, which translates requests expressed in embedded SQL into function calls. At runtime, these function calls connect to the database using an API that provides dynamic SQL facilities but may be specific to the database that is being used.

❀ **JDBC:** Is a standard that defines an application program interface (API) that Java programs can use to connect to database servers. The first step in accessing a database from a Java program is to open a connection to the database. A connection is opened using the getConnection() method of the DriverManager class (within java.sql). This method takes three parameters:

- ➔ 1st parameter: String that specifies the URL
- ➔ 2nd parameter: database user identifier (string)
- ➔ 3rd parameter: Password (string)

Connections and statements: Once the connection is open, the program can use it to send SQL statements. A Statement object is not the SQL statement itself, but rather an object that allows the Java program to invoke methods that ship an SQL statement given as an argument. To execute a statement, we invoke either the **executeQuery()** method or the **executeUpdate()** method.

Exceptions: The try { … } catch { … } construct permits us to catch any exceptions (error conditions) that arise when JDBC calls are made and take appropriate action.

Programmers must take care to ensure that programs close all such resources. Failure to do so may cause the database system's resource pools to become exhausted, rendering the system inaccessible or inoperative until a time-out period expires. This protects us from leaving connections or statements unclosed.

Retrieving the result of a query: Execute a query by using stmt.executeQuery(). It retrieves the set of tuples in the result into a ResultSet object rset and fetches them one tuple at a time. The method getString() can retrieve any of the basic SQL data types.

Prepared statements: We can create a prepared statement in which some values are replaced by "?", thereby specifying that actual values will be provided later. The prepareStatement() method of the Connection class defines a query that may contain parameter values. The method returns an object of class PreparedStatement. Prepared statements allow for more efficient execution in cases where the same query can be compiled once and then run multiple times with different parameter values.

SQL injection: ?

Programmers must pass user-input strings to the database only through parameters of prepared statements; creating SQL queries by concatenating strings with user-input values is an extremely serious security risk and should never be done in any program.

Callable statements: Allows invocation of SQL stored procedures and functions:

```
CallableStatement cStmt2 = conn.prepareCall("{call some procedure(?,?)}");
```

Metadata: The interface ResultSet has a method, getMetaData(), that returns a ResultSetMetaData object that contains metadata about the result set. The DatabaseMetaData interface in turn has a very large number of methods to get metadata about the database and the database system to which the application is connected.

The method getColumns() takes four arguments: a catalog name (null signifies that the catalog name is to be ignored), a schema name pattern, a table name pattern, and a column name pattern.

The getTables() method allows you to get a list of all tables in the database. The first three parameters to getTables() are the same as for getColumns(). The fourth parameter can be used to restrict the types of tables returned.

❀ **Access from python:**

```
import psycopg2
def PythonDatabaseExample(userid, passwd)
        try:
                conn = psycopg2.connect( host="db.yale.edu", port=5432, dbname="univdb",
        user=userid, password=passwd)
                cur = conn.cursor()
                try:
                        cur.execute("insert into instructor values(%s, %s, %s, %s)",
                ("77987","Kim","Physics",98000))
                        conn.commit();
                except Exception as sqle:
                        print("Could not insert tuple. ", sqle)
                        conn.rollback()
                cur.execute( ("select dept name, avg (salary) " " from instructor group by dept
                name"))
                for dept in cur:
```

```
                print dept[0], dept[1]
except Exception as sqle:
        print("Exception : ", sqle)
```

❀ **ODBC (open database connectivity):** defines an API that applications can use to open a connection with a database, send queries and updates, and get back results.

the program can send SQL commands to the database by using SQLExecDirect.
SQLFetch: SQL SUCCESS cuando encuentra resultado
SQLBindCol
A negative value returned for the length field indicates that the value is null
SQLSetConnectOption(conn, SQL AUTOCOMMIT, 0)
SQLTransact(conn, SQL COMMIT)

❀ **Embedded SQL:** A language in which SQL queries are embedded is referred to as a host language, and the SQL structures permitted in the host language constitute embedded SQL.

EXEC SQL <embedded SQL statement>;
Variables of the host language can be used within embedded SQL statements, but they must be preceded by a colon (:) to distinguish them from SQL variables.
The exact syntax for embedded SQL requests depends on the language in which SQL is embedded.

Functions and procedures

Procedures and functions allow “business logic” to be stored in the database and executed from SQL statements.

Most databases implement nonstandard versions of this syntax (SQL standard).

❀ **Declaring and invoking SQL functions and procedures:** Suppose that we want a function that, given the name of a department, returns the count of the number of instructors in that department. This function can be used in a query that returns names and budgets of all departments with more than 12 instructors:

```
select dept name, budget
from department
where dept count(dept name) > 12;
```

```
create function instructor_of (dept_name varchar(20))
    returns table (
        ID varchar (5),
        name varchar (20),
        dept_name varchar (20),
        salary numeric (8,2))
return table
    (select ID, name, dept_name, salary
    from instructor
    where instructor.dept_name = instructor_of.dept_name);
```

Performance problems have been observed on many database systems when invoking complex user-defined functions within a query. Programmers should therefore take performance into consideration when deciding whether to use user-defined functions in a query.

Table functions: functions that can return tables as results.

La función de la imagen puede ser usada en una query de la siguiente manera:

```
select *
from table(instructor of('Finance'));
```

Table-valued functions can be thought of as **parameterized views** that generalize the regular notion of views by allowing parameters.
The dept count function could instead be written as a procedure:

```
create procedure dept_count_proc(in dept_name varchar(20),
                                 out d_count integer)
    begin
        select count( *) into d_count
        from instructor
        where instructor.dept_name= dept_count_proc.dept_name
    end
```

The keywords **in** and **out** indicate, respectively, parameters that are expected to have values assigned to them and parameters whose values are set in the procedure in order to return results.

Procedures can be invoked either from an SQL procedure or from embedded SQL by the **call** statement:

```
declare d_count integer;
call dept_count_proc('Physics', d count);
```

❀ **Language Constructs for Procedures and Functions:** SQL supports constructs that give it almost all the power of a general-purpose programming language. (**Persistent storage module** deals with these constructs)

Variables are declared using a **declare** statement and can have any valid SQL data type.
A compound statement is of the form **begin** … **end.**

The syntax for while statements and repeat statements is:

```
while boolean expression do
        sequence of statements;
end while

repeat
        sequence of statements;
until boolean expression
end repeat
```

There is also a for loop that permits iteration over all the results of a query:

```
declare n integer default 0;
for r as
        select budget from department
        where dept_name = 'Music'
do
    set n = n− r.budget
end for
```

The statement leave can be used to exit the loop.
The conditional statements supported by SQL include if-then-else statements by using this syntax:

```
if boolean expression
    then statement or compound statement
elseif boolean expression
    then statement or compound statement
else statement or compound statement
end if
```

The SQL procedural language also supports the signaling of exception conditions and declaring of handlers that can handle the exception, as in this code:

```
declare out_of_classroom_seats condition
declare exit handler for out_of_classroom_seats
begin
sequence of statements
end
```

The statements between the **begin** and the **end** can raise an exception by executing **signal** out of classroom seats. The handler says that if the condition arises, the action to be taken is to exit the enclosing begin end statement.

Triggers

after update of takes **on** grade
Triggers can be activated before the event (insert, delete, or update) instead of after the event. Triggers that execute before an event can serve as extra constraints that can prevent invalid updates, inserts, or deletes.

create trigger *credits_earned* **after update of** *takes* **on** *grade*

Recursive Queries

Capitulo 12

Review Terms

- Physical storage media
  - Cache
  - Main memory
  - Flash memory
  - Magnetic disk
  - Optical storage
  - Tape storage
- Volatile storage
- Non-volatile storage
- Sequential-access
- Direct-access
- Storage interfaces
  - Serial ATA (SATA)
  - Serial Attached SCSI (SAS)
  - Non-Volatile Memory Express (NVMe)
  - Storage area network (SAN)
  - Network attached storage (NAS)
- Magnetic disk
  - Platter
  - Hard disks
  - Tracks
  - Sectors
  - Read–write head
  - Disk arm
  - Cylinder
  - Disk controller
  - Checksums
  - Remapping of bad sectors
  - Level 6 (block striping, P + Q redundancy)
- Rebuild performance
- Software RAID
- Hardware RAID
- Hot swapping
- Optimization of disk-block access
- Disk block
- Performance measures of disks
  - Access time
  - Seek time
  - Latency time
  - I/O operations per second (IOPS)
  - Rotational latency
  - Data-transfer rate
  - Mean time to failure (MTTF)
- Flash Storage
  - Erase Block
  - Wear leveling
  - Flash translation table
  - Flash Translation Layer
- Storage class memory
  - 3D-XPoint
- Redundant arrays of independent disks (RAID)
  - Mirroring
  - Data striping
  - Bit-level striping
  - Block-level striping
- RAID levels
  - Level 0 (block striping, no redundancy)
  - Level 1 (block striping, mirroring)
  - Level 5 (block striping, distributed parity)
  - Disk-arm scheduling
  - Elevator algorithm
  - File organization
  - Defragmenting
  - Non-volatile write buffers
  - Log disk

Physical storage systems

The goal of a database system is to simplify and facilitate access to data; users of the system should not be burdened unnecessarily with the physical details of the implementation of the system.

Overview of physical storage media

These storage media are classified by the speed with which data can be accessed, by the cost per unit of data to buy the medium, and by the medium's reliability.

- ➔ Cache: The cache is the fastest and most costly form of storage. Cache memory is relatively small; its use is managed by the computer system hardware

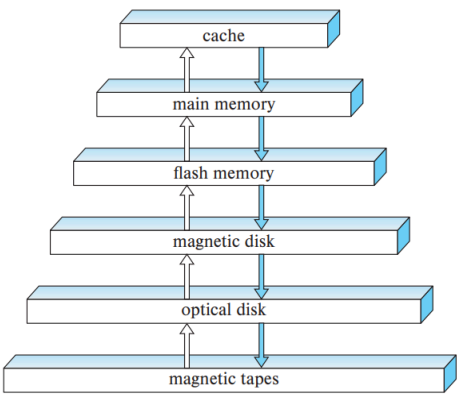
- ➔ Main memory: The storage medium used for data that are available to be operated on is the main memory. It is generally too small (or too expensive) for storing the entire database for very large databases, but many enterprise databases can fit in main memory. However, the contents of main memory are lost in the event of a power failure or system crash; main memory is therefore said to be volatile.

- ➔ Flash memory: Flash memory differs from main memory in that stored data are retained even if power is turned off (or fails)—that is, it is non-volatile. Flash memory has a lower cost per byte than main memory, but a higher cost per byte than magnetic disks. A solid-state drive (SSD) uses flash memory internally to store data but provides an interface similar to a magnetic disk, allowing data to be stored or retrieved in units of a block.

- ➔ Optical storage: The digital video disk (DVD) is an optical storage medium, with data written and read back using a laser light source.

- ➔ Tape storage: Tape storage is used primarily for backup and archival data. Archival data refers to data that must be stored safely for a long period of time.

However, access to data is much slower because the tape must be accessed sequentially from the beginning of the tape; tape storage is referred to as sequential-access storage. The various storage media can be organized in a hierarchy (Figure 12.1) according to their speed and their cost. The higher levels are expensive, but fast. As we move down the hierarchy, the cost per bit decreases, whereas the access time increases.



The fastest storage media—for example, cache and main memory—are referred to as primary storage. The media in the next level in the hierarchy—for example, flash memory and magnetic disks—are referred to as secondary storage, or online storage. The media in the lowest level in the hierarchy—for example, magnetic tape and optical disk jukeboxes—are referred to as tertiary storage, or offline storage.

Storage interfaces

Disks typically support either the Serial ATA (SATA) interface, or the Serial Attached SCSI (SAS) interface.

The NonVolatile Memory Express (NVMe) interface is a logical interface standard developed to better support SSDs and is typically used with the PCIe interface.

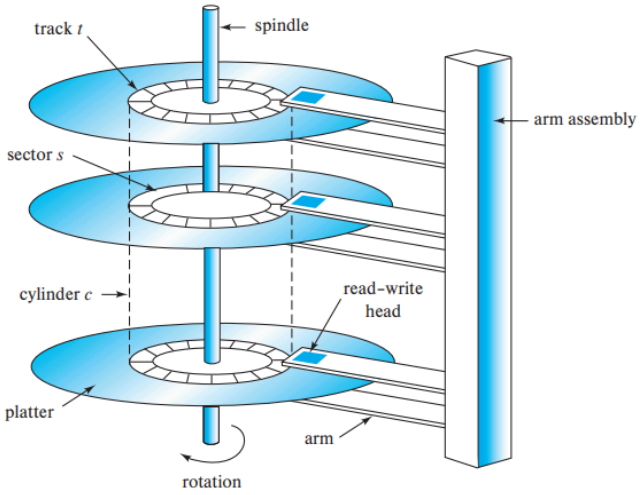
In the storage area network (SAN) architecture, large numbers of disks are connected by a high-speed network to a number of server computers. Network attached storage (NAS) is an alternative to SAN. NAS is much like SAN, except that instead of the networked storage appearing to be a large disk, it provides a file system interface using networked file system protocols such as NFS or CIFS

Magnetic disks

Magnetic disks provide the bulk of secondary storage for modern computer systems.

❀ Physical characteristics of disks

Each disk platter has a flat, circular shape. Its two surfaces are covered with a magnetic material, and information is recorded on the surfaces. Platters are made from rigid metal or glass.



The disk surface is logically divided into tracks, which are subdivided into sectors. A sector is the smallest unit of information that can be read from or written to the disk.

The read–write head stores information on a sector magnetically as reversals of the direction of magnetization of the magnetic material.

A disk typically contains many platters, and the read-write heads of all the tracks are mounted on a single assembly called a disk arm and move together. The disk platters mounted on a spindle and the heads mounted on a disk arm are together known as head-disk assemblies.

The ith tracks of all the platters together are called the ith cylinder. The read-write heads are kept as close as possible to the disk surface to increase the recording density.

Head crashes can be a problem. If the head contacts the disk surface, the head can scrape the recording medium off the disk, destroying the data that had been there; in older-generation disks, a head crash could thus result in failure of the entire disk. Current-generation disk drives use a thin film of magnetic metal as recording medium. They are much less susceptible to failure of the entire disk, but are susceptible to failure of individual sectors.

A disk controller interfaces between the computer system and the actual hardware of the disk drive;

A disk controller accepts high-level commands to read or write a sector, and initiates actions, such as moving the disk arm to the right track and actually reading or writing the data. Disk controllers also attach checksums to each sector that is written (el checksum sirve para comprobar si los datos fueron corrompidos).

Another interesting task that disk controllers perform is remapping of bad sectors. If the controller detects that a sector is damaged when the disk is initially formatted, or when an attempt is made to write the sector, it can logically map the sector to a different physical location

❀ **Performance measures of disks**

The main measures of the qualities of a disk are capacity, access time, data-transfer rate, and reliability.

Access time: Time from when a read or write request is issued to when data transfer begins.

The time for repositioning the arm is called the seek time, and it increases with the distance that the arm must move. The average seek time (⅓ of the worst seek time) is the average of the seek times, measured over a sequence of random requests.

Once the head has reached the desired track, the time spent waiting for the sector to be accessed to appear under the head is called the rotational latency time. The average latency time of the disk is one-half the time for a full rotation of the disk.

Once the first sector of the data to be accessed has come under the head, data transfer begins. The data-transfer rate is the rate at which data can be retrieved from or stored to the disk.

A disk block is a logical unit of storage allocation and retrieval, and block sizes today typically range from 4 to 16 kb. Data are transferred between disk and main memory in units of blocks. The term page is often used to refer to blocks

A sequence of requests for blocks from disk may be classified as a sequential access pattern or a random access pattern. In a sequential access pattern, successive requests are for successive block numbers, which are on the same track.

In contrast, in a random access pattern, successive requests are for blocks that are randomly located on disk. Each request would require a seek.

The number of I/O operations per second (IOPS), that is, the number of random block accesses that can be satisfied by a disk in a second, depends on the access time, and the block size, and the data transfer rate of the disk.

The final commonly used measure of a disk is the mean time to failure (MTTF), which is a measure of the reliability of the disk. The mean time to failure of a disk (or of any other system) is the amount of time that, on average, we can expect the system to run continuously without any failure. According to vendors' claims, the mean time to failure of disks today ranges from 500,000 to 1,200,000 hours—about 57 to 136 years.

Flash memory

There are two types of flash memory:

1. NOR
2. NAND: predominantly used for data storage. Reading from NAND flash requires an entire page of data to be fetched from NAND into main memory (similar to sectors in magnetic disks).

Solid-state disks (SSDs) are built using NAND flash and provide the same block-oriented interface as disk storage. SSDs can provide much faster random access.

The data transfer rate of SSDs is higher than that of magnetic disks and is usually limited by the interconnect technology.

The power consumption of SSDs is also significantly lower than that of magnetic disks.

Writes to flash memory are a little more complicated. However, once written, a page of flash memory cannot be directly overwritten; it has to be erased and rewritten. The erase operation must be performed on a group of pages, called an erase block.

Flash memory systems limit the impact of both the slow erase speed and the update limits by mapping logical page numbers to physical page numbers. When a logical page is updated, it can be remapped to any already erased physical page, and the original location can be erased later. The logical-to-physical page mapping is replicated in an in-memory translation table for quick access.

Cold y hot data para las páginas que fueron borradas/actualizadas muchas o pocas veces. This principle of evenly distributing erase operations across physical blocks is called wear leveling and is usually performed transparently by flash-memory controllers. All the above

actions are carried out by a layer of software called the flash translation layer; above this layer, flash storage looks identical to magnetic disk storage; but flash is much faster.

❀ **Performance**

- ➔ Number of random block reads per second: Typical values in 2018 are about 10,000 random reads per second. Unlike magnetic disks, SSDs can support multiple random requests in parallel, with 32 parallel requests being commonly supported.
- ➔ The data transfer rate for sequential reads and sequential writes: 400 to 500 megabytes per second.
- ➔ The number of random block writes per second: Typical values in 2018 are about 40,000 random 4-kilobyte writes per second.

Hybrid disk drives are hard-disk systems that combine magnetic storage with a smaller amount of flash memory, which is used as a cache for frequently accessed data.

RAID

Having a large number of disks in a system presents opportunities for improving the rate at which data can be read or written, if the disks are operated in parallel. Potential for improving the reliability of data storage, because redundant information can be stored on multiple disks. Thus, failure of one disk does not lead to loss of data.

A variety of disk-organization techniques, collectively called redundant arrays of independent disks (RAID), have been proposed to achieve improved performance and reliability. RAID systems are used for their higher reliability and higher performance rate, rather than for economic reasons. Another key justification for RAID use is easier management and operations.

❀ **Improvement of reliability via redundancy**

The chance that at least one disk out of a set of N disks will fail is much higher than the chance that a specific single disk will fail. If we store only one copy of the data, then each disk failure will result in loss of a significant amount of data.

The solution to the problem of reliability is to introduce redundancy; that is, we store extra information that is not needed normally but that can be used in the event of failure of a disk to rebuild the lost information.

The simplest (but most expensive) approach to introducing redundancy is to duplicate every disk. This technique is called mirroring. Data will be lost only if the second disk fails before the first failed disk is repaired.

The mean time to failure (where failure is the loss of data) of a mirrored disk depends on the mean time to failure of the individual disks, as well as on the mean time to repair, which is the time it takes (on an average) to replace a failed disk and to restore the data on it.

Then, if the mean time to failure of a single disk is 100,000 hours, and the mean time to repair is 10 hours, the mean time to data loss of a mirrored disk system is aprox 57.000 years.

The assumption of independence (no connection between the failure of one disk and the failure of the other) of disk failures is not valid. As disks age, the probability of failure increases. Power failures are a particular source of concern, since they occur far more frequently than do natural disasters. Power failures are not a concern if there is no data transfer to disk in progress when they occur.

❀ **Improvement in performance via parallelism**

With disk mirroring, the rate at which read requests can be handled is doubled, since read requests can be sent to either disk. The transfer rate of each read is the same as in a single-disk system, but the number of reads per unit time has doubled.

With multiple disks, we can improve the transfer rate as well (or instead) by striping data across multiple disks. In its simplest form, data striping consists of splitting the bits of each byte across multiple disks; such striping is called bit-level striping.

Block-level striping stripes blocks across multiple disks. With an array of n disks, block-level striping assigns logical block i of the disk array to disk (i mod n) + 1.

In summary, there are two main goals of parallelism in a disk system:

1. Load-balance multiple small accesses (block accesses), so that the throughput of such accesses increases.
2. Parallelize large accesses so that the response time of large accesses is reduced.

❀ **RAID levels**

.Mirroring provides high reliability, but it is expensive. Striping provides high data-transfer rates, but does not improve reliability.

Blocks in a RAID system are partitioned into sets. For a given set of blocks, a parity block can be computed and stored on disk.

The ith bit of the parity block is computed as the “exclusive or” (XOR) of the ith bits of the all blocks in the set. If the contents of any one of the blocks in a set is lost due to a failure, the block contents can be recovered by computing the bitwise-XOR of the remaining blocks in the set, along with the parity block. Whenever a block is written, the parity block for its set must be recomputed and written to disk.

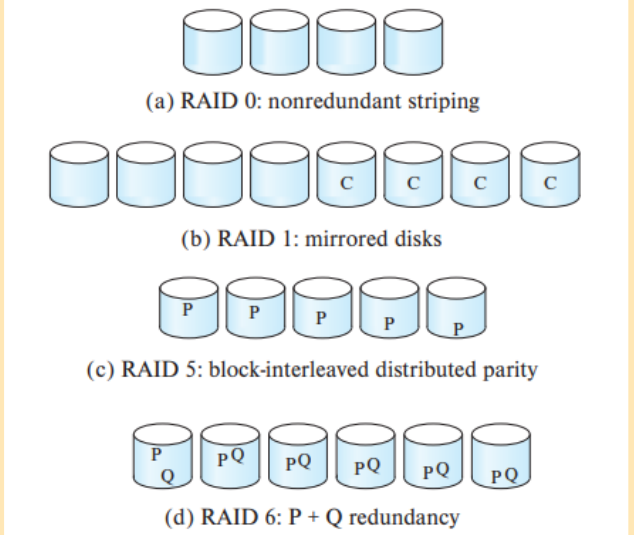
¿Cómo se calcula la paridad?

Los bloques de datos de los discos se agrupan en conjuntos. Para cada conjunto, el sistema calcula un bloque especial llamado "bloque de paridad". Este cálculo se realiza bit por bit aplicando la operación lógica **XOR (O exclusivo)** entre todos los bloques de datos del conjunto.

Recuperación ante fallos

Si un disco falla y se pierde un bloque de datos, ¡no se pierde la información! El sistema puede reconstruir exactamente el contenido del bloque perdido aplicando de nuevo la operación **XOR** entre los bloques que sobrevivieron y el bloque de paridad.

The schemes are classified into RAID levels. P indicates error-correcting bits, and C indicates a second copy of the data.



level 0: Disk arrays with striping at the level of blocks, but without any redundancy.

level 1: Disk mirroring with block striping. En la imagen vemos los discos con datos y sus respectivas copias

level 5: Block-interleaved distributed parity. P es el bloque de paridad

level 6: Is much like RAID level 5, but it stores extra redundant information to guard against multiple disk failures.

❀ **Hardware issues**

RAID can be implemented with no change at the hardware level, using only software modification. Such RAID implementations are called software RAID. Systems with special hardware support are called hardware RAID systems.

Hardware RAID implementations can use non-volatile RAM to record writes before they are performed. In case of power failure, when the system comes back up, it retrieves information about any incomplete writes from non-volatile RAM and then completes the writes. In contrast, with software RAID extra work needs to be done to detect blocks that may have been partially written before power failure.

Even if all writes are completed properly, there is a small chance of a sector in a disk becoming unreadable at some point, even though it was successfully written earlier. Reasons for loss of data on individual sectors could range from manufacturing defects to data corruption on a track when an adjacent track is written repeatedly. When such a failure

happens, if it is detected early the data can be recovered from the remaining disks in the RAID organization.

To minimize the chance of such data loss, good RAID controllers perform scrubbing; that is, during periods when disks are idle, every sector of every disk is read, and if any sector is found to be unreadable, the data are recovered from the remaining disks in the RAID organization, and the sector is written back.

Server hardware is often designed to permit hot swapping; that is, faulty disks can be removed and replaced by new ones without turning power off. The RAID controller can detect that a disk was replaced by a new one and can immediately proceed to reconstruct the data that was on the old disk, and write it to the new disk.

Many RAID implementations assign a spare disk for each array (or for a set of disk arrays). If a disk fails, the spare disk is immediately used as a replacement.

Good RAID implementations have multiple redundant power supplies. Such RAID systems have multiple disk interfaces and multiple interconnections to connect the RAID system to the computer system (or to a network of computer systems). Thus, failure of any single component will not stop the functioning of the RAID system.

❀ **Choice of RAID level**

The factors to be taken into account in choosing a RAID level are:

- ➔ Monetary cost of extra disk-storage requirements.
- ➔ Performance requirements in terms of number of I/O operations per second.
- ➔ Performance when a disk has failed.
- ➔ Performance during rebuild (i.e., while the data in a failed disk are being rebuilt on a new disk).

The rebuild performance of a RAID system may be an important factor if continuous availability of data is required, as it is in high-performance database systems.

- ➔ Rebuilding is easiest for RAID level 1, since data can be copied from another disk;
- ➔ RAID level 0 is used in a few high-performance applications where data safety is not critical, but not anywhere else.
- ➔ RAID level 1 is popular for applications such as storage of log files in a database system, since it offers the best write performance.
- ➔ RAID level 5 has a lower storage overhead than level 1, but it has a higher time overhead for writes.
- ➔ For applications where data are read frequently, and written rarely, level 5 is the preferred choice.
- ➔ RAID level 5 has a significant overhead for random writes, since a single random block write requires 2 block reads (to get the old values of the block and parity block) and 2 block writes to write these blocks back.
- ➔ RAID level 1 is therefore the RAID level of choice for many applications with moderate storage requirements and high random I/O requirements.

- ➔ RAID level 6 offers better reliability than level 1 or 5. Higher storage cost than RAID level 5.

Armar una tabla con la característica especial de cada uno, y lugar de implementación/uso. También una tabla para comparar todos los medios de almacenamiento

Disk-block access

.As we saw earlier, a sequence of requests for blocks from disk may be classified as a sequential access pattern or a random access pattern. A number of techniques have been developed for improving the speed of access to blocks, by minimizing the number of accesses, and in particular minimizing the number of random accesses.

Buffering: Blocks that are read from disk are stored temporarily in an in-memory buffer, to satisfy future requests.

Read-ahead: When a disk block is accessed, consecutive blocks from the same track are read into an in-memory buffer even if there is no pending request for the blocks. In the case of sequential access, such read-ahead ensures that many blocks are already in memory when they are requested, and it minimizes the time wasted in disk seeks and rotational latency per block read.

Scheduling: If several blocks from a cylinder need to be transferred from disk to main memory, we may be able to save access time by requesting the blocks in the order in which they will pass under the heads. Disk-arm–scheduling algorithms attempt to order accesses to tracks in a fashion that increases the number of accesses that can be processed. A commonly used algorithm is the elevator algorithm (lee de adentro hacia afuera, detecta si hay requests en algún track y sigue hacia afuera hasta que termina el escaneo, vuelve a ir para el centro, hasta que no ve ninguna request en los tracks que van al centro y empieza a ir hacia afuera nuevamente).

File organization: To reduce block-access time, we can organize blocks on disk in a way that corresponds closely to the way we expect data to be accessed. For example, if we expect a file to be accessed sequentially, then we should ideally keep all the blocks of the file sequentially on adjacent cylinders. Over time, a sequential file that has multiple small appends may become fragmented; that is, its blocks become scattered all over the disk. To reduce fragmentation, the system can make a backup copy of the data on disk and restore the entire disk.
scheduling = pedirlos como están ordenados; FO = Ordenarlos como los van a pedir
Non-volatile write buffers: Since the contents of main memory are lost in a power failure, information about database updates has to be recorded on disk to survive possible system crashes. The idea is that, when the database system (or the operating system) requests that a block be written to disk, the disk controller writes the block to a non-volatile write buffer and immediately notifies the operating system that the write completed successfully. The controller can subsequently write the data to their destination on disk in a way that minimizes disk arm movement.

Capitulo 13

Data storage structures

In this chapter, we focus on the organization of data stored on the underlying storage media, and how data are accessed.

Database storage architecture

Persistent data are stored on non-volatile storage. Magnetic disks as well as SSDs are block structured devices, that is, data are read or written in units of a block. In contrast, databases deal with records, which are usually much smaller than a block.

Given a set of records, the next decision lies in how to organize them in the file structure; for example, they may be stored in sorted order, in the order they are created, or in an arbitrary order.

For a CPU to access data, it must be in main memory, whereas persistent data must be resident on non-volatile storage such as magnetic disks or SSDs.

Some applications need very fast access to data and have small enough data sizes that the entire database can fit into the main memory of a database server machine. In such cases, we can keep a copy of the entire database in memory. Databases that store the entire database in memory and optimize in-memory data structures as well as query processing and other algorithms used by the database to exploit the memory residency of data are called main-memory databases.

File organization

1. A database is mapped into a number of different files that are maintained by the underlying operating system.
2. These files reside permanently on disks.
3. A file is organized logically as a sequence of records.
4. These records are mapped onto disk blocks.
5. Files are provided as a basic construct in operating systems, so we shall assume the existence of an underlying file system.

We need to consider ways of representing logical data models in terms of files.

Each file is also logically partitioned into fixed-length storage units called blocks, which are the units of both storage allocation and data transfer. Many databases allow the block size to be specified when a database instance is created. Larger block sizes can be useful in some database applications.

A block may contain several records; the exact set of records that a block contains is determined by the form of physical data organization being used. We shall assume that no record is larger than a block. Each record is entirely contained in a single block;

In a relational database, tuples of distinct relations are generally of different sizes. One approach to mapping the database to files is to use several files and to store records of only one fixed length in any given file. An alternative is to structure our files so that we can accommodate multiple lengths for records; however, files of fixed-length records are easier to implement than are files of variable-length records.

❀ Fixed-length records

```
type instructor = record
                    ID varchar (5);
                    name varchar(20);
                    dept_name varchar (20);
                    salary numeric (8,2);
                  end
```

En este caso tendríamos 53 bytes por tupla

However, there are two problems with this simple approach:

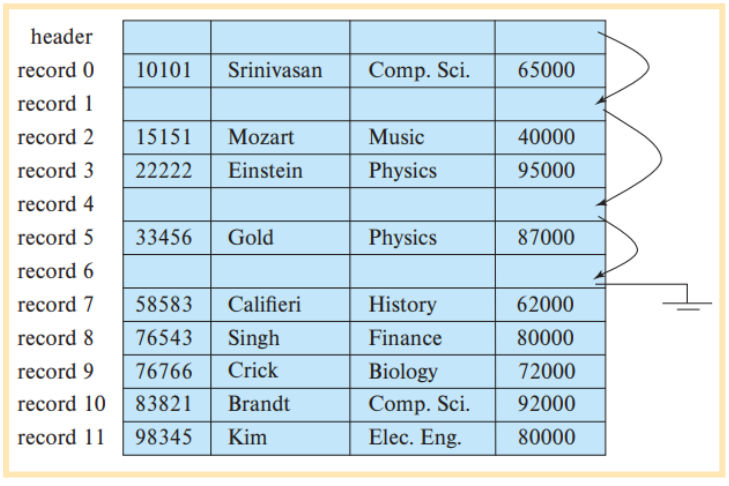
1. Unless the block size happens to be a multiple of 53 (which is unlikely), some records will cross block boundaries (two block accesses).
2. It is difficult to delete a record from this structure. The space occupied by the record to be deleted must be filled with some other record of the file, or we must have a way of marking deleted records so that they can be ignored

To avoid the first problem, we allocate only as many records to a block as would fit entirely in the block.

When a record is deleted, we could move the record that comes after it into the space formerly occupied by the deleted record, and so on, until every record following the deleted record has been moved ahead (It might be easier to move the final record of the file into the space occupied by the deleted record).

Since insertions tend to be more frequent than deletions, it is acceptable to leave open the space occupied by the deleted record and to wait for a subsequent insertion before reusing the space. A simple marker on a deleted record is not sufficient, since it is hard to find this available space when an insertion is being done. Thus, we need to introduce an additional structure.

At the beginning of the file, we allocate a certain number of bytes as a file header to store the address of the first record whose contents are deleted (and other info). We can think of these stored addresses as pointers, since they point to the location of a record.



header				
record 0	10101	Srinivasan	Comp. Sci.	65000
record 1				
record 2	15151	Mozart	Music	40000
record 3	22222	Einstein	Physics	95000
record 4				
record 5	33456	Gold	Physics	87000
record 6				
record 7	58583	Califieri	History	62000
record 8	76543	Singh	Finance	80000
record 9	76766	Crick	Biology	72000
record 10	83821	Brandt	Comp. Sci.	92000
record 11	98345	Kim	Elec. Eng.	80000

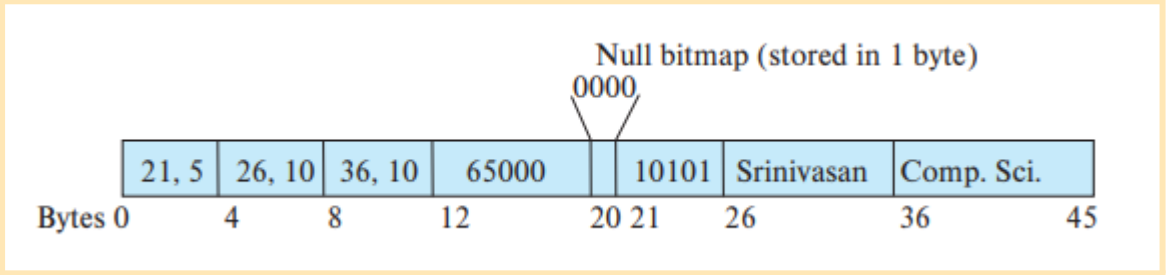
❀ Variable-length records

Variable-length records arise in database systems due to several reasons. The most common reason is the presence of variable length fields, such as strings. Different techniques for implementing variable-length records exist. Two different problems must be solved by any such technique:

1. How to represent a single record in such a way that individual attributes can be extracted easily, even if they are of variable length
2. How to store variable-length records within a block, such that records in a block can be extracted easily

The representation of a record with variable-length attributes typically has two parts: an initial part with fixed-length information, whose structure is the same for all records of the same relation, followed by the contents of variable-length attributes.

Variable-length attributes, such as varchar types, are represented in the initial part of the record by a pair (offset, length), where offset denotes where the data for that attribute begins within the record, and length is the length in bytes of the variable-sized attribute.

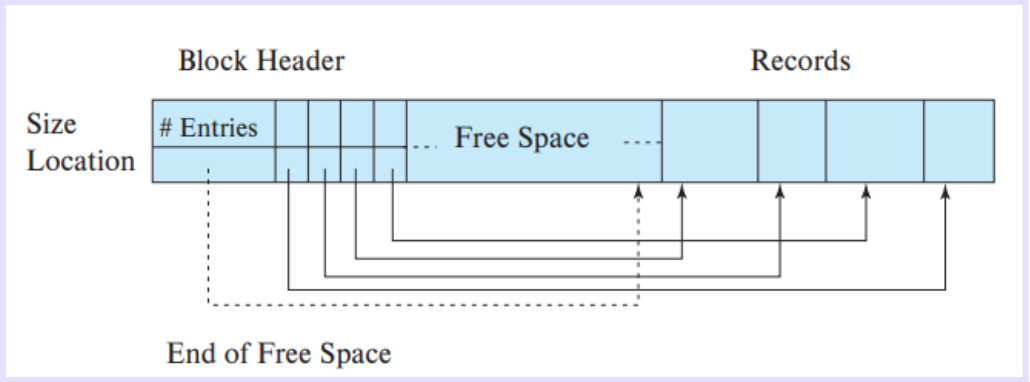


Instructor record whose first three attributes *ID, name, and dept_name* are variable-length strings, and whose fourth attribute salary is a fixed-sized number.

The figure also illustrates the use of a null bitmap, which indicates which attributes of the record have a null value. In this particular record, if the salary were null, the fourth bit of the bitmap would be set to 1, and the salary value stored in bytes 12 through 19 would be ignored.

The slotted-page structure is commonly used for organizing records within a block. There is a header at the beginning of each block, containing the following information:

➔ The number of record entries in the header
➔ The end of free space in the block
➔ An array whose entries contain the location and size of each record



The free space in the block is contiguous between the final entry in the header array and the first record. If a record is deleted, the space that it occupies is freed, and its entry is set to *deleted*.

The slotted-page structure requires that there be no pointers that point directly to records. Instead, pointers must point to the entry in the header that contains the actual location of the record.

❀ **Storing large objects**

Databases often store data that can be much larger than a disk block. Many databases internally restrict the size of a record to be no larger than the size of a block.

Large objects may be stored either as files in a file system area managed by the database, or as file structures stored in and managed by the database.

There are some performance issues with storing very large objects in databases. The efficiency of accessing large objects via database interfaces is one concern. A second concern is the size of database backups. Storing large objects in the database can result in a large increase in the size of the database dumps.

Storing data in files outside the database can result in file names in the database pointing to files that do not exist, perhaps because they have been deleted, which results in a form of foreign-key constraint violation. Some databases support file system integration with the database, to ensure that constraints are satisfied, and to ensure that access authorizations are enforced.

Organization of records in files

Several of the possible ways of organizing records in files are:

- ➔ Heap file organization: Any record can be placed anywhere in the file where there is space for the record. There is no ordering of records.

- ➔ Sequential file organization: Records are stored in sequential order, according to the value of a “search key” of each record.

- ➔ Multitable clustering file organization: Generally, a separate file or set of files is used to store the records of each relation. However, in a multitable clustering file organization, records of several different relations are stored in the same file, and in fact in the same block within a file, to reduce the cost of certain join operations.

- ➔ B+-tree file organization: The B+-tree file organization is related to the B+-tree index structure described in that chapter and can provide efficient ordered access to records even if there are a large number of insert, delete, or update operations. Further, it supports very efficient access to specific records, based on the search key.

➔ Hashing file organization: A hash function is computed on some attribute of each record. The result of the hash function specifies in which block of the file the record should be placed.

❀ **Heap file organization**

Once placed in a particular location, the record is not usually moved. (se pone donde haya espacio)

When a record is inserted in a file, one option for choosing the location is to always add it at the end of the file. However, if records get deleted, it makes sense to use the space thus freed up to store new records. It is important for a database system to be able to efficiently find blocks that have free space, without having to sequentially search through all the blocks of the file.

Most databases use a space-efficient data structure called a free-space map to track which blocks have free space to store records. The free-space map is commonly represented by an array containing 1 entry for each block in the relation. Each entry represents a fraction f such that at least a fraction f of the space in the block is free.

4	2	1	4	7	3	6	5	1	2	0	1	1	0	5	6

A. Estructura de un Array (Arreglo)

El mapa de espacio libre se representa comúnmente como un gran arreglo en el que **cada posición (índice) corresponde exactamente a un bloque del archivo de datos.** Si tu archivo tiene 16 bloques, el mapa tendrá 16 entradas. PDF + 1

B. Almacenamiento en Fracciones (f)

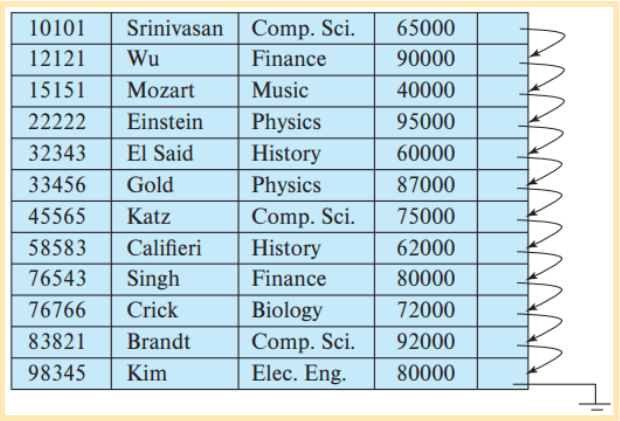
Cada entrada en este mapa no guarda el número exacto de bytes libres, sino una **fracción (f)** que representa el espacio disponible estimado. PDF

- **Ejemplo teórico del libro:** Imagina que el sistema utiliza **3 bits** para almacenar esta fracción en cada entrada. Al usar 3 bits, los números posibles van del 0 al 7, y para saber la fracción real de espacio libre, el valor de la entrada se divide por 8. PDF + 1
  - Si una entrada tiene el valor `7`, significa que al menos $\frac{7}{8}$ del bloque está libre. PDF
  - Si tiene un `4`, significa que al menos $\frac{4}{8}$ (la mitad) del bloque está libre.

The free-space map is written periodically; as a result, the free-space map on disk may be outdated, and when a database starts up, it may get outdated data about available free space.

❀ **Sequential file organization**

A sequential file is designed for efficient processing of records in sorted order based on some search key. A search key is any attribute or set of attributes; it need not be the primary key, or even a superkey.



10101	Srinivasan	Comp. Sci.	65000	
12121	Wu	Finance	90000	
15151	Mozart	Music	40000	
22222	Einstein	Physics	95000	
32343	El Said	History	60000	
33456	Gold	Physics	87000	
45565	Katz	Comp. Sci.	75000	
58583	Califieri	History	62000	
76543	Singh	Finance	80000	
76766	Crick	Biology	72000	
83821	Brandt	Comp. Sci.	92000	
98345	Kim	Elec. Eng.	80000	

To permit fast retrieval of records in search-key order, we chain together records by pointers. The pointer in each record points to the next record in search-key order.

To minimize the number of block accesses in sequential file processing, we store records physically in search-key order, or as close to search-key order as possible.

It is difficult, however, to maintain physical sequential order as records are inserted and deleted, since it is costly to move many records as a result of a single insertion or deletion. We can manage deletion by using pointer chains

Two rules for managing insertion:

1. Locate the record in the file that comes before the record to be inserted in searchkey order.
2. If there is a free record within the same block as this record, insert the new record there. Otherwise, insert the new record in an overflow block.

❀ Multitable Clustering File Organization

Most relational database systems store each relation in a separate file, or a separate set of files. Thus, each file, and as a result, each block, stores records of only one relation, in such a design. However, in some cases it can be useful to store records of more than one relation in a single block.

This structure allows for efficient processing of the join (between two relations).

A multitable clustering file organization is a file organization that stores related records of two or more relations in each block. The cluster key is the attribute that defines which records are stored together; in our preceding example, the cluster key is dept_name.

dept_name	*building*	*budget*
Comp. Sci.	Taylor	100000
Physics	Watson	70000

ID	*name*	*dept_name*	*salary*
10101	Srinivasan	Comp. Sci.	65000
33456	Gold	Physics	87000
45565	Katz	Comp. Sci.	75000
83821	Brandt	Comp. Sci.	92000

Comp. Sci.	Taylor	100000	
10101	Srinivasan	Comp. Sci.	65000
45565	Katz	Comp. Sci.	75000
83821	Brandt	Comp. Sci.	92000
Physics	Watson	70000	
33456	Gold	Physics	87000

❀ **Partitioning**

Many databases allow the records in a relation to be partitioned into smaller relations that are stored separately. Such table partitioning is typically done on the basis of an attribute value.

The cost of some operations, such as finding free space for a record, increases with relation size; by reducing the size of each relation, partitioning helps reduce such overheads.

Data-dictionary storage

In general, such “data about data” are referred to as metadata. Relational schemas and other metadata about relations are stored in a structure called the data dictionary or system catalog. We store:

- ➔ Names of the relations
- ➔ Names of the attributes of each relation
- ➔ Domains and lengths of attributes
- ➔ Names of views defined on the database, and definitions of those views
- ➔ Integrity constraints
- ➔ Names of users, the default schemas of the users, and passwords or other information to authenticate users
- ➔ Information about authorizations for each user DATA ON USERS OF THE SYSTEM

The data dictionary may also note the storage organization of relations.

También almacena donde están almacenadas las relaciones, información sobre los índices de cada relación.

All this metadata information constitutes a miniature database. By using database relations to store system metadata, we simplify the overall structure of the system and harness the full power of the database for fast access to system data. The exact choice of how to represent system metadata by relations must be made by the system designers.

Database buffer

Since data access from disk is much slower than in-memory data access, a major goal of the database system is to minimize the number of block transfers between the disk and memory.

One way to reduce the number of disk accesses is to keep as many blocks as possible in main memory. The buffer is that part of main memory available for storage of copies of disk blocks. The subsystem responsible for the allocation of buffer space is called the buffer manager.

❀ Buffer manager

Programs in a database system make requests on the buffer manager when they need a block from disk. If the block is already in the buffer, the buffer manager passes the address of the block in main memory to the requester. If the block is not in the buffer, the buffer manager first allocates space in the buffer for the block, throwing out some other block, if necessary, to make space for the new block.

Replacement strategy

When there is no room left in the buffer, a block must be evicted, that is, removed, from the buffer before a new one can be read in. Most operating systems use a least recently used (LRU) scheme.

Pinned blocks

The process executes a pin operation on the block; the buffer manager never evicts a pinned block. When it has finished reading data, the process should execute an unpin operation, allowing the block to be evicted when required.

The block cannot be evicted until all processes that have executed a pin have then executed an unpin operation. A simple way to ensure this property is to keep a pin count for each buffer block.

Shared and exclusive locks on buffer

The locking system provided by the buffer manager allows a database process to lock a buffer block either in shared mode or in exclusive mode before accessing the block, and to release the lock later, after the access is completed.

Output of blocks

It makes sense to not wait until the buffer space is needed, but to rather write out (output) updated blocks ahead of such a need. Then, when space is required in the buffer, a block that has already been written out can be evicted.

Forced output of blocks

There are situations in which it is necessary to write a block to disk, to ensure that certain data on disk are in a consistent state. Such a write is called a forced output of a block.

❀ Buffer replacement strategies

The goal of a replacement strategy for blocks in the buffer is to minimize accesses to the disk. It is not possible to predict accurately which blocks will be referenced. If a block must be replaced, the least recently referenced block is replaced. This approach is called the least recently used (LRU) block-replacement scheme.

```
select *
from instructor natural join department;
```

The buffer manager should be instructed to free the space occupied by an instructor block as soon as the final tuple has been processed. This buffer-management strategy is called the toss-immediate strategy. For some systems, the optimal strategy for block replacement for the above procedure is the most recently used (MRU) strategy.

The strategy that the buffer manager uses for block replacement is influenced by factors other than the time at which the block will be referenced again.

Reordering of writes and recovery

Database buffers allow writes to be performed in-memory and output to disk at a later time, possibly in an order different from the order in which the writes were performed. Such reordering can lead to inconsistent data on disk in the event of a system crash.

However, most disks do not come with a non-volatile write buffer; instead, modern file systems assign a disk for storing a log of the writes in the order that they are performed. Such a disk is called a log disk.

For each write, the log contains the block number to be written to, and the data to be written, in the order in which the writes were performed. File systems that support log disks as above are called journaling file systems.

Column oriented storage

In contrast, in column-oriented storage, also called a columnar storage, each attribute of a relation is stored separately, with values of the attribute from successive tuples stored at successive positions in the file.

In the simplest form of column-oriented storage, each attribute is stored in a separate file.

Column-oriented storage thus has the drawback that fetching multiple attributes of a single tuple requires multiple I/O operations.

However, column-oriented storage is well suited for data analysis queries, which process many rows of a relation, but often only access some of the attributes. The reasons are as follows:

- ➔ Reduced I/O: When a query needs to access only a few attributes of a relation with a large number of attributes, the remaining attributes need not be fetched from disk into memory.
- ➔ Improved CPU cache performance: When the query processor fetches the contents of a particular attribute, with modern CPU architectures multiple consecutive bytes, called a cache line, are fetched from memory to CPU cache.

- ➔ Improved compression: Storing values of the same type together significantly increases the effectiveness of compression, when compared to compressing data stored in row format.
- ➔ Vector processing: Many modern CPU architectures support vector processing, which allows a CPU operation to be applied in parallel on a number of elements of an array.

Databases that use column-oriented storage are referred to as column stores, while databases that use row-oriented storage are referred to as row stores.
Drawbacks: Cost of tuple reconstruction, Cost of tuple deletion and update, Cost of decompression.

A sequence of tuples occupying several hundred megabytes is broken up into a columnar representation called a stripe.

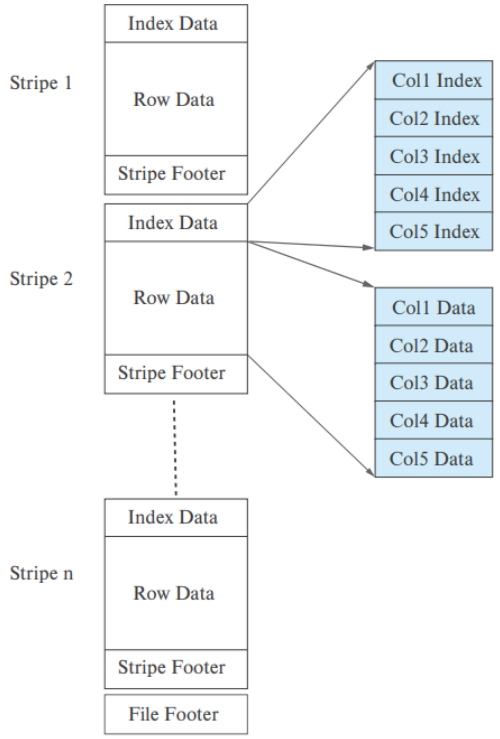


Figure 13.15 Columnar data representation in the ORC file format.

Some column-store systems allow groups of columns that are often accessed together to be stored together, instead of breaking up each column into a different file.

The choice of which attributes to store together depends on the query workload.

Using column-oriented storage within the block provides the benefits of more efficient memory access and cache usage, as well as the potential for using vector processing on the data.

Storage organization in main-memory databases

Today, main-memory sizes are large enough, and main memory is cheap enough, that many organizational databases fit entirely in memory.

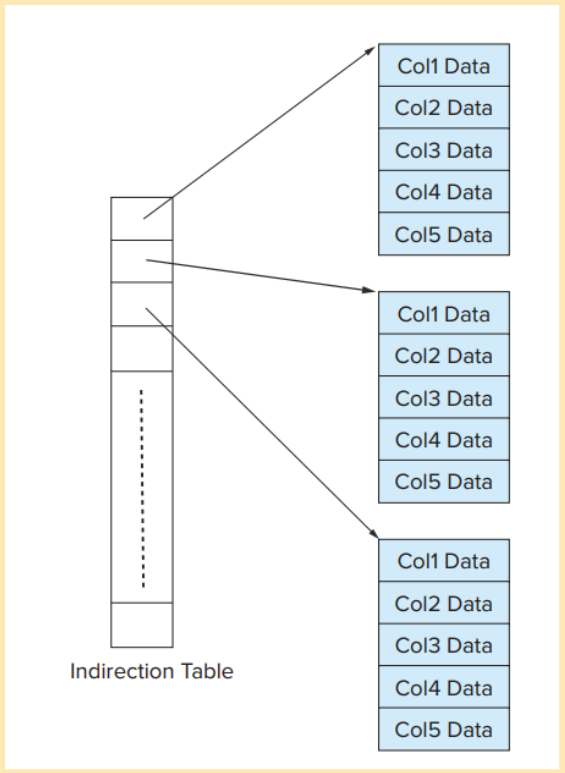
Such large main memories can be used by allocating a large amount of memory to the database buffer, which will allow the entire database to be loaded into buffer, avoiding disk I/O operations for reading data; updated blocks still have to be written back to disk for persistence.
A main-memory database is a database where all data reside in memory.

In contrast, in a main-memory database, it is possible to keep direct pointers to records in memory, and accessing a record is just an in-memory pointer traversal, which is a very efficient operation. This is possible as long as records are not moved around.

Many main-memory databases do not use a slotted-page structure for allocating records. Instead they directly allocate records in main memory, and ensure that records never get moved due to updates to other records.

However, a problem with direct allocation of records is that memory may get fragmented if variable sized records are repeatedly inserted and deleted. The database must ensure that main memory does not get fragmented over time.

For a column-oriented storage scheme the logical array for a column may be divided into multiple physical arrays. An indirection table stores pointers to all the physical arrays.



Capitulo 14

Review Terms

- Index type
  - Ordered indices
  - Hash indices
- Evaluation factors
  - Access types
  - Access time
  - Insertion time
  - Deletion time
  - Space overhead
- Search key
- Ordered indices
  - Ordered index
  - Clustering index
  - Primary indices;
  - Nonclustering indices
  - Secondary indices
  - Index-sequential files
- Index entry
- Index record
- Dense index
- Sparse index
- Multilevel indices
- Nonunique search key
- Composite search key
- B^+-tree index files
  - Balanced tree
  - Leaf nodes
  - Nonleaf nodes
  - Internal nodes
  - Range queries
  - Node split
  - Node coalesce
  - Redistribute of pointers
  - Uniquifier
- B^+-tree extensions
  - Prefix compression
  - Bulk loading
  - Bottom-up B^+-tree construction
- B-tree indices
- Hash file organization
  - Hash function
  - Bucket
  - Overflow chaining
  - Closed addressing
  - Closed hashing
  - Bucket overflow
  - Skew
  - Static hashing
  - Dynamic hashing
- Multiple-key access
- Covering indices
- Write-optimized index structure
  - Log-structured merge (LSM) tree
  - Stepped-merge index
  - Buffer tree
- Bitmap index
- Bitmap intersection
- Indexing of spatial data
  - Range queries
  - Nearest neighbor queries
  - k-d tree
  - k-d-B tree
  - Quadtrees
  - R-tree
  - Bounding box
- Temporal indices
- Time interval
- Closed interval
- Open interval

Indexing

It is inefficient for the system to read every tuple in the instructor relation to check if the *dept_name* value matches our query. Ideally, the system should be able to locate these records directly.

Basic concepts

An index for a file in a database system works in much the same way as the index in this textbook. The words in the index are in sorted order, making it easy to find the word we want.

Without indices, every query would end up reading the entire contents of every relation that it uses; doing so would be unreasonably expensive for queries that only fetch a few records.

There are two basic kinds of indices:

- ➔ Ordered indices: Based on a sorted ordering of the values.
- ➔ Hash indices: Based on a uniform distribution of values across a range of buckets. The bucket to which a value is assigned is determined by a function, called a hash function.

Each technique is best suited to particular database applications. Each technique must be evaluated on the basis of these factors:

- ➔ Access types: The types of access that are supported efficiently. Access types can include finding records with a specified attribute value and finding records whose attribute values fall in a specified range.
- ➔ Access time: The time it takes to find a particular data item, or set of items, using the technique in question.
- ➔ Insertion time: The time it takes to insert a new data item. This value includes the time it takes to find the correct place to insert the new data item, as well as the time it takes to update the index structure.
- ➔ Deletion time: The time it takes to delete a data item. This value includes the time it takes to find the item to be deleted, as well as the time it takes to update the index structure.
- ➔ Space overhead: The additional space occupied by an index structure. Provided that the amount of additional space is moderate, it is usually worthwhile to sacrifice the space to achieve improved performance.

An attribute or set of attributes used to look up records in a file is called a search key.

Ordered indices

An ordered index stores the values of the search keys in sorted order and associates with each search key the records that contain it.

If the file containing the records is sequentially ordered, a clustering index is an index whose search key also defines the sequential order of the file. Clustering indices are also called primary indices; the term primary index may appear to denote an index on a primary key.

Indices whose search key specifies an order different from the sequential order of the file are called nonclustering indices, or secondary indices.

Files that are ordered sequentially, and have a clustering index on the search key, are called index-sequential files.

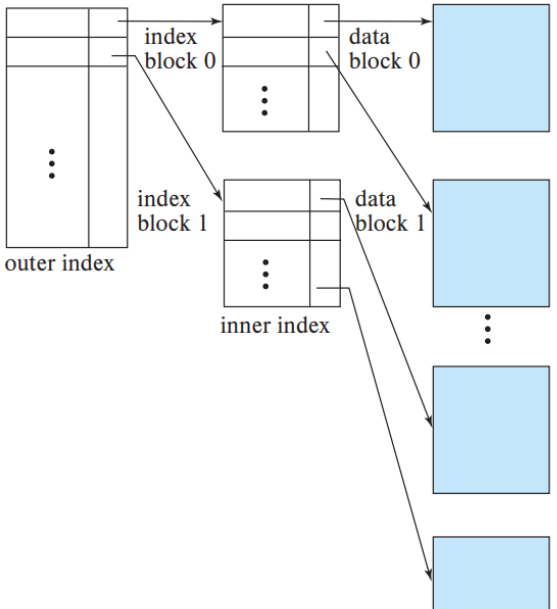
❀ **Dense and sparse indices:** An index entry, or index record, consists of a search-key value and pointers to one or more records with that value as their search-key value.

- ➔ Dense index: An index entry appears for every search-key value in the file. In a dense clustering index, the index record contains the search-key value and a pointer to the first data record with that search-key value. The rest of the records with the same search-key value would be stored sequentially after the first record, since, because the index is a clustering one, records are sorted on the same search key. In a dense nonclustering index, the index must store a list of pointers to all records with the same search-key value.
- ➔ Sparse index: An index entry appears for only some of the searchkey values. Sparse indices can be used only if the relation is stored in sorted order of the search key; that is, if the index is a clustering index. As is true in dense indices, each index entry contains a search-key value and a pointer to the first data record with that search-key value. To locate a record, we find the index entry with the largest search-key value that is less than or equal to the search-key value for which we are looking. We start at the record pointed to by that index entry and follow the pointers in the file until we find the desired record.

BUSCAR DIAGRAMAS EN EL LIBRO (pag 657 reader)

❀ **Multilevel indices:** A sequential search requires b sequential block reads, which may take even longer (although in some cases the lower cost of sequential block reads may result in sequential search being faster than a binary search). Thus, the process of searching a large index may be costly. To deal with this problem, we treat the index just as we would treat any other sequential file, and we construct a sparse outer index on the original index, which we now call the inner index.

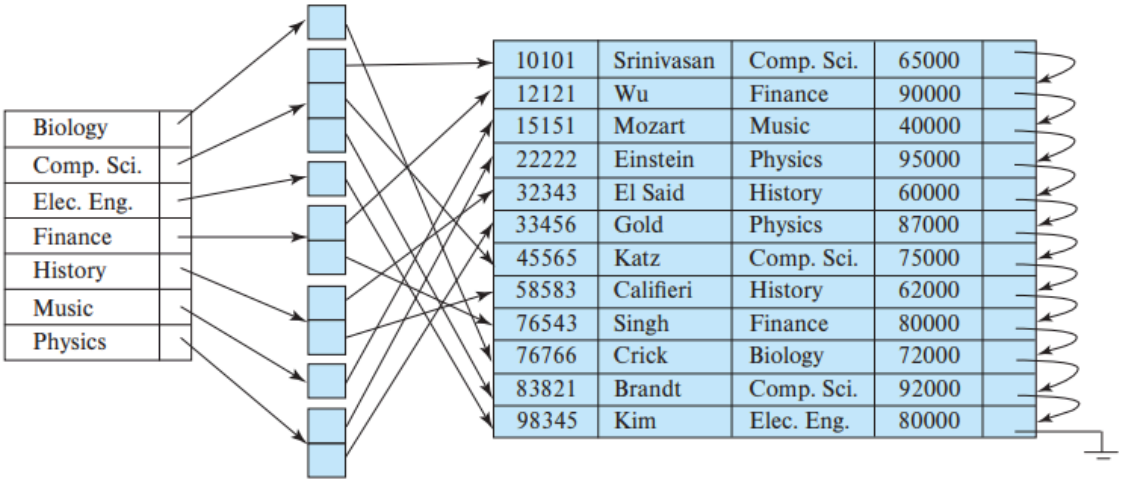
Indices with two or more levels are called multilevel indices.



❀**Index update:** Every index must be updated whenever a record is either inserted into or deleted from the file. We only need to consider insertion and deletion on an index. BUSCAR INSERT Y DELETE PARA CASOS DE DENSE Y SPARSE

❀ **Secondary indices:** Secondary indices must be dense, with an index entry for every search-key value, and a pointer to every record in the file. In general, secondary indices may have a different structure from clustering indices.

If a relation can have more than one record containing the same search key value, the search key is said to be a nonunique search key. Secondary indices improve the performance of queries that use keys other than the search key of the clustering index.

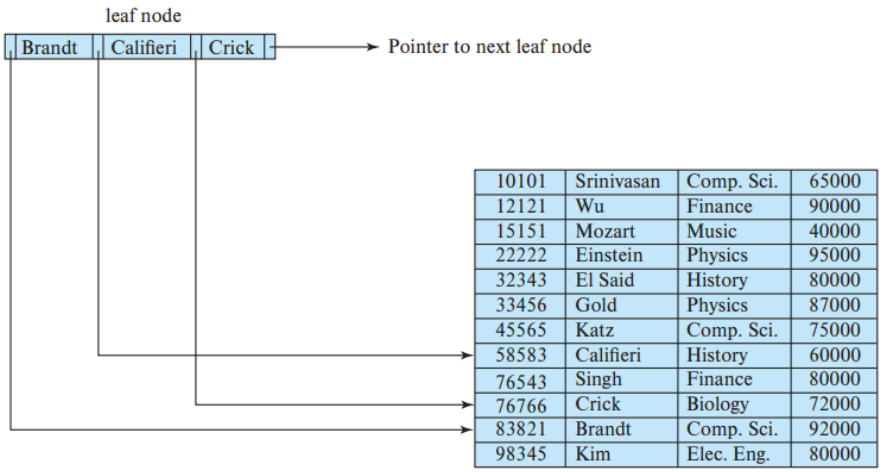


❀ **Indices on multiple keys:** A search key containing more than one attribute is referred to as a composite search key. The structure of the index is a list of attributes (represented as a tuple of values, of the form (a1, …, an)).

B+ Tree index files

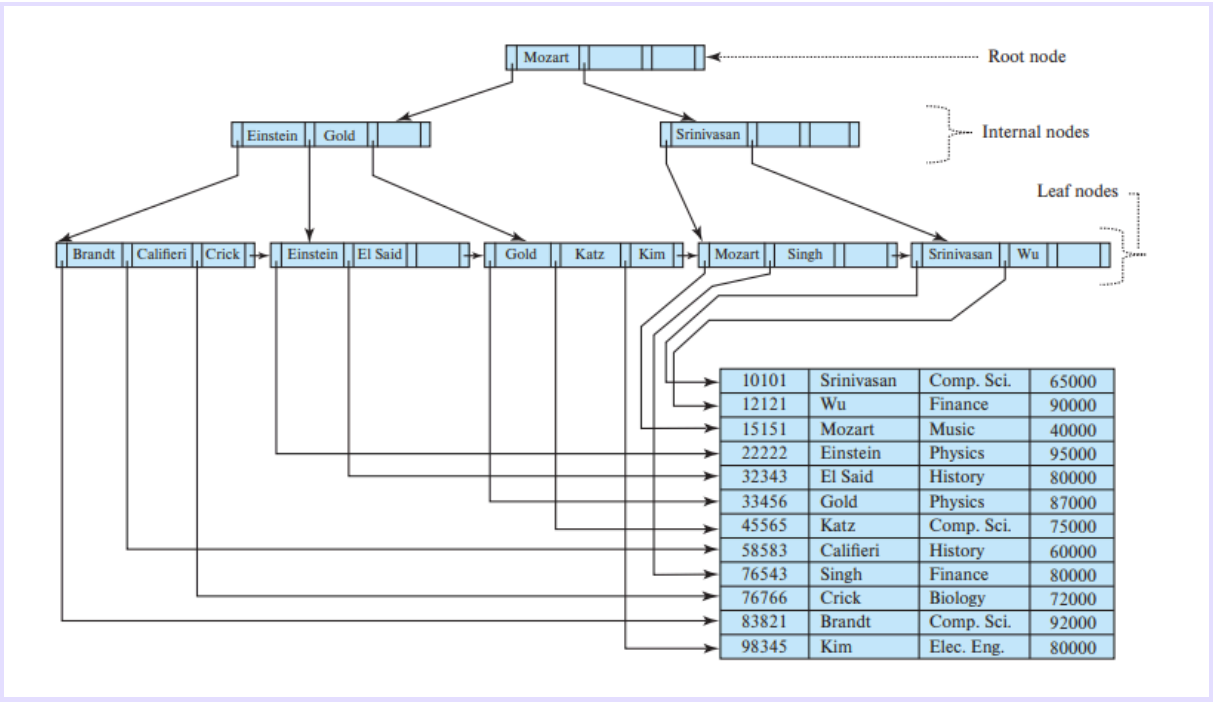
A B+tree index takes the form of a balanced tree in which every path from the root of the tree to a leaf of the tree is of the same length. Each nonleaf node in the tree (other than the root) has between $\lceil n/2 \rceil$ and n children (n is fixed for a particular tree).

❀ **Structure of a B+tree:** A B+-tree index is a multilevel index. It contains up to $n - 1$ search-key values $K_1, K_2, \ldots, K_{n-1}$, and n pointers $P_1, P_2, \ldots, P_n$. The search-key values within a node are kept in sorted order; thus, if $i < j$, then $K_i < K_j$.



We consider first the structure of the leaf nodes → pointer Pi points to a file record with search-key value Ki

The nonleaf nodes of the B+-tree form a multilevel (sparse) index on the leaf nodes.



One way to handle the case of nonunique search keys is to modify the tree structure to store each search key at a leaf node as many times as it appears in records. The condition that Ki < Kj if i < j will need to be modified to Ki ≤ Kj. This approach can result in duplicate search key values at internal nodes, making the insertion and deletion procedures more complicated and expensive. This approach is more complicated and can result in inefficient access, especially if the number of record pointers for a particular key is very large.

Most database implementations instead make search keys unique as follows: the unique composite search key (ai , Ap) is used instead of ai when building the index.

❀ **Queries on B+trees:** The function starts at the root of the tree and traverses the tree down until it reaches a leaf node that would contain the specified value if it exists in the tree.

For range queries: first traverses to a leaf in a manner similar to find(lb); the leaf may or may not actually contain value lb. It then steps through records in that and subsequent leaf nodes collecting pointers to all records with key values C.Ki s.t. lb ≤ C.Ki ≤ ub into a set resultSet.

Cost of querying on a B+-tree index: In processing a query, we traverse a path in the tree from the root to some leaf node. If there are N records in the file, the path is no longer than ⌈log⌈n/2⌉(N)⌉.

❀ **Updates, insertions and deletions on B+trees:** Estas nos las jugamos al truco pa

❀ **Complexity on B+trees:** It is the speed of operation on B+-trees that makes them a frequently used index structure in database implementations. Although B+-trees only guarantee that nodes will be at least half full, if entries are inserted in random order, nodes can be expected to be more than two-thirds full on average. If entries are inserted in sorted order, on the other hand, nodes will be only half full.

❀ **Nonunique search keys:** *"make search keys unique by creating a composite search key containing the original search key and extra attributes"*. The extra attribute is called a uniquifier attribute.

One alternative is to store each key value only once in the tree, and to keep a bucket (or list) of record pointers with a search-key value. Another option is to store the search key value once per record.
A major problem with both these approaches, as compared to the unique searchkey approach, lies in the efficiency of record deletion.

B+ Tree extensions

In a B+-tree file organization, the leaf nodes of the tree store records, instead of storing pointers to records.
When we use a B+-tree for file organization, space utilization is particularly important, since the space occupied by the records is likely to be much more than the space occupied by keys and pointers.

❀ **Secondary Indices and Record Relocation:** Some file organizations, may change the location of records even when the records have not been updated. As an example, when a leaf node is split in a B+-tree file organization, a number of records are moved to a new node. In such cases, all secondary indices that store pointers to the relocated records would have to be updated.

❀ **Indexing strings:** With variable-length search keys, different nodes can have different fanouts even if they are full. The fanout of nodes can be increased by using a technique called prefix compression(we do not store the entire search key value at nonleaf nodes).

❀ **Bulk loading of indices:** Insertion of a large number of entries at a time into an index is referred to as bulk loading of the index. An efficient way to perform bulk loading of an index is as follows: Create a temporary file containing index entries for the relation, then sort the file on the search key of the index being constructed, and finally scan the sorted file and insert the entries into the index.

❀ **B-tree index files:** B-tree indices are similar to B+-tree indices. The primary distinction between the two approaches is that a B-tree eliminates the redundant storage of search-key values. A B-tree allows search-key values to appear only once. Every search-key value appears in some leaf node.

B-tree has a smaller fanout and therefore may have depth greater than that of the corresponding B+-tree. Deletion in a B-tree is more complicated. The space advantages of B-trees are marginal for large indices and usually do not outweigh the disadvantages that we have noted. Pretty much all database-system implementations use the B+-tree data structure.

❀ **Indexing on flash storage:**

- ➔ SSDs provide much faster random I/O operations than magnetic disks (lookups run much faster with data on SSDs, compared to data on magnetic disks.)
- ➔ The performance of write operations is more complicated with flash storage.
- ➔ An important difference between flash storage and magnetic disks is that flash storage does not permit in-place updates to data at the physical level. Every update turns into a copy+write of an entire flash-storage page.
- ➔ The optimum B+-tree node size for flash storage is smaller than that with magnetic disk, since flash pages are smaller than disk blocks (it makes sense for tree-node sizes to match to flash pages.
- ➔ Bulk loading provides significant performance benefits.

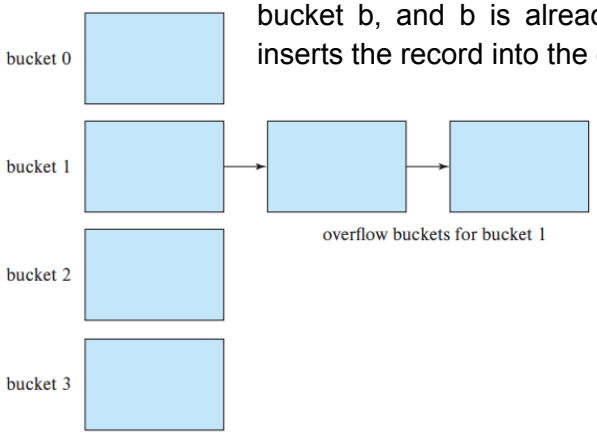
❀ **Indexing on main memory:**

- ➔ Since memory is costlier than disk space, internal data structures in main memory databases have to be designed to reduce space requirements.
- ➔ Data structures that require traversal of multiple pointers are acceptable for in-memory data. Thus, tree structures in main memory databases can be relatively deep, unlike B+-trees.
- ➔ Instead of scanning data linearly or using binary search within a node, the tree-structure within the large B+-tree node is used to access the data with a minimal number of cache misses.

Hash indices

We shall use the term bucket to denote a unit of storage that can store one or more records.

- ➔ In a hash file organization, instead of record pointers, buckets store the actual records; such structures only make sense with disk-resident data.
- ➔ No usamos direccionamiento abierto porque las operaciones de borrado no son eficientes.
- ➔ If the bucket does not have enough space, a bucket overflow is said to occur. We handle bucket overflow by using overflow buckets. If a record must be inserted into a bucket b, and b is already full, the system provides an overflow bucket for b and inserts the record into the overflow bucket.



Hash indexing as described above, where the number of buckets is fixed when the index is created, is called static hashing.

Several schemes have been proposed that allow the number of buckets to be increased in a more incremental fashion. Such schemes are called dynamic hashing techniques;

Multiple-key access

For certain types of queries, it is advantageous to use multiple indices if they exist, or to use an index built on a multiattribute search key.

❀ **Using Multiple Single-Key Indices:** Consider the following query: “Find all instructors in the Finance department with salary equal to $80,000.”

There are three strategies possible for processing this query:

1. Use the index on *dept_name* to find all records pertaining to the Finance department. Examine each such record to see whether *salary* = 80000.
2. Use the index on *salary* to find all records pertaining to instructors with *salary* of $80,000. Examine each such record to see whether the *dept_name* is “Finance”.
3. Use the index on *dept_name* to find pointers to all records pertaining to the Finance department. Also, use the index on *salary* to find pointers to all records pertaining to instructors with a salary of $80,000. Take the intersection of these two sets of pointers. Those pointers that are in the intersection point to records pertaining to instructors of the Finance department and with *salary* of $80,000.

La última estrategia no es tan buena si ambos atributos tienen muchos o pocos registros.

❀ **Indices on multiple keys:** An alternative strategy for this case is to create and use an index on a composite search key (dept name, salary).

❀ **Covering Indices:** Covering indices are indices that store the values of some attributes (other than the search-key attributes) along with the pointers to the record.
Storing extra attribute values is useful with secondary indices, since they allow us to answer some queries using just the index, without even looking up the actual records.

Creation of indices

Indices can be created using the following syntax, which is supported by most databases:

create index <index-name> **on** <relation-name> (<attribute-list>);
and **drop index** <index-name>; to drop it

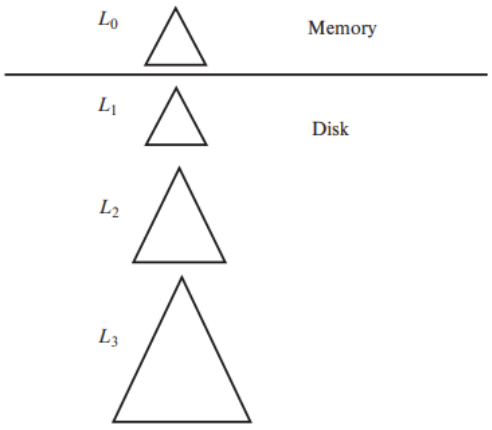
The attribute-list is the list of attributes of the relations that form the search key for the index.
To declare that an attribute or list of attributes is a candidate key, we can use the syntax create unique index in place of create index above.

However, indices do have a cost, since they have to be updated whenever there is an update to the underlying relation. Creating too many indices would slow down update processing, since each update would have to also update all affected indices.
It is important when building an application to figure out which indices are important for performance and to create them before the application goes live.

Although most database systems do not automatically create them, it is often a good idea to create indices on foreign-key attributes, too.

Write-optimized index structures

One of the drawbacks of the B+-tree index structure is that performance can be quite poor with random writes.
Now suppose writes or inserts are done in an order that does not match the sort order of the index. Then, each write/insert is likely to touch a different leaf node; if the number of leaf nodes is significantly larger than the buffer size, most of these leaf accesses would require a random read operation, as well as a subsequent write operation to write the updated leaf page back to disk.
The log-structured merge tree or LSM tree and its variants are write-optimized index structures that have seen very significant adoption.



❀ **LSM trees:** An LSM tree consists of several B+-trees, starting with an in-memory tree, called L0, and on-disk trees L1, L2, …, Lk (where k is called the level). When a record is first inserted into an LSM tree, it is inserted into the in-memory B+-tree structure L0. A fairly large amount of memory space is allocated for this tree.
The tree grows as more inserts are processed, until it fills the memory allocated to it. At this point, we need to move data from the in-memory structure to a B+-tree on disk.
(pág 696 si querés saber más xd)

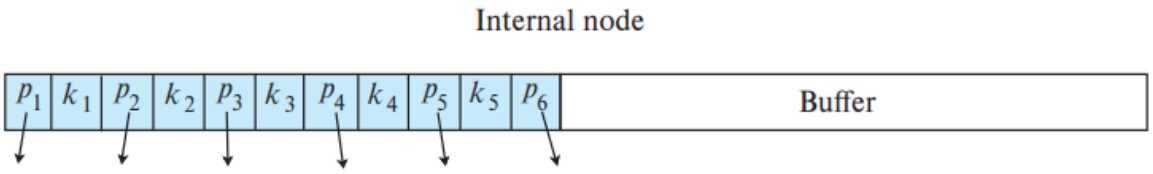
2. Each level (other than L_0) can have up to some number b of trees, instead of just 1 tree. When an L_0 tree is written to disk, a new L_1 tree is created instead of merging it with an existing L_1 tree. When there are b such L_1 trees, they are merged into a single new L_2 tree. Similarly, when there are b trees at level L_i they are merged into a new L_{i+1} tree.

   This variant of the LSM tree is called a **stepped-merge index**. The stepped-merge index decreases the insert cost significantly compared to having only one tree per level, but it can result in an increase in query cost, since multiple trees may need to be searched. Bitmap-based structures called *Bloom filters*, described in Section 24.1, are used to reduce the number of lookups by efficiently detecting that a search key is not present in a particular tree. Bloom filters occupy very little space, but they are quite effective at reducing query cost.

These systems are built on top of distributed file systems that allow appends to files but do not support updates to existing data. The fact that LSM trees do not perform in-place update made LSM trees a very good fit for these systems. These systems are built on top of distributed file systems that allow appends to files but do not support updates to existing data. Subsequently, a large number of BigData storage systems such as Apache

Cassandra, Apache AsterixDB, and MongoDB added support for LSM trees, with most implementing versions with multiple trees in each layer.

❀ **Buffer tree:** The key idea behind the buffer tree is to associate a buffer with each internal node of a B+-tree, including the root node.



When an index record is inserted into the buffer tree, instead of traversing the tree to the leaf, the index record is inserted into the buffer of the root. If the buffer becomes full, each index record in the buffer is pushed one level down the tree to the appropriate child node.
child node = internal node → index record added to the child node's buffer
child node = leaf node → index records are inserted into the leaf as usual

Buffer trees provide better worst-case complexity bounds on the number of I/O operations than do LSM tree variants. In terms of read cost, buffer trees are significantly faster than LSM trees. However, write operations on buffer trees involve random I/O, 670 Chapter 14 Indexing requiring more seeks, in contrast to sequential I/O operations with LSM tree variants

Buffer trees have been implemented as part of the Generalized Search Tree (GiST) index structure in PostgreSQL. The GiST index allows user-defined code to be executed to implement search, update, and split operations on nodes and has been used to implement R-trees and other spatial index structures.

Bitmap indices

Bitmap indices are a specialized type of index designed for easy querying on multiple keys, although each bitmap index is built on a single key.
For bitmap indices to be used, records in a relation must be numbered sequentially, starting from, say, 0.
A bitmap is simply an array of bits. In its simplest form, a bitmap index on the attribute A of relation r consists of one bitmap for each value that A can take. Each bitmap has as many bits as the number of records in the relation.
In fact, bitmap indices are useful for selections mainly when there are selections on multiple keys.

record number	*ID*	*gender*	*income_level*
0	76766	m	L1
1	22222	f	L2
2	12121	f	L1
3	15151	m	L4
4	58583	f	L3

Bitmaps for *gender*

m	10010
f	01101

Bitmaps for *income_level*

L1	10100
L2	01000
L3	00001
L4	00010
L5	00000

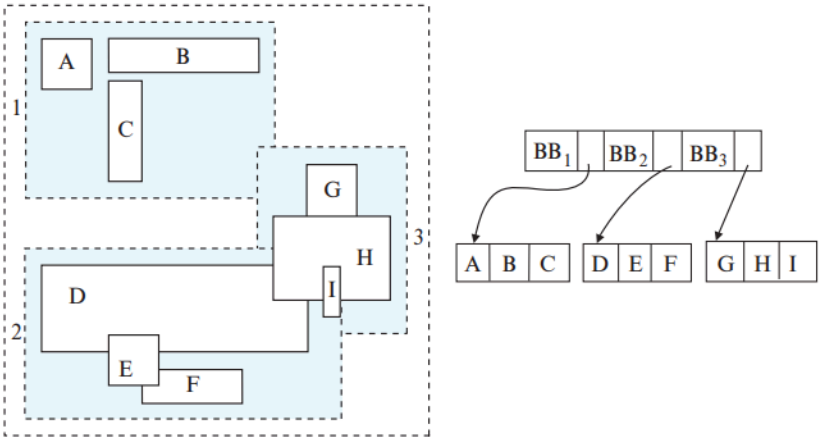
Indexing of spatial and temporal data

When tuples have temporal intervals associated with them, and queries may specify time points or time intervals, the traditional index structures may result in poor performance.

❀ **Indexing of spatial data ¿:** Spatial data refers to data referring to a point or a region in two or higher dimensional space(coordenadas latitud y longitud por ej.). The goal of spatial indexing is to support different forms of spatial queries, with range and nearest neighbor queries being of particular interest, since they are widely used.

A tree structure called a k-d tree was one of the early structures used for indexing in multiple dimensions. Each level of a k-d tree partitions the space into two. The partitioning is done along one dimension at the node at the top level of the tree, along another dimension in nodes at the next level, and so on, cycling through the dimensions. Partitioning stops when a node has less than a given maximum number of points.

A storage structure called an R-tree is useful for indexing of objects spanning regions of space, such as line segments, rectangles, and other polygons, in addition to points. Instead of a range of values, a rectangular bounding box is associated with each tree node.
Each internal node stores the bounding boxes of the child nodes along with the pointers to the child nodes. Each leaf node stores the indexed objects.



R-tree is at the right side

❀ **Indexing temporal data:** Temporal data refers to data that has an associated time period. The time period associated with a tuple indicates the period of time for which the tuple is valid. A time interval has a start time and an end time. Any valid period can be represented by multiple intervals; thus, a tuple with any valid period can be represented by multiple tuples, each of which has a valid period that is a single time interval.

- ➔ A better solution is to use a spatial index such as an R-tree, with the indexed tuple treated as having two dimensions; one issue that complicates the use of a spatial index such as an R-tree is that the end time interval may be infinity.

- ➔ Specialized indices, such as the interval B+-tree, are designed to index intervals in a single dimension, and provide better complexity guarantees than R-tree indices.

➔ Most database implementations find it simpler to use R-tree indices instead of implementing yet another type of index for time intervals.

➔ Recall that with temporal data, more than one tuple may have the same value for a primary key, as long as the tuples with the same primary-key value have non-overlapping time intervals.

Capitulo 15

Review Terms

- Query processing
- Evaluation primitive
- Query-execution plan
- Query-evaluation plan
- Query-execution engine
- Measures of query cost
- Sequential I/O
- Random I/O
- File scan
- Linear search
- Selections using indices
- Access paths
- Index scans
- Conjunctive selection
- Hash-join
  - Build
  - Probe
  - Build input
  - Probe input
  - Recursive partitioning
  - Hash-table overflow
  - Skew
  - Fudge factor
  - Overflow resolution
  - Overflow avoidance
- Hybrid hash-join
- Disjunctive selection
- Composite index
- Intersection of identifiers
- External sorting
- External sort–merge
- Runs
- *N*-way merge
- Equi-join
- Nested-loop join
- Block nested-loop join
- Indexed nested-loop join
- Merge join
- Sort-merge join
- Hybrid merge join
- Spatial join
- Operator tree
- Materialized evaluation
- Double buffering
- Pipelined evaluation
  - Demand-driven pipeline (lazy, pulling)
  - Producer-driven pipeline (eager, pushing)
  - Iterator
  - Pipeline stages
- Double-pipelined join
- Continuous query evaluation

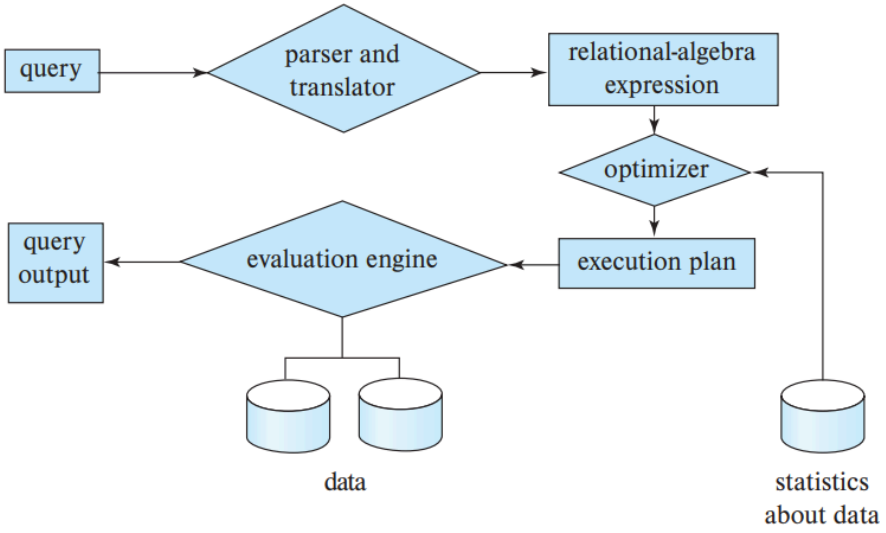
Query processing

Query processing refers to the range of activities involved in extracting data from a database. The activities include translation of queries in high-level database languages into expressions that can be used at the physical level of the file system, a variety of query-optimizing transformations, and actual evaluation of queries.

The basic steps involved in processing are:

1. Parsing and translation
2. Optimization
3. Evaluation

Before query processing can begin, the system must translate the query into a usable form. Thus, the first action the system must take in query processing is to translate a given query into its internal form. This translation process is similar to the work performed by the parser of a compiler.



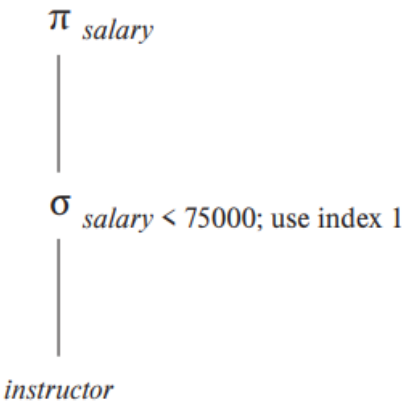
To specify fully how to evaluate a query, we need not only to provide the relational algebra expression, but also to annotate it with instructions specifying how to evaluate each operation. Annotations may state the algorithm to be used for a specific operation or the particular index or indices to use.

A relational-algebra operation annotated with instructions on how to evaluate it is called an evaluation primitive. A sequence of primitive operations that can be used to evaluate a query is a query-execution plan or query-evaluation plan.

The query-execution engine takes a query-evaluation plan, executes that plan, and returns the answers to the query. It is the responsibility of the system to construct a query-evaluation plan that minimizes the cost of query evaluation; this task is called *query optimization*.

In order to optimize a query, a query optimizer must know the cost of each operation.

Several operations may be grouped together into a **pipeline**, in which each of the operations starts working on its input tuples even as they are being generated by another operation.



Measures of query cost

The cost of query evaluation can be measured in terms of a number of different resources, including disk accesses, CPU time to execute a query, and, in parallel and distributed database systems, the cost of communication.

The database has default values for each of these costs (CPU cost per tuple, for processing each index entry and per operator/function), which are multiplied by the number of tuples processed, the number of index entries processed, and the number of operators and functions executed, respectively. The defaults can be changed as a configuration parameter.

We use the number of blocks transferred from storage and the number of random I/O accesses, each of which will require a disk seek on magnetic storage, as two important factors in estimating the cost of a query-evaluation plan.

If the disk subsystem takes an average of tT seconds to transfer a block of data and has an average block-access time (disk seek time plus rotational latency) of tS seconds, then an operation that transfers b blocks and performs S random I/O accesses would take:
b ∗ tT + S ∗ tS seconds.

The values of tT and tS must be calibrated for the disk system used.

Acá el libro hace una banda de cálculos, pero vos cerrá los ojos y confiá

Given the wide diversity of speeds of different storage devices, database systems must ideally perform test seeks and block transfers to estimate tS and tT for specific systems/storage devices, as part of the software installation process. Databases that do not automatically infer these numbers often allow users to specify the numbers as part of configuration files.

We can refine our cost estimates further by distinguishing block reads from block writes. Block writes are typically about twice as expensive as reads on magnetic disks. On PCIe flash, write throughput may be about 50 percent less than read throughput, but the difference is masked by the limited speed of SATA interfaces, leading to write throughput matching read throughput.

The cost estimates we give do not include the cost of writing the final result of an operation back to disk. These are taken into account separately where required.

The costs of all the algorithms that we consider depend on the size of the buffer in main memory.
Best case → Data fits in the buffer and can be read into it (menos acceso a disco)
Worst case → Buffer can only hold a few blocks of data
However, with large main memories available today, such worst-case assumptions are overly pessimistic.

The response time for a query-evaluation plan (that is, the wall-clock time required to execute the plan), would account for all these costs, and could be used as a measure of the cost of the plan. Unfortunately, the response time of a plan is very hard to estimate without actually executing the plan, for the following two reasons:

1. The response time depends on the contents of the buffer when the query begins execution; this information is not available when the query is optimized.
2. In a system with multiple disks, the response time depends on how accesses are distributed among disks

As a result, instead of trying to minimize the response time, optimizers generally try to minimize the total resource consumption of a query plan.

Selection operation

In query processing, the file scan is the lowest-level operator to access data. File scans are search algorithms that locate and retrieve records that fulfill a selection condition. A file scan allows an entire relation to be read in those cases where the relation is stored in a single, dedicated file.

❀ **Selections using file scans and indices:** The most straightforward way of performing a selection is using a linear search.

A1 (linear search). In a linear search, the system scans each file block and tests all records to see whether they satisfy the selection condition. An initial seek is required to access the first block of the file. In case blocks of the file are not stored contiguously, extra seeks may be required, but we ignore this effect for simplicity. The linear-search algorithm can be applied to any file

- ➔ Most real-life optimizers assume that the internal nodes of the tree are present in the in-memory buffer since they are frequently accessed
- ➔ Index structures are referred to as access paths, since they provide a path through which data can be located and accessed.
- ➔ Search algorithms that use an index are referred to as index scans. We use the selection predicate to guide us in the choice of the index to use in processing the query.

Esta tabla por toda la cara. Por favor que la sergiada no abarque esto
We use h_i to represent the height of the B+-tree, and assume a random I/O operation is required for each node in the path from the root to a leaf.

	Algorithm	Cost	Reason
A1	Linear Search	$t_S + b_r * t_T$	One initial seek plus b_r block transfers, where b_r denotes the number of blocks in the file.
A1	Linear Search, Equality on Key	Average case $t_S+(b_r/2) * t_T$	Since at most one record satisfies the condition, scan can be terminated as soon as the required record is found. In the worst case, b_r block transfers are still required.
A2	Clustering B^+-tree Index, Equality on Key	$(h_i + 1) * (t_T + t_S)$	(Where h_i denotes the height of the index.) Index lookup traverses the height of the tree plus one I/O to fetch the record; each of these I/O operations requires a seek and a block transfer.
A3	Clustering B^+-tree Index, Equality on Non-key	$h_i * (t_T + t_S) + t_S + b * t_T$	One seek for each level of the tree, one seek for the first block. Here b is the number of blocks containing records with the specified search key, all of which are read. These blocks are leaf blocks assumed to be stored sequentially (since it is a clustering index) and don't require additional seeks.
A4	Secondary B^+-tree Index, Equality on Key	$(h_i + 1) * (t_T + t_S)$	This case is similar to clustering index.
A4	Secondary B^+-tree Index, Equality on Non-key	$(h_i + n) * (t_T + t_S)$	(Where n is the number of records fetched.) Here, cost of index traversal is the same as for A3, but each record may be on a different block, requiring a seek per record. Cost is potentially very high if n is large.
A5	Clustering B^+-tree Index, Comparison	$h_i * (t_T + t_S) + t_S + b * t_T$	Identical to the case of A3, equality on non-key.
A6	Secondary B^+-tree Index, Comparison	$(h_i + n) * (t_T + t_S)$	Identical to the case of A4, equality on non-key.

A2 (clustering index, equality on key). For an equality comparison on a key attribute with a clustering index, we can use the index to retrieve a single record that satisfies the corresponding equality condition.

A3 (clustering index, equality on non-key). We can retrieve multiple records by using a clustering index when the selection condition specifies an equality comparison on a non-key attribute.

A4 (secondary index, equality). Selections specifying an equality condition can use a secondary index. This strategy can retrieve a single record if the equality condition is on a key; multiple records may be retrieved if the indexing field is not a key.

In the first case, only one record is retrieved. Cost (in this case) = A2. In the second case, each record may be resident on a different block, which may result in one I/O operation per retrieved record, with each I/O operation requiring a seek and a block transfer.

It is possible to construct an estimate of the average or expected cost of the selection by taking into account the probability of the block containing the record already being in the buffer.

"secondary indices store the values of the attributes used as the search key in a B+-tree file organization". Accessing a record through such a secondary index is then more expensive: First the secondary index is searched to find the B+-tree file organization search-key values, then the B+- tree file organization is looked up to find the records. The cost formulae described for secondary indices have to be modified appropriately if such indices are used.

❀ **Selections involving comparisons:** Consider a selection of the form($\sigma_{A \leq v}(r)$). We can implement the selection either by using linear search or by using indices in one of the following ways:

A5 (clustering index, comparison). A clustering ordered index (for example, a clustering B+-tree index) can be used when the selection condition is a comparison. For comparison conditions of the form A > v or A ≥ v, a clustering index on A can be used to direct the retrieval of tuples

A6 (secondary index, comparison). We can use a secondary ordered index to guide retrieval for comparison conditions involving <,≤,≥, or >. The lowest-level index blocks are scanned, either from the smallest value up to v, or from v up to the maximum value (for > and ≥).

As long as the number of matching tuples is known ahead of time, a query optimizer can choose between using a secondary index or using a linear scan based on the cost estimates. However, if the number of matching tuples is not known accurately at compilation time, either choice may lead to bad performance, depending on the actual number of matching tuples.

To deal with the above situation, PostgreSQL uses a hybrid algorithm that it calls a bitmap index scan, when a secondary index is available, but the number of matching records is not known precisely. Algoritmo:

1. First creates a bitmap with as many bits as the number of blocks in the relation.
2. Then uses the secondary index to find index entries for matching tuples

3. As each index entry is found, the algorithm gets the block number from the index entry, and sets the corresponding bit in the bitmap to 1.
4. Once all index entries have been processed, the bitmap is scanned to find all blocks whose bit is set to 1. These are exactly the blocks containing matching records.

In the worst case, this algorithm is only slightly more expensive than linear scan, but in the best case it is much cheaper than linear scan.

❀ Implementation of complex selections:

- **Conjunction:** A **conjunctive selection** is a selection of the form:

$$\sigma_{\theta_1 \wedge \theta_2 \wedge \cdots \wedge \theta_n}(r)$$

- **Disjunction:** A **disjunctive selection** is a selection of the form:

$$\sigma_{\theta_1 \vee \theta_2 \vee \cdots \vee \theta_n}(r)$$

A disjunctive condition is satisfied by the union of all records satisfying the individual, simple conditions θ_i.

- **Negation:** The result of a selection $\sigma_{\neg\theta}(r)$ is the set of tuples of r for which the condition θ evaluates to false. In the absence of nulls, this set is simply the set of tuples in r that are not in $\sigma_{\theta}(r)$.

We can implement a selection operation involving either a conjunction or a disjunction of simple conditions by using one of the following algorithms:

A7 (conjunctive selection using one index). We first determine whether an access path is available for an attribute in one of the simple conditions. If one is, one of the selection algorithms A2 through A6 can retrieve records satisfying that condition.

A8 (conjunctive selection using composite index). An appropriate composite index (index on multiple attributes) may be available for some conjunctive selections. If the selection specifies an equality condition on two or more attributes, and a composite index exists on these combined attribute fields, then the index can be searched directly.

A9 (conjunctive selection by intersection of identifiers). This algorithm requires indices with record pointers, on the fields involved in the individual conditions. The algorithm scans each index for pointers to tuples that satisfy an individual condition. The intersection of all the retrieved pointers is the set of pointers to tuples that satisfy the conjunctive condition. The algorithm then uses the pointers to retrieve the actual records.

A10 (disjunctive selection by union of identifiers). If access paths are available on all the conditions of a disjunctive selection, each index is scanned for pointers to tuples that satisfy the individual condition. The union of all the retrieved pointers yields the set of pointers to all tuples that satisfy the disjunctive condition. We then use the pointers to retrieve the actual records.

However, if even one of the conditions does not have an access path, we have to perform a linear scan of the relation to find tuples that satisfy the condition.

Sorting

Sorting of data plays an important role in database systems for two reasons. First, SQL queries can specify that the output be sorted. Second, and equally important for query processing, several of the relational operations, such as joins, can be implemented efficiently if the input relations are first sorted.

We can sort a relation by building an index on the sort key and then using that index to read the relation in sorted order. This process orders the relation only logically rather than physically. Hence, the reading of tuples in the sorted order may lead to a disk access for each record, which can be very expensive, since the number of records can be much larger than the number of blocks.

Here, we are going to discuss the case where we have to sort relations bigger than memory.

❀ **External sort-merge algorithm:** Sorting of relations that do not fit in memory is called external sorting. The most commonly used technique for external sorting is the external sort–merge algorithm.

Let M denote the number of blocks in the main memory buffer available for sorting (number of disk blocks whose contents can be buffered in available main memory).

1. A number of sorted runs are created; each run is sorted but contains only some of the records of the relation.
2. In the second stage, the runs are merged. Suppose, for now, that the total number of runs, N, is less than M, so that we can allocate one block to each run and have space left to hold one block of output. The merge stage operates as follows: Read one block of each of the N files Ri into a buffer block in memory;

```
repeat
    choose the first tuple (in sort order) among all buffer blocks;
    write the tuple to the output, and delete it from the buffer block;
    if the buffer block of any run R_i is empty and not end-of-file(R_i)
        then read the next block of R_i into the buffer block;
until all input buffer blocks are empty
```

   The output of the merge stage is the sorted relation. The preceding merge operation is a generalization of the two-way merge used by the standard in-memory sort–merge algorithm; it merges N runs, so it is called an N-way merge.

   If the relation is much larger than memory, there may be M or more runs generated in the first stage. Each pass reduces the number of runs by a factor of M − 1. The passes repeat as many times as required, until the number of runs is less than M; a final pass then generates the sorted output.

❀ **Cost analysis of external sort-merge:** Let br denote the number of blocks containing records of relation r. Let M denote the number of blocks in the main memory buffer available for sorting. Let bb denote the size of the block buffer (SEGUN LA IA)

➔ The total number of merge passes required is $\lceil \log_{LM / bbJ-1}(br / M) \rceil$

Finally:

$$2[br / M] + [br / bb](2[\log_{LM / bbJ-1}(br / M)] - 1)$$

Applying this equation to the example in Figure 15.4, we get a total of 8 + 12 ∗ (2 ∗ 2 − 1) = 44 disk seeks if we set the number of buffer blocks per run bb to 1

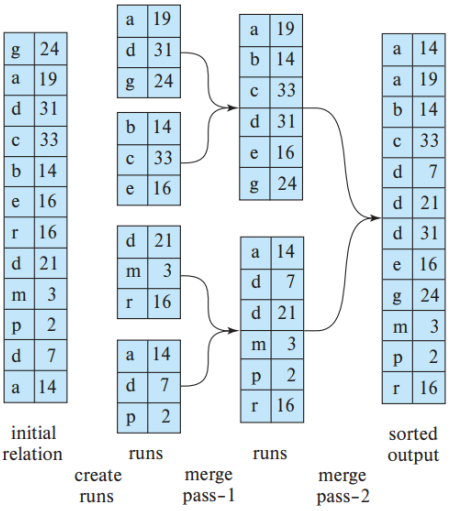


Figure 15.4 External sorting using sort-merge.

Join operation

We use the term equi-join to refer to a junta natural.

❀ **Nested-loop join:** This algorithm is called the nested-loop join

```
for each tuple t_r in r do begin
    for each tuple t_s in s do begin
        test pair (t_r, t_s) to see if they satisfy the join condition θ
        if they do, add t_r · t_s to the result;
    end
end
```

Relation r is called the outer relation and relation s the inner relation of the join.

- ➔ The nested-loop join algorithm requires no indices, and it can be used regardless of what the join condition is. The only change required is an extra step of deleting repeated attributes from the tuple tr · ts, before adding it to the result.
- ➔ The nested-loop join algorithm is expensive, since it examines every pair of tuples in the two relations
- ➔ If one of the relations fits entirely in main memory, it is beneficial to use that relation as the inner relation, since the inner relation would then be read only once.

❀ **Block nested-loop join:** Variant of the nested-loop join where every block of the inner relation is paired with every block of the outer relation. Within each pair of blocks, every tuple in one block is paired with every tuple in the other block, to generate all pairs of tuples.

In the worst case, each block in the inner relation s is read only once for each block in the outer relation, instead of once for each tuple in the outer relation. Thus, in the worst case, there will be a lot of block transfers.

- ➔ If the join attributes in a natural join or an equi-join form a key on the inner relation, then for each outer relation tuple the inner loop can terminate as soon as the first match is found.
- ➔ In the block nested-loop algorithm we can use the biggest size that can fit in memory, while leaving enough space for the buffers of the inner relation and the output.

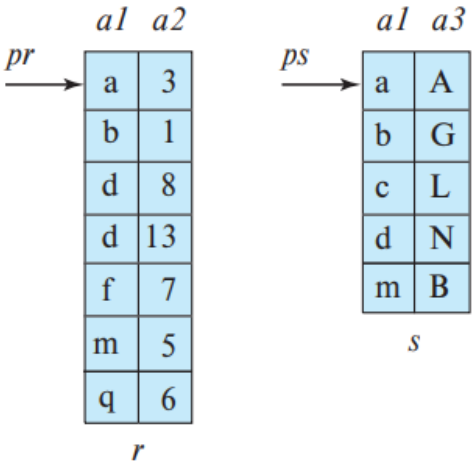
➔ We can scan the inner loop alternately forward and backward. This scanning method orders the requests for disk blocks so that the data remaining in the buffer from the previous scan can be reused (reducing the number of disk accesses needed).

❀ **Index nested-loop join:** If an index is available on the inner loop's join attribute, index lookups can replace file scans: indexed nested-loop join

The cost of an indexed nested-loop join can be computed as follows:
$br(tT + tS) + nr * c$, where nr is the number of records in relation r, and c is the cost of a single selection on s using the join condition. (tT y tS son tuplas de t y s respectivamente y br es la cantidad de bloques que tienen registros de r).

The cost formula indicates that, if indices are available on both relations r and s, it is generally most efficient to use the one with fewer tuples as the outer relation.

❀ **Merge join:** It can be used to compute natural joins and equi-joins.



Algorithm

The merge-join algorithm associates one pointer with each relation. These pointers point initially to the first tuple of the respective relations. As the algorithm proceeds, the pointers move through the relation. A group of tuples of one relation with the same value on the join attributes is read into S. Then, the corresponding tuples (if any) of the other relation are read in and are processed as they are read.

Cost analysis

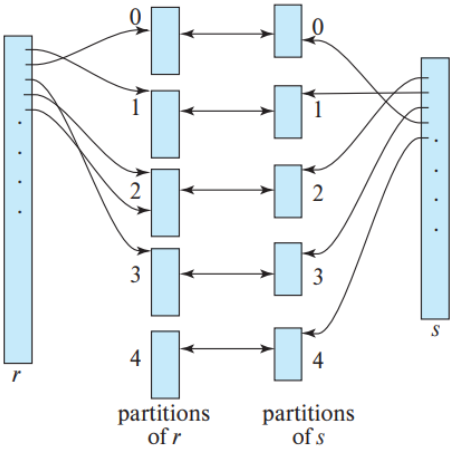
Each tuple in the sorted order needs to be read only once, and, as a result, each block is also read only once. Since it makes only a single pass through both files (assuming all sets Ss fit in memory), the merge-join method is efficient; the number of block transfers is equal to the sum of the number of blocks in both files, br + bs.

Assuming that bb buffer blocks are allocated to each relation, the number of disk seeks required would be ⌈br / bb⌉ + ⌈bs / bb⌉ disk seeks. Since seeks are much more expensive than data transfer, it makes sense to allocate multiple buffer blocks to each relation, provided extra memory is available.

Hybrid merge join

Suppose that one of the relations is sorted; the other is unsorted, but has a secondary B+-tree index on the join attributes. The hybrid merge-join algorithm merges the sorted relation with the leaf entries of the secondary B+-tree index.

❀ **Hash join:** The hash-join algorithm can be used to implement natural joins and equi-joins. A hash function h is used to partition tuples of both relations. The basic idea is to partition the tuples of each of the relations into sets that have the same hash value on the join attributes.



Suppose that an r tuple and an s tuple satisfy the join condition; then, they have the same value for the join attributes.

…To do so, it first builds a hash index on each s_i , and then probes (that is, looks up s_i) with tuples from ri . The relation s is the build input, and r is the probe input.

- ➔ The hash index on s_i is built in memory, so there is no need to access the disk to retrieve the tuples.
- ➔ The build and probe phases require only a single pass through both the build and probe inputs.

If the size of the build relation is b_s blocks, then, for each of the n_h partitions to be of size less than or equal to M, n_h must be at least [bs / M].

Recursive partitioning: Partitioning has to be done in repeated passes. In one pass, the input can be split into at most as many partitions as there are blocks available for use as output buffers. The hash function used in a pass is different from the one used in the previous pass. The system repeats this splitting of the input until each partition of the build input fits in memory.

Hash table overflow: Occurs in partition *i* of the build relation *s* if the hash index on s_i is larger than main memory. Some of the partitions will have more tuples than the average, whereas others will have fewer; partitioning is then said to be skewed (sesgada, desvirtuada).

We can handle a small amount of skew by increasing the number of partitions: that small increased value is called fudge factor.

Overflow resolution: Performed during the build phase if a hash-index overflow is detected. Overflow resolution proceeds in this way: If any partition is found to be too large, it is further partitioned into smaller partitions by using a different hash function.

Overflow avoidance: performs the partitioning carefully, so that overflows never occur during the build phase.

A hash join is estimated to require:

3(br + bs) + 4nh → BLOCK TRANSFERS
2(br + bs)⌈logL(M/bb)−1(bs/M)⌉ + br + bs → '' WITH RECURSIVE PARTITIONING
2(⌈br / bb⌉+⌈bs / bb⌉)+ 2nh → SEEKS.
2(⌈br/bb⌉ + ⌈bs/bb⌉)⌈logL(M/bb)−1(bs/M)⌉ → '' WITH RECURSIVE PARTITIONING

The hash join can be improved if the main memory size is large

The hybrid hash-join algorithm performs another optimization; To reduce the impact of seeks, a larger number of blocks would be used as a buffer.

❀ **Complex joins:** Nested-loop and block nested-loop joins can be used regardless of the join conditions. We can implement joins with complex join conditions, such as conjunctions and disjunctions, by using the efficient join techniques.

❀**Joins over spatial data:** The presented join algorithms assume the use of standard comparison operations such as equality, less than, or greater than, where the values are linearly ordered.

Selection and join conditions on spatial data involve comparison operators that check if one region contains or overlaps another, or whether a region contains a particular point; and the regions may be multi-dimensional.

Other operations

❀ **Duplicate elimination:** We can implement duplicate elimination easily by sorting. Identical tuples will appear adjacent to each other as a result of sorting, and all but one copy can be removed. The worst-case cost estimate for duplicate elimination is the same as the worst-case cost estimate for sorting of the relation. (external sort-merge and hashing)

❀ **Projection:** We can implement projection easily by performing projection on each tuple, which gives a relation that could have duplicate records, and then removing duplicate records.

❀ **Set operations:** We can implement the union, intersection, and set-difference operations by first sorting both relations, and then scanning once through each of the sorted relations to produce the result.

- $r \cap s$
  1. Build an in-memory hash index on r_i.
  2. For each tuple in s_i, probe the hash index and output the tuple to the result only if it is already present in the hash index.

- $r \cup s$
  1. Build an in-memory hash index on r_i.
  2. Add the tuples in s_i to the hash index only if they are not already present.
  3. Add the tuples in the hash index to the result.

- $r - s$
  1. Build an in-memory hash index on r_i.
  2. For each tuple in s_i, probe the hash index, and, if the tuple is present in the hash index, delete it from the hash index.
  3. Add the tuples remaining in the hash index to the result.

❀ **Outer join:** We can implement the outer-join operations by using one of two strategies.

1_ Compute the corresponding join, and then add further tuples to the join result to get the outer-join result

2_ It is easy to extend the nested-loop join algorithms to compute the left outer join: Tuples in the outer relation that do not match any tuple in the inner relation are written to the output after being padded with null values. However, it is hard to extend the nested-loop to compute the full outer join.

Natural outer joins and outer joins with an equi-join condition can be computed by extensions of the merge-join and hash-join algorithms.

❀ **Aggregation:** The aggregation operation can be implemented in the same way as duplicate elimination. We use either sorting or hashing, just as we did for duplicate elimination, but based on the grouping attributes (dept_name in the preceding example).

Instead of eliminating tuples with the same value for the grouping attribute, we gather them into groups and apply the aggregation operations on each group to get the result.
Instead of gathering all the tuples in a group and then applying the aggregation operations, we can implement the aggregation operations sum, min, max, count, and avg on the fly as the groups are being constructed.

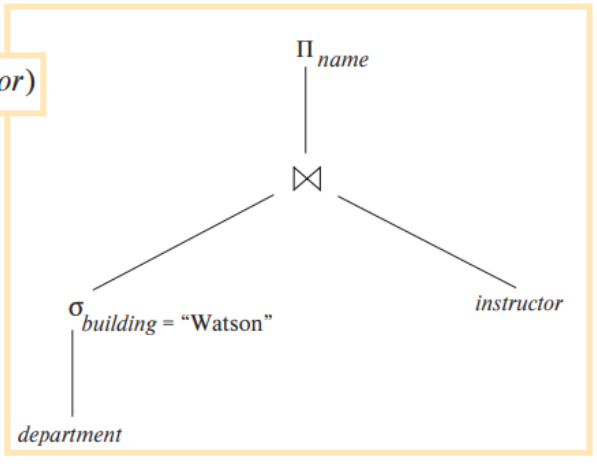
Evaluation of expressions

Evaluating an expression at a time (in appropriate order): the result of each evaluation is materialized in a temporary relation for subsequent use. The temporary relations must be written to disk.

Evaluating several operations simultaneously: We use a pipeline, with the results of one operation passed on to the next, without the need to store a temporary relation.

❀ **Materialization:** It is easiest to understand intuitively how to evaluate an expression by looking at a pictorial representation of the expression in an operator tree.

$$\Pi_{name}(\sigma_{building = \text{“Watson”}}(department) \bowtie instructor)$$



Evaluation as just described is called materialized evaluation, since the results of each intermediate operation are created (materialized) and then are used for evaluation of the next-level operations.

Double buffering (using two buffers, with one continuing execution of the algorithm while the other is being written out) allows the algorithm to execute more quickly by performing CPU activity in parallel with I/O activity.

❀ **Pipelining:** Reduction of number of produced temporary files by combining several relational operations into a pipeline of operations, in which the results of one operation are passed along to the next operation in the pipeline (pipelined evaluation).

Benefits:

1. It eliminates the cost of reading and writing temporary relations, reducing the cost of query evaluation.
2. It can start generating query results quickly, if the root operator of a query evaluation plan is combined in a pipeline with its inputs.

As a result of pipelining, the inputs to the operations are not available all at once for processing.

Ways to execute:

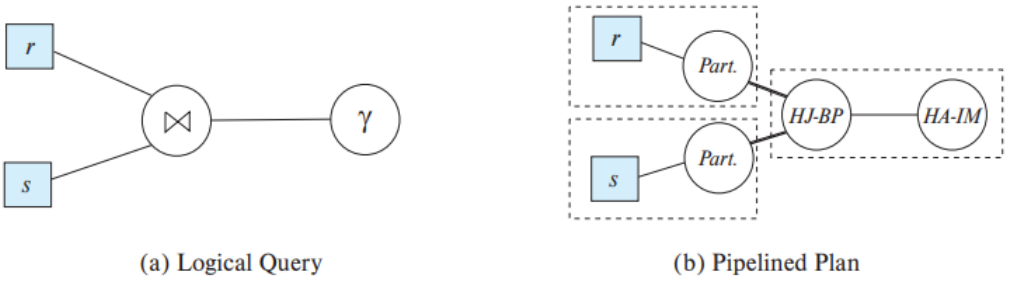
1. Demand-driven pipeline: The system makes repeated requests for tuples from the operation at the top of the pipeline. Each time that an operation receives a request for tuples, it computes the next tuple to be returned and then returns that tuple. Implemented as an iterator that provides the following functions: open(), next(), and close(). Tuples are generated eagerly.
2. Producer-driven pipeline: Operations do not wait for requests to produce tuples, but instead generate the tuples eagerly. Implemented in a way that, for each pair of adjacent operations, the system creates a buffer to hold tuples being passed from one operation to the next. Tuples are generated lazily.

➔ Using producer-driven pipelining can be thought of as pushing data up an operation tree from below, whereas using demand-driven pipelining can be thought of as pulling data up an operation tree from the top.

➔ Query plans can be annotated to mark edges that are pipelined; such edges are called pipelined edges. In contrast, non-pipelined edges are referred to as blocking edges or materialized edges. The two operators connected by a pipelined edge must be executed concurrently.

➔ A query plan can be divided into subtrees such that each subtree has only pipelined edges, and the edges between the subtrees are nonpipelined. Each such subtree is called a pipeline stage.

➔ Some operations, such as sorting, are inherently blocking operations, that is, they may not be able to output any results until all tuples from their inputs have been examined.



(a) Logical Query (b) Pipelined Plan

Figure 15.12 Query plan with pipelining.

Double-pipelined join: The algorithm assumes that the input tuples for both input relations, r and s, are pipelined.

❀ **Pipelines for continuous stream data:** Queries may be written over stream data in order to respond to data as they arrive. Such queries are called continuous queries. The operations in a continuous query should be implemented using pipelined algorithms, so that results from the pipeline can be output without blocking.

Many such queries perform aggregation with windowing; tumbling windows which divide time into fixed size intervals.

Query processing in memory

In this section, we discuss extensions to the query processing techniques that help minimize memory access costs by using cache-conscious query processing algorithms and query compilation.

❀ **Cache conscious algorithms:** The speed difference between cache memory and main memory, results in a situation where the relationship between cache and main memory is not dissimilar to the relationship between main memory and disk.

While the contents of the main memory buffers disk-based data are controlled by the database system, CPU cache is controlled by the algorithms built into the computer hardware. Thus, the database system cannot directly control what is kept in cache.

Ways we can design a query processing algorithm in a way that makes the best use of cache (pág 732 del libro / 761 pdf reader)

Primero habla de cómo ordenar relaciones usando external merge-sort y de el algoritmo de hash-join. Está enfocado en los tamaños de las relaciones en relación a la RAM (memory).
…
Attributes in a tuple can be arranged such that attributes that tend to be accessed together are laid out consecutively.

❀ **Query compilation:** To avoid overhead due to interpretation, modern main-memory databases compile query plans into machine code or intermediate level byte-code. The

compiler can also combine the code for multiple functions in a way that minimizes function calls.

❀ **Column oriented storage:** *"only a few attributes of a large schema may be needed, and that in such cases, storing a relation by column instead of by row may be advantageous"*.

Selection operations on a single attribute have significantly lower cost in a column store since only the relevant attributes need to be accessed. However, since accessing each attribute requires its own data access, the cost of retrieving many attributes is higher and may incur additional seeks if data are stored on disk.

Capitulo 16

Review Terms

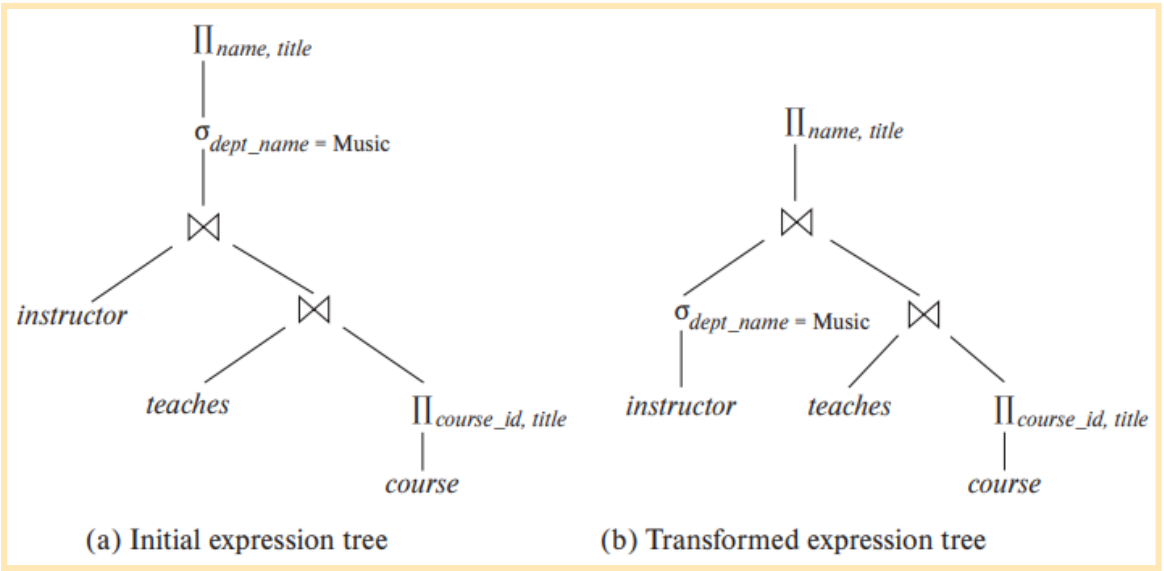
- Query optimization
- Transformation of expressions
- Equivalence of expressions
- Equivalence rules
  - Join commutativity
  - Join associativity
- Minimal set of equivalence rules
- Enumeration of equivalent expressions
- Statistics estimation
- Catalog information
- Size estimation
  - Selection
  - Selectivity
  - Join
- Correlated evaluation
- Decorrelation
- Semijoin
- Anti-semijoin
- Materialized views
- Materialized view maintenance
  - Recomputation
  - Incremental maintenance
  - Insertion
- Histograms
- Distinct value estimation
- Random sample
- Choice of evaluation plans
- Interaction of evaluation techniques
- Cost-based optimization
- Join-order optimization
  - Dynamic-programming algorithm
  - Left-deep join order
  - Interesting sort order
- Heuristic optimization
- Plan caching
- Access-plan selection
  - Deletion
  - Updates
- Query optimization with materialized views
- Index selection
- Materialized view selection
- Top-K optimization
- Join minimization
- Halloween problem
- Multiquery optimization

Query optimization

Query optimization is the process of selecting the most efficient query-evaluation plan from among the many strategies usually possible for processing a given query, especially if the query is complex. We expect the system to construct a query-evaluation plan that minimizes the cost of query evaluation.

One aspect of optimization occurs at the relational-algebra level; another aspect is selecting a detailed strategy for processing the query, such as choosing the algorithm to use for executing an operation, choosing the specific indices to use, and so on.

It is worthwhile for the system to spend a substantial amount of time on the selection of a good strategy for processing a query.



An evaluation plan defines exactly what algorithm should be used for each operation and how the execution of the operations should be coordinated.

Given a relational-algebra expression, it is the job of the query optimizer to come up with a query-evaluation plan that computes the same result as the given expression, and is the least costly way of generating the result.

Generation of query-evaluation plans involves three steps:

1. Generating expressions that are logically equivalent to the given expression.
2. Annotating the resultant expressions in alternative ways to generate alternative query-evaluation plans
3. Estimating the cost of each evaluation plan, and choosing the one whose estimated cost is the least.

As evaluation plans are generated, their costs are estimated by using statistical information about the relations, such as relation sizes and index depths.

Since the cost is an estimate, the selected plan is not necessarily the least costly plan; however, as long as the estimates are good, the plan is likely to be the least costly one.

Transformation of relational expressions

Two relational-algebra expressions are said to be equivalent if, on every legal database instance, the two expressions generate the same set of tuples.

❀ **Equivalence rules:** An equivalence rule says that expressions of two forms are equivalent.

We can replace an expression of the first form with an expression of the second form, or vice versa since the two expressions generate the same result on any valid database.

Now, let θ denote predicates, L denote lists of attributes and E denote relational-algebra expressions:

1. Conjunctive selection operations can be deconstructed into a sequence of individual selections. This transformation is referred to as a cascade of σ.
$$\sigma_{\theta 1 \wedge \theta 2}(E) \equiv \sigma_{\theta 1}(\sigma_{\theta 2}(E))$$

2. Selection operations are commutative.
$$\sigma_{\theta 1}(\sigma_{\theta 2}(E)) \equiv \sigma_{\theta 2}(\sigma_{\theta 1}(E))$$

3. Only the final operations in a sequence of projection operations are needed; the others can be omitted. This transformation can also be referred to as a cascade of Π.
$$\Pi_{L1}(\Pi_{L2}(\ldots(\Pi_{Ln}(E))\ldots)) \equiv \Pi_{L1}(E)$$
where $L_1 \subseteq L_2 \subseteq \ldots \subseteq L_n$.

4. Selections can be combined with Cartesian products and theta joins.
   a. $\sigma_{\theta}(E_1 \times E_2) \equiv E_1 \bowtie_{\theta} E_2$
   This expression is just the definition of the theta join.
   b. $\sigma_{\theta 1}(E_1 \bowtie_{\theta 2} E_2) \equiv E_1 \bowtie_{\theta 1 \wedge \theta 2} E_2$

5. Theta-join operations are commutative.
$$E_1 \bowtie_{\theta} E_2 \equiv E_2 \bowtie_{\theta} E_1$$

   The order of attributes differs between the left-hand side and right-hand side of the commutativity rule, so the equivalence does not hold if the order of attributes is taken into account.

6. a. Natural-join operations are associative.
$$(E1 \bowtie E2) \bowtie E3 \equiv E1 \bowtie (E2 \bowtie E3)$$
   b. Theta joins are associative in the following manner:
$$(E1 \bowtie_{\theta 1} E2) \bowtie_{\theta 2 \wedge \theta 3} E3 \equiv E1 \bowtie_{\theta 1 \wedge \theta 3} (E2 \bowtie_{\theta 2} E3)$$

7. The selection operation distributes over the theta-join operation under the following two conditions:

   a. Selection distributes over the theta-join operation when all the attributes in selection condition θ1 involve only the attributes of one of the expressions (say, E1) being joined.

   $$\sigma_{\theta 1}(E_1 \bowtie_\theta E_2) \equiv (\sigma_{\theta 1}(E_1)) \bowtie_\theta E2$$

   b. Selection distributes over the theta-join operation when selection condition θ1 involves only the attributes of E1 and θ2 involves only the attributes of E2.

   $$\sigma_{\theta 1 \wedge \theta 2}(E1 \bowtie_\theta E2) \equiv (\sigma_{\theta 1}(E1)) \bowtie_\theta (\sigma_{\theta 2}(E2))$$

8. The projection operation distributes over the theta-join operation under the following conditions.

   a. Let L1 and L2 be attributes of E1 and E2, respectively. Suppose that the join condition θ involves only attributes in L1 ∪ L2. Then,

   $$\Pi_{L1 \cup L2}(E1 \bowtie_\theta E2) \equiv (\Pi_{L1}(E1)) \bowtie_\theta (\Pi_{L2}(E2))$$

   b. Consider a join E1 ⋈θ E2. Let L1 and L2 be sets of attributes from E1 and E2, respectively. Let L3 be attributes of E1 that are involved in join condition θ, but are not in L1 and let L4 be attributes of E2 that are involved in join condition θ, but are not in L2. Then,

   $$\Pi L1 \cup L2 (E1 \bowtie \theta\ E2) \equiv \Pi L1 \cup L2 ((\Pi L1 \cup L3 (E1)) \bowtie \theta (\Pi L2 \cup L4 (E2)))$$

9. The set operations union and intersection are commutative.

   a. $E1 \cup E2 \equiv E2 \cup E1$

   b. $E1 \cap E2 \equiv E2 \cap E1$

   Set difference is not commutative.

10. Set union and intersection are associative.

    a. $(E1 \cup E2) \cup E3 \equiv E1 \cup (E2 \cup E3)$

    b. $(E1 \cap E2) \cap E3 \equiv E1 \cap (E2 \cap E3)$

11. The selection operation distributes over the union, intersection, and set difference operations.

    a. $\sigma\theta(E1 \cup E2) \equiv \sigma\theta(E1) \cup \sigma\theta(E2)$

    b. $\sigma\theta(E1 \cap E2) \equiv \sigma\theta(E1) \cap \sigma\theta(E2)$

    c. $\sigma\theta(E1 - E2) \equiv \sigma\theta(E1) - \sigma\theta(E2)$

    d. $\sigma\theta(E1 \cap E2) \equiv \sigma\theta(E1) \cap E2$

    e. $\sigma\theta(E1 - E2) \equiv \sigma\theta(E1) - E2$

    The preceding equivalence does not hold if − is replaced by ∪.

12. The projection operation distributes over the union operation

    $$\Pi L(E1 \cup E2) \equiv (\Pi L(E1)) \cup (\Pi L(E2))$$

    provided E1 and E2 have the same schema.

13. Selection distributes over aggregation under the following conditions. Let G be a set of group by attributes, and A a set of aggregate expressions. When θ only involves attributes in G, the following equivalence holds.

    $$\sigma\theta(G\gamma A(E) \equiv G\gamma A(\sigma\theta(E))$$

14.

    a. Full outer join is commutative.

    $$E1 \bowtie E2 \equiv E2 \bowtie E2$$

    b. Left and right outer join are not commutative. However, left outer join and right outer join can be exchanged as follows.

    $$E1 \bowtie E2 \equiv E2 \bowtie E1$$

15. Selection distributes over left and right outer join under some conditions. Specifically, when the selection condition θ1 involves only the attributes of one of the expressions being joined, say E1, the following equivalences hold.

    a. $\sigma_{\theta 1}(E1 \rtimes_{\theta} E2) \equiv (\sigma_{\theta 1}(E1)) \rtimes_{\theta} E2$

    b. $\sigma_{\theta 1}(E2 \ltimes_{\theta} E1) \equiv (E2 \ltimes_{\theta} (\sigma_{\theta 1}(E1)))$

16. Outer joins can be replaced by inner joins under some conditions. Specifically, if θ1 has the property that it evaluates to false or unknown whenever the attributes of E2 are null, then the following equivalences hold.

    a. $\sigma_{\theta 1}(E1 \rtimes_{\theta} E2) \equiv \sigma_{\theta 1}(E1 \bowtie_{\theta} E2)$

    b. $\sigma_{\theta 1}(E2 \ltimes_{\theta} E1) \equiv \sigma_{\theta 1}(E2 \bowtie_{\theta} E2)$

Some equivalences that hold for joins do not hold for outer joins.

❀ **Examples of transformations:** pagina 753 del libro/781 pdf reader

❀ **Join ordering:** A good ordering of join operations is important for reducing the size of temporary results; hence, most query optimizers pay a lot of attention to the join order.

$$(r_1 \bowtie r_2) \bowtie r_3 \equiv r_1 \bowtie (r_2 \bowtie r_3)$$

❀ **Enumeration of equivalent expressions:** Query optimizers can use equivalence rules to systematically generate expressions equivalent to the given query expression.

The process of enumeration proceeds as follows:

1. Given a query expression E, the set of equivalent expressions EQ initially contains only E
2. Each expression in EQ is matched with each equivalence rule
3. If a subexpression e_j of any expression $E_i \in EQ$ matches one side of an equivalence rule, the optimizer generates a copy E_k of E_i , in which e_j is transformed to match the other side of the rule, and adds E_k to EQ.

This process continues until no more new expressions can be generated. With a properly chosen set of equivalence rules, the set of equivalent expressions is finite, and the process can be guaranteed to terminate. Eventually applying any equivalence rule will only generate expressions that were already generated earlier.

The preceding process is extremely costly both in space and in time, but optimizers can greatly reduce both the space and time cost, using two key ideas:

1. If we generate an expression E′ from an expression E_1 by using an equivalence rule on subexpression e_i , then E′ and E_1 have identical subexpressions except for e_i and its transformation.
2. It is not always necessary to generate every expression that can be generated with the equivalence rules.

With these and other techniques to reduce the optimization time, equivalence rules can be used to enumerate alternative plans, whose costs can be computed; the lowest-cost plan

amongst the alternatives is then chosen. Although this approach can be implemented to execute quite fast, there is no guarantee that it will find the optimal plan.

Estimating statistics of expression results

The cost of an operation depends on the size and other statistics of its inputs.

We first list some statistics about database relations that are stored in database-system catalogs, and then show how to use the stored statistics to estimate statistics on the results of various relational operations.

Given a query expression, we consider it as a tree; we can start from the bottomlevel operations, and estimate their statistics, and continue the process on higher-level operations, till we reach the root of the tree. The size estimates that we compute as part of these statistics can be used to compute the cost of algorithms for individual operations in the tree, and these costs can be added up to find the cost of an entire query plan.

Estimates are not very accurate, since they are based on assumptions that may not hold exactly.

The plans with the lowest estimated costs usually have actual execution costs that are either the lowest actual execution costs or are close to the lowest actual execution costs.

❀ **Catalog information:** The database-system catalog stores the following statistical information about database relations:

- n_r, the number of tuples in the relation r.
- b_r, the number of blocks containing tuples of relation r.
- l_r, the size of a tuple of relation r in bytes.
- f_r, the blocking factor of relation r—that is, the number of tuples of relation r that fit into one block.
- $V(A, r)$, the number of distinct values that appear in the relation r for attribute A. This value is the same as the size of $\Pi_A(r)$. If A is a key for relation r, $V(A, r)$ is n_r.

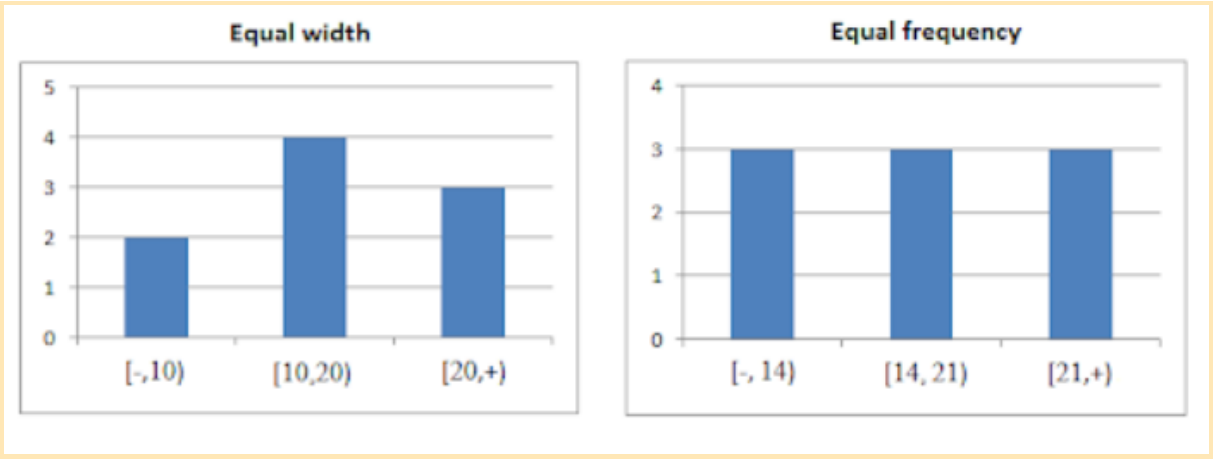
If we assume that the tuples of relation r are stored together physically in a file, the following equation holds:

$$b_r = \left\lceil \frac{n_r}{f_r} \right\rceil$$

If we wish to maintain accurate statistics, then every time a relation is modified, we must also update the statistics. This update incurs a substantial amount of overhead.

Because of this, most systems update the statistics during periods of light system load.

Real-world optimizers often maintain further statistical information to improve the accuracy of their cost estimates of evaluation plans. For instance, most databases store the distribution of values for each attribute as a histogram: in a histogram, the values for the attribute are divided into a number of ranges, and with each range the histogram associates the number of tuples whose attribute value lies in that range.



An equi-width histogram divides the range of values into equal sized ranges

An equi-depth histogram adjusts the boundaries of the ranges such that each range has the same number of values

Histograms used in database systems can also record the number of distinct values in each range, in addition to the number of tuples with attribute values in that range.

In many database applications, some values are very frequent, compared to other values. To get better estimates for queries that specify these values, many databases store a list of n most frequent values for some n (say 5 or 10), along with the number of times each value appears. The histogram then only stores statistics for values other than these on the list, since now we have exact counts for these values.

A histogram takes up only a little space, so histograms on several different attributes can be stored in the system catalog.

❀ **Selection size estimation:** The size estimate of the result of a selection operation depends on the selection predicate.

- ➔ $\sigma_{A=a}(r)$: If 'a' is a frequently occurring value for which the occurrence count is available, we can use that value directly as the size estimate for the selection. Otherwise if there is no histogram available, we assume uniform distribution of values. Despite the fact that the uniform-distribution assumption is often not correct, it is a reasonable approximation of reality in many cases

- ➔ $\sigma_{A\leq v}(r)$: Assuming that values are uniformly distributed, we can estimate the number of records that will satisfy the condition $A \leq v$ as:

- 0 if $v < \min(A, r)$
- n_r if $v \geq \max(A, r)$, and,
- $n_r \cdot \frac{v - \min(A,r)}{\max(A,r) - \min(A,r)}$, otherwise.

➔ Complex selections: conjunction, disjunction, negation

❀ **Join size estimation:** Estimating the size of a natural join is somewhat more complicated than estimating the size of a selection or of a Cartesian product. Let r(R) and s(S) be relations.

➔ If R ∩ S = ∅, then r ⋈ s is the same as r × s
➔ If R ∩ S is a key for R, then we know that a tuple of s will join with at most one tuple from r. Therefore, the number of tuples in r ⋈ s is no greater than the number of tuples in s.
➔ The most difficult case is when R ∩ S is a key for neither R nor S. In this case, we assume, as we did for selections, that each value appears with equal probability.

❀ **Size estimation for other operations:**

➔ Projection($\Pi_A(r)$): V (A,r), since projection eliminates duplicates.
➔ Aggregation: V (G,r), since there is one tuple in the ag for each distinct value of G.
➔ Set operations: If the two inputs to a set operation are selections on the same relation, we can rewrite the set operation as disjunctions, conjunctions, or negations. Union or intersection (sizes of r and s and minimum of the sizes of r and s respectively)
➔ Outer join: The estimated size is r ⋈ s plus the size of the outer relation.

❀ **Estimation of number of distinct values:**
For selections, V (A, σθ(r)) can be estimated in these ways:

➔ If the selection condition θ forces A to take on a specified value: 1
➔ If θ forces A to take on one of a specified set of values: Set to the number of specified values
➔ If the selection condition θ is of the form A op v, where op is a comparison operator: V (A,r) ∗ s (where s is the selectivity of the selection)
➔ In all other cases, we assume: min(V (A,r), $n_{\sigma\theta(r)}$)

For joins, V (A, r ⋈ s) can be estimated in these ways:

➔ If all attributes in A are from r/s, V (A,r ⋈ s) is estimated as min(V (A,r/s), $n_{r \bowtie s}$)
➔ If A contains attributes A1 from r and A2 from s, then V (A,r ⋈ s) is estimated as: min(V (A1,r) ∗ V (A2 − A1,s), V (A1 − A2,r) ∗ V (A2,s), $n_{r \bowtie s}$)

Choice of evaluation plans

A cost-based optimizer explores the space of all query-evaluation plans that are equivalent to the given query, and chooses the one with the least estimated cost.

Cost-based optimization with arbitrary equivalence rules is fairly complicated.

Exploring the space of all possible plans may be too expensive for complex queries.

❀**Cost-based join-order selection:** In this section we consider the problem of choosing the optimal join order for a query with a couple joins and selections specified in the where clause.

Consider the expression: $r_1 \bowtie r_2 \bowtie \cdots \bowtie r_n$. With n = 3, there are 12 different join orderings:

$$
\begin{array}{llll}
r_1 \bowtie (r_2 \bowtie r_3) & r_1 \bowtie (r_3 \bowtie r_2) & (r_2 \bowtie r_3) \bowtie r_1 & (r_3 \bowtie r_2) \bowtie r_1 \\
r_2 \bowtie (r_1 \bowtie r_3) & r_2 \bowtie (r_3 \bowtie r_1) & (r_1 \bowtie r_3) \bowtie r_2 & (r_3 \bowtie r_1) \bowtie r_2 \\
r_3 \bowtie (r_1 \bowtie r_2) & r_3 \bowtie (r_2 \bowtie r_1) & (r_1 \bowtie r_2) \bowtie r_3 & (r_2 \bowtie r_1) \bowtie r_3
\end{array}
$$

with n relations, there are (2(n − 1))! / (n − 1)! different join orders.

Dynamic-programming algorithms store results of computations and reuse them, a procedure that can reduce execution time greatly.

We now consider how to find the optimal join order for a set of n relations S = {r1,r2, …,rn}, where each relation may have selection conditions, and a set of join conditions between the relations r_i is provided. We assume that relations have unique names.

The procedure applies selections on individual relations at the earliest possible point, that is, when the relations are accessed.

The procedure stores the evaluation plans it computes in an associative array bestplan, which is indexed by sets of relations. Each element of the associative array contains two components: the cost of the best plan of S, and the plan itself. The value of bestplan[S].cost is assumed to be initialized to ∞ if bestplan[S] has not yet been computed.

- ➔ Checks if the best plan for computing the join of the given set of relations S has been computed already; if so, it returns the already computed plan.
- ➔ If S contains only one relation, the best way of accessing S is recorded in bestplan.
- ➔ If S contains more than one relation, the procedure tries every way of dividing S into two disjoint subsets. For each division, the procedure recursively finds the best plans for each of the two subsets. It then considers all possible algorithms for joining the results of the two subsets.

The cost of each alternative is considered, and the least cost option chosen.

The order in which tuples are generated by the join of a set of relations is important for finding the best overall join order, since it can affect the cost of further joins.

A particular sort order of the tuples is said to be an interesting sort order if it could be useful for a later operation. Hence, it is not sufficient to find the best join order for each subset of the set of n given relations. Instead, we have to find the best join order for each subset, for each interesting sort order of the join result for that subset.

❀ **Cost-based optimization with equivalence rules:** In this section we outline how to create a general-purpose cost-based optimizer based on equivalence rules.

The procedure for generating equivalent expressions can be modified to generate all possible evaluation plans as follows: A new class of equivalence rules, called physical equivalence rules, is added that allows a logical operation, such as a join, to be transformed to a physical operation, such as a hash join, or a nested-loops join.
But this is very expensive; to make the approach work efficiently requires the following:

1. A space-efficient representation of expressions that avoids making multiple copies of the same subexpressions when equivalence rules are applied.
2. Efficient techniques for detecting duplicate derivations of the same expression.
3. A form of dynamic programming based on memoization, which stores the optimal query evaluation plan for a subexpression when it is optimized for the first time; subsequent requests to optimize the same subexpression are handled by returning the already memorized plan.
4. Techniques that avoid generating all possible equivalent plans by keeping track of the cheapest plan generated for any subexpression up to any point of time, and pruning away any plan that is more expensive than the cheapest plan found so far for that subexpression.

❀ **Heuristics in optimization:** Optimizers use heuristics to reduce the cost of optimization. Here is an example of an heuristics rule:

“Perform selection operations as early as possible”

We say that the preceding rule is a heuristic because it usually, but not always, helps to reduce the cost.

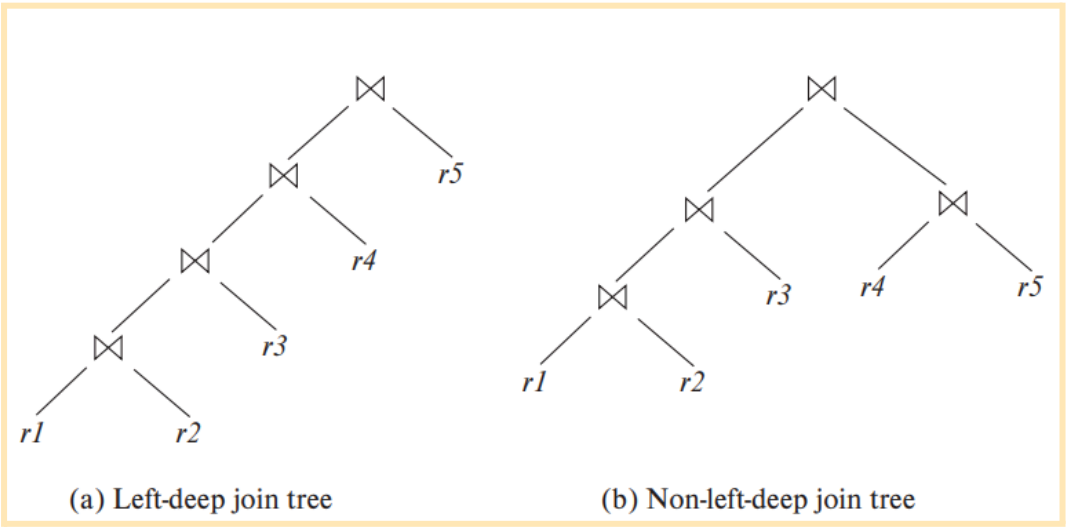
Whenever we need to generate a temporary relation, it is advantageous to apply immediately any projections that are possible (because projection reduces the size of relations).

“Perform projections early”

Optimizers based on join-order enumeration typically use heuristic transformations to handle constructs other than joins, and applying the cost-based join-order selection algorithm to subexpressions involving only joins and selections.

Most practical query optimizers have further heuristics to reduce the cost of optimization. For example, many query optimizers, such as the System R optimizer, do not consider all join orders. The System R optimizer considers only those join orders where the right operand of

each join is one of the initial relations r1, …,rn. Such join orders are called left-deep join orders.



(a) Left-deep join tree (b) Non-left-deep join tree

The time it takes to consider all left-deep join orders is **O(n!)**, which is much less than the time to consider all join orders.
With the use of dynamic-programming optimizations, the System R optimizer can find the best join order in time **$O(n2^n)$**. Contrast this cost with the $O(3^n)$ time required to find the best overall join order.

Un enfoque heurístico para reducir el costo del join-order selection, funciona algo así:

1. Para un n-way join, considera n planes de evaluación. Cada plan utiliza un left-deep join order, comenzando con una de las n relaciones diferentes.
2. La heurística construye el orden de unión para cada uno de los n planes de evaluación seleccionando repetidamente la "mejor" relación para unir a continuación, sobre la base de una clasificación de las rutas de acceso disponibles.
3. Se elige la unión de nested-loop o la de sort-merge para cada una de las uniones, dependiendo de las rutas de acceso disponibles.
4. Finalmente, la heurística elige uno de los n planes de evaluación de manera heurística, sobre la base de minimizar el número de uniones de bucles anidados que no tienen un índice disponible en la relación interna y en el número de uniones de ordenación-fusión.

Los enfoques de optimización de querys que han adoptado elecciones heurísticas de plan ya se han adoptado en varios sistemas.

Most optimizers allow a cost budget to be specified for query optimization. The search for the optimal plan is terminated when the optimization cost budget is exceeded, and the best plan found up to that point is returned. The budget itself may be set dynamically.

If the best plan found so far is expensive, it makes sense to invest more time in optimization, which could result in a significant reduction in execution time; optimizers usually first apply cheap heuristics to find a plan and then start full cost-based optimization with a budget based on the heuristically chosen plan.

Caching and reuse of query plans is referred to as plan caching.

Even with the use of heuristics, cost-based query optimization imposes a substantial overhead on query processing. However, the added cost of cost-based query optimization is usually more than offset by the saving at query-execution time, which is dominated by slow disk accesses.

The difference in execution time between a good plan and a bad one may be huge, making query optimization essential.

The achieved saving is magnified in those applications that run on a regular basis. Therefore, most commercial systems include relatively sophisticated optimizers.

❀ **Optimizing nested subqueries:** SQL conceptually treats nested subqueries in the **where** clause as functions that take parameters and return either a single value or a set of values.

The parameters are the variables from an outer-level query that are used in the nested subquery (these variables are called correlation variables).

```
select name
from instructor
where exists (select *
              from teaches
              where instructor.ID = teaches.ID
                    and teaches.year = 2019);
```

Conceptually, the subquery can be viewed as a function that takes a parameter (here, *instructor.ID*) and returns the set of all courses taught in 2019 by instructors (with the same *ID*).

The technique of evaluating a nested subquery by invoking it as a function is called correlated evaluation. Correlated evaluation is not very efficient, since the subquery is separately evaluated for each tuple in the outer level query. A large number of random disk I/O operations may result. (SQL optimizers therefore attempt to transform nested subqueries into joins).

En esta parte habla de semijoin y anti semijoin pero yo voy a fingir demencia (por favor que no entre)(pág 777 del libro/806 pdf reader)

The process of replacing a nested query by a query with a join, semijoin, or antisemijoin is called decorrelation.

Decorrelation is clearly more complicated when the nested subquery uses aggregation, or when the nested subquery is used as a scalar subquery. In fact, decorrelation is not possible for certain cases of subqueries.

Optimization of complex nested subqueries is a complicated task, as you can infer from the preceding discussion, and many optimizers do only a limited amount of decorrelation. It is better to avoid using complex nested subqueries, where possible, since we cannot be sure that the query optimizer will succeed in converting them to a form that can be evaluated efficiently.

Materialized views

A materialized view is a view whose contents are computed and stored. It is much cheaper in many cases to read the contents of a materialized view than to compute the contents of the view by executing the query defining the view.

❀ **View maintenance:** Materialized views must be kept up-to-date when the data used in the view definition changes. The task of keeping a materialized view up-to-date with the underlying data is known as view maintenance.

- ➔ Manually written code: Every piece of code that updates salary must also update the total salary amount for the corresponding department, but this approach is error prone, since it is easy to miss some places where the salary is updated.

- ➔ Define triggers: On insert, delete, and update of each relation in the view definition. The triggers must modify the contents of the materialized view, to take into account the change that caused the trigger to fire.

- ➔ Incremental view maintenance: Modifying only the affected parts of the materialized view

- ➔ Immediate view maintenance: Incremental view maintenance is performed as soon as an update occurs

- ➔ Deferred view maintenance: View maintenance is deferred to a later time

In modern database systems, once a view is declared to be materialized, the DBS computes the contents of the view and incrementally updates the contents when the underlying data change.

❀ **Incremental view maintenance:** To understand how to maintain materialized views incrementally, we start off by considering individual operations, and then we see how to handle a complete expression.

The changes (inserts and deletes) to a relation or expression are referred to as its differential.

1) Join: Supongamos que tenemos una vista $v = r \bowtie s$, pero hacemos una inserción de tuplas i en r, por lo tanto v ahora es: $v^{new} = v^{old} \cup (i_r \bowtie s)$. Si en su lugar hubiéramos borrado las tuplas d en r, v ahora sería: $v^{new} = v^{old} - (d_r \bowtie s)$

2) Selection: Nuestra vista v = $\sigma_\theta(r)$. Si hacemos una inserción de un conjunto de tuplas i: $v^{new} = v^{old} \cup \sigma_\theta(i_r)$; Si borramos un conjunto de tuplas d: $v^{new} = v^{old} - \sigma_\theta(d_r)$.

3) Projection: For each tuple in a projection such as $\Pi_A(r)$, we will keep a count of how many times it was derived. Si queremos borrar un conjunto de tuplas d en r, para cada tupla en d tenemos que:
   a) Encontrar t.A (proyección de t en el atributo A) en la vista materializada y decrementar el contador de "almacenamiento"(count stored) en 1.
   b) Si la cuenta se hace 0, borramos la tupla
   c) Para insertar: Si la tupla a insertar ya está en la vista materializada, aumentamos el contador de "almacenamiento" en 1, sino, la agregamos a la vista con el contador en 1.

4) Agregación: count, sum, avg, min y max tienen procedimientos parecidos a la proyección. Para avg nos basamos en sum y count

5) Intersección, unión y diferencia: En la inserción, vemos si la agregamos si se cumple la condición (según la operación). Podemos borrar directamente.

❀ **Handling expressions:** To handle an entire expression, we can derive expressions for computing the incremental change to the result of each subexpression, starting from the smallest subexpressions

❀ **Query optimization and materialized views:** Query optimization can be performed by treating materialized views just like regular relations. However, materialized views offer further opportunities for optimization:

Rewriting queries to use materialized views: Using v⋈ t instead of r ⋈ s ⋈ t (where v = r ⋈ s)
Replacing a use of a materialized view with the view definition: Case of $\sigma_{A=10}(v)$ (where v = r ⋈ s) but r has an index on A and s has an index on B. The best plan for this query may be to replace v with r ⋈ s.

❀ **Materialized views and index selection:** Another related optimization problem is that of materialized view selection, namely, "What is the best set of views to materialize?". This decision must be made on the basis of the system workload, which is a sequence of queries and updates that reflects the typical load on the system.

Indices are just like materialized views, in that they too are derived data, can speed up queries, and may slow down updates.

Advanced topics in query optimization

❀ **Top-K optimization:** Many queries fetch results sorted on some attributes, and require only the top K results for some K (sometimes the bound K is specified explicitly).

One approach is to use pipelined plans that can generate the results in sorted order. Another approach is to estimate what is the highest value on the sorted attributes that will appear in the top-K output

❀ **Join minimization:** When queries are generated through views, sometimes more relations are joined than are needed for computation of the query. Dropping a relation from a join as above is an example of join minimization (el ejemplo habla de usar solo los atributos de una relación en un join, por lo que podríamos no hacer el join e ignorar una de las relaciones)

❀ **Update optimization:** Update queries often involve subqueries in the set and where clauses, which must also be taken into account in optimizing the update. If the update is done while the selection is being evaluated by an index scan, an updated tuple may be reinserted in the index ahead of the scan and seen again by the scan.

The problem of an update affecting the execution of a query associated with the update is known as the Halloween problem. The problem can be avoided by executing the queries defining the update first, creating a list of affected tuples, and updating the tuples and indices as the last step.

Update plans can be optimized by checking if the Halloween problem can occur, and if it cannot occur, updates can be performed while the query is being processed, reducing the update overheads.

❀ **Multiquery optimization and shared scans:** When a batch of queries are submitted together, a query optimizer can potentially exploit common subexpressions between the different queries, evaluating them once and reusing them where required. Such optimization is known as multiquery optimization.

Common subexpression elimination optimizes subexpressions shared by different expressions in a program by computing and storing the result and reusing it wherever the subexpression occurs.

Sharing of relation scans between queries is another limited form of multiquery optimization that is implemented in some databases. The shared-scan optimization works as follows: Instead of reading the relation repeatedly from disk, data are read once from disk and pipelined to each of the queries.

❀ **Parametric query optimization:** Reuse of plans by plan caching is reasonable if the optimal query plan is not significantly affected by the exact value of the constants in the query. However, if the plan is affected by the value of the constants, parametric query optimization is an alternative.

In parametric query optimization, a query is optimized without being provided specific values for its parameters. The optimizer then outputs several plans, each optimal for a different parameter value. A plan would be output by the optimizer only if it is optimal for some possible value of the parameters. The set of alternative plans output by the optimizer are stored.

❀**Adaptive query processing: The** optimizer may choose a plan that turns out to perform very badly; adaptive operators that choose the specific operator at execution time provide a partial solution to this problem.

For example, SQL Server supports an adaptive join algorithm that checks the size of its outer input, and chooses either nested loops join, or hash join depending on the size of the outer input. Many systems also include the ability to monitor the behavior of a plan during query execution, and adapt the plan accordingly.

Capitulo 17

Review Terms

- Transaction
- ACID properties
  - Atomicity
  - Consistency
  - Isolation
  - Durability
- Inconsistent state
- Storage types
  - Volatile storage
  - Non-volatile storage
  - Stable storage
- Concurrency-control system
- Recovery system
- Transaction state
  - Active
  - Partially committed
  - Failed
  - Aborted
  - Committed
  - Terminated
- compensating transaction
- Transaction
  - Restart
  - Kill
- Observable external writes
- Concurrent executions
- Serial execution
- Schedules
- Conflict of operations
- Conflict equivalence
- Conflict serializability
- Serializability testing
- Precedence graph
- Serializability order
- Recoverable schedules
- Cascading rollback
- Cascadeless schedules
- Isolation levels
  - Serializable
  - Repeatable read
  - Read committed
  - Read uncommitted
- Dirty writes
- Automatic commit
- Concurrency control
- Locking
- Timestamp ordering
- Snapshot isolation
- Phantom phenomenon
- Predicate locking

Transactions

It is essential that all these operations occur, or that, in case of a failure, none occur. It would be unacceptable if the checking account were debited but the savings account not credited.

Transaction concept

A transaction is a unit of program execution that accesses and possibly updates various data items. Usually, a transaction is initiated by a user program written in a high-level data-manipulation language.

ACID Properties

- ➔ Atomicity: Either all operations of the transaction are reflected properly in the database, or none are (all-or-none).
- ➔ Consistency: Execution of a transaction in isolation (i.e., with no other transaction executing concurrently) preserves the consistency of the database. (good data-integrity constraints)
- ➔ Isolation: Even though multiple transactions may execute concurrently, the system guarantees that, for every pair of transactions Ti and Tj , it appears to Ti that either Tj finished execution before Ti started or Tj started execution after Ti finished. Thus, each transaction is unaware of other transactions executing concurrently in the system.
- ➔ Durability: After a transaction completes successfully, the changes it has made to the database persist, even if there are system failures.

A simple transaction model

We begin our study of transactions with a simple database language that focuses on when data are moved from disk to main memory and from main memory to disk.

Transactions access data using two operations:

- ➔ Read(X): transfers the data item X from the database to a variable, also called X, in a buffer in main memory belonging to the transaction that executed the read operation.
- ➔ Write(X): transfers the value in the variable X in the main-memory buffer of the transaction that executed the write to the data item X in the database. The write operation may be temporarily stored elsewhere and executed on the disk later.

Let us now consider each of the ACID properties for:

T_i: read(A);
$A := A - 50$;
write(A);
read(B);
$B := B + 50$;
write(B).

➔ Atomicity: Suppose that the failure happened after the write(A) operation but before the write(B) operation. In this case, the values of accounts A and B reflected in the database are $950 and $2000. Because of the failure, the state of the system no longer reflects a real state of the world that the database is supposed to capture. We term such a state an inconsistent state. We must ensure that such inconsistencies are not visible in a database system.

> The basic idea behind ensuring atomicity is this: The database system keeps track (on disk) of the old values of any data on which a transaction performs a write. This information is written to a file called the log.
>
> *"If the transaction does not complete its execution, the database system restores the old values from the log to make it appear as though the transaction never executed."*

➔ Consistency: The consistency requirement here is that the sum of A and B be unchanged by the execution of the transaction. Without the consistency requirement, money could be created or destroyed by the transaction.

➔ Isolation: A way to avoid the problem of concurrently executing transactions is to execute transactions serially. However, concurrent execution of transactions provides significant performance benefits. Ensuring the isolation property is the responsibility of a component of the database system called the concurrency-control system.

➔ Durability: Once the execution of the transaction completes successfully, it must be the case that no system failure can result in a loss of data corresponding to this transfer of funds. Once a transaction completes successfully, all the updates that it carried out on the database persist. We can guarantee durability by ensuring that either:

- Updates carried out by the transaction have been written to disk before the transaction completes.
- Information about the updates carried out by the transaction is written to disk, and such information is sufficient to enable the database to reconstruct the updates when the database system is restarted after the failure.

Storage structure

To understand how to ensure the atomicity and durability properties of a transaction, we must gain a better understanding of how the various data items in the database may be stored and accessed.

In Chapter 12, we saw that storage media can be distinguished by their relative speed, capacity, and resilience to failure, and classified as volatile storage or non-volatile storage; we are going to introduce another class of storage called, stable storage.

- ➔ Volatile storage: Information does not usually survive system crashes. Examples of such storage are main memory and cache memory. Access to volatile storage is extremely fast, both because of the speed of the memory access itself and because it is possible to access any data item in volatile storage directly.

- ➔ Non-volatile storage: Information survives system crashes. Examples of non-volatile storage include secondary storage devices such as magnetic disk and flash storage, used for online storage, and tertiary storage devices such as optical media and magnetic tapes, used for archival storage. Non-volatile storage is slower than volatile storage, particularly for random access. Both secondary and tertiary storage devices, however, are susceptible to failures that may result in loss of information.

- ➔ Stable storage: Information is never lost (it is possible, although extremely unlikely). Although stable storage is theoretically impossible to obtain, it can be closely approximated by techniques that make data loss extremely unlikely. To implement stable storage, we replicate the information in several non-volatile storage media (usually disk) with independent failure modes. Updates must be done with care to ensure that a failure during an update to stable storage does not cause a loss of information.

For a transaction to be durable, its changes need to be written to stable storage.
For a transaction to be atomic, log records need to be written to stable storage before any changes are made to the database on disk.
The degree to which a system ensures durability and atomicity depends on how stable its implementation of stable storage really is.

Transaction atomicity and durability

A transaction that does not complete its execution successfully is termed aborted.

An aborted transaction must have no effect on the state of the database. Once the changes caused by an aborted transaction have been undone, we say that the transaction has been rolled back. It is part of the responsibility of the recovery scheme to manage transaction aborts (typically using logs).

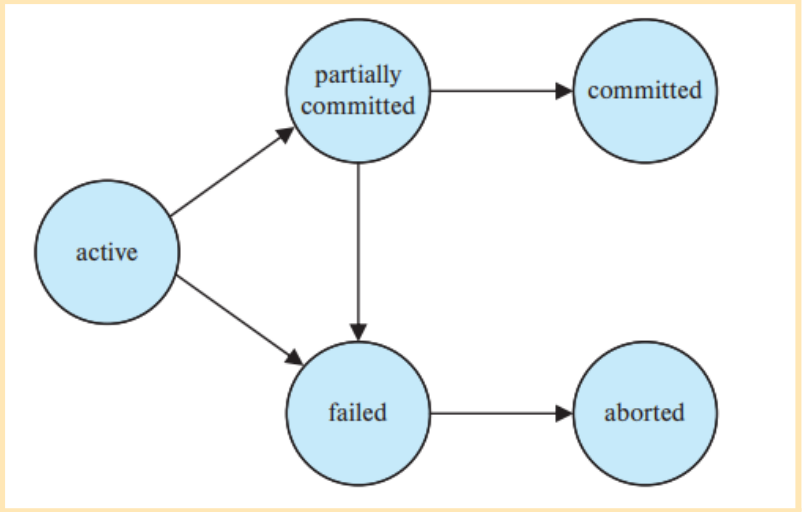
A transaction that completes its execution successfully is said to be committed. A committed transaction that has performed updates transforms the database into a new consistent state, which must persist even if there is a system failure. The only way to undo the effects of a committed transaction is to execute a compensating transaction.

However, it is not always possible to create such a compensating transaction. Therefore, the responsibility of writing and executing a compensating transaction is left to the user and is not handled by the database system.

A transaction must be in one of the following states (abstract transaction model):

- ➔ Active: The initial state; the transaction stays in this state while it is executing.
- ➔ Partially committed: After the final statement has been executed.
- ➔ Failed: After the discovery that normal execution can no longer proceed.
- ➔ Aborted: After the transaction has been rolled back and the database has been restored to its state prior to the start of the transaction.
- ➔ Committed: After successful completion.

A transaction is said to have terminated if it has either committed or aborted.



A transaction enters the failed state after the system determines that the transaction can no longer proceed with its normal execution. Such a transaction must be rolled back. Then, it enters the aborted state. At this point, the system has two options:

1. restart (new) the transaction, but only if the transaction was aborted as a result of some hardware or software error that was not created through the internal logic of the transaction.
2. kill the transaction. It usually does so because of some internal logical error that can be corrected only by rewriting the application program

Once an observable external write has occurred, it cannot be erased, since it may have been seen external to the database system. Most systems allow such writes to take place only after the transaction has entered the committed state.

For certain applications, it may be desirable to allow active transactions to display data to users, particularly for long-duration transactions that run for minutes or hours. Unfortunately, we cannot allow such output of observable data unless we are willing to compromise transaction atomicity.

Transaction Isolation

Allowing multiple transactions to update data concurrently causes several complications with consistency of the data, as we saw earlier. Ensuring consistency in spite of concurrent execution of transactions requires extra work; it is far easier to insist that transactions run serially.

There are two good reasons for allowing concurrency:

- ➔ Improved throughput and resource utilization: The parallelism of the CPU and the I/O system can be exploited to run multiple transactions in parallel. (Ejemplo: Un disco puede estar haciendo lectura/escritura de una transacción en un disco, otra transacción puede estar corriendo en la CPU y otro disco puede estar ejecutando una tercera transacción). All of this increases the throughput (rendimiento) of the system; the processor and disk utilization also increase.

- ➔ Reduced waiting time: If transactions run serially, a short transaction may have to wait for a preceding long transaction to complete, which can lead to unpredictable delays in running a transaction. Concurrent execution reduces the unpredictable delays in running transactions. It also reduces the average response time: the average time for a transaction to be completed after it has been submitted.

The motivation for using concurrent execution in a database is essentially the same as the motivation for using multiprogramming in an operating system.

The database system must control the interaction among the concurrent transactions to prevent them from destroying the consistency of the database. It does so through a variety of mechanisms called concurrency-control schemes.

The execution sequences just described are called schedules. They represent the chronological order in which instructions are executed in the system. A schedule for a set of transactions must consist of all instructions of those transactions and they must preserve the order in which the instructions appear in each individual transaction. We include in our schedules the commit operation to indicate that the transaction has entered the committed state.

These schedules are serial: Each serial schedule consists of a sequence of instructions from various transactions, where the instructions belonging to one single transaction appear together in that schedule.

T_1	T_2
	read(A)
	$temp := A * 0.1$
	$A := A - temp$
	write(A)
	read(B)
	$B := B + temp$
	write(B)
	commit
read(A)	
$A := A - 50$	
write(A)	
read(B)	
$B := B + 50$	
write(B)	
commit	

T_1	T_2
read(A)	
$A := A - 50$	
write(A)	
	read(A)
	$temp := A * 0.1$
	$A := A - temp$
	write(A)
read(B)	
$B := B + 50$	
write(B)	
commit	
	read(B)
	$B := B + temp$
	write(B)
	commit

When the database system executes several transactions concurrently, the corresponding schedule no longer needs to be serial. Several execution sequences are possible, since the various instructions from both transactions may now be interleaved. In general, it is not possible to predict exactly how many instructions of a transaction will be executed before the CPU switches to another transaction.

It is the job of the database system to ensure that any schedule that is executed will leave the database in a consistent state. The concurrency-control component of the database system carries out this task.

That is, the schedule should, in some sense, be equivalent to a serial schedule. Such schedules are called serializable schedules.

Serializability

Before we can consider how the concurrency-control component of the database system can ensure serializability, we consider how to determine when a schedule is serializable.

Certainly, serial schedules are serializable, but if steps of multiple transactions are interleaved, it is harder to determine whether a schedule is serializable. It is difficult to determine exactly what operations a transaction performs and how operations of various transactions interact.

We discuss different forms of schedule equivalence but focus on a particular form called conflict serializability.

Analizamos el caso donde, en un schedule S, tenemos dos operaciones I y J consecutivas en las transacciones T_i y T_j respectivamente (i != j). Si I y J se refieren al mismo data item Q, su orden nos va a importar.
Since we are dealing with only read and write instructions, there are four cases that we need to consider:

1. I = read(Q), J = read(Q). The order of I and J does not matter, since the same value of Q is read by T_i and T_j , regardless of the order.
2. I = read(Q), J = write(Q). If I comes before J, then T_i does not read the value of Q that is written by T_j in instruction J. If J comes before I, then Ti reads the value of Q that is written by Tj . Thus, the order of I and J matters.
3. I = write(Q), J = read(Q). The order of I and J matters for reasons similar to those of the previous case.
4. I = write(Q), J = write(Q). Since both instructions are write operations, the order of these instructions does not affect either T_i or T_j . However, the value obtained by the

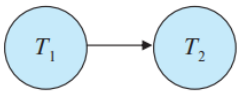
next read(Q) instruction of S is affected, since the result of only the latter of the two write instructions is preserved in the database. If there is no other write(Q) instruction after I and J in S, then the order of I and J directly affects the final value of Q in the database state that results from schedule S.

We say that I and J conflict if they are operations by different transactions on the same data item, and at least one of these instructions is a write operation.

If a schedule S can be transformed into a schedule S′ by a series of swaps of nonconflicting instructions, we say that S and S′ are conflict equivalent. We say that a schedule S is conflict serializable if it is conflict equivalent to a serial schedule.

We now present a simple and efficient method for determining the conflict serializability of a schedule. Consider a schedule S. We construct a directed graph, called a precedence graph, from S. This graph consists of a pair G = (V, E), where V is a set of transactions and E is the set of edges $T_i \rightarrow T_j$ for which one of three conditions hold:

1. T_i executes write(Q) before T_j executes read(Q)
2. T_i executes read(Q) before T_j executes write(Q).
3. T_i executes write(Q) before T_j executes write(Q).



If an edge $T_i \rightarrow T_j$ exists in the precedence graph, then, in any serial schedule S′ equivalent to S, Ti must appear before Tj . If the graph contains no cycles, then the schedule S is conflict serializable.

A serializability order of the transactions can be obtained by finding a linear order consistent with the partial order of the precedence graph. This process is called topological sorting. Thus, to test for conflict serializability, we need to construct the precedence graph and to invoke a cycle-detection algorithm.

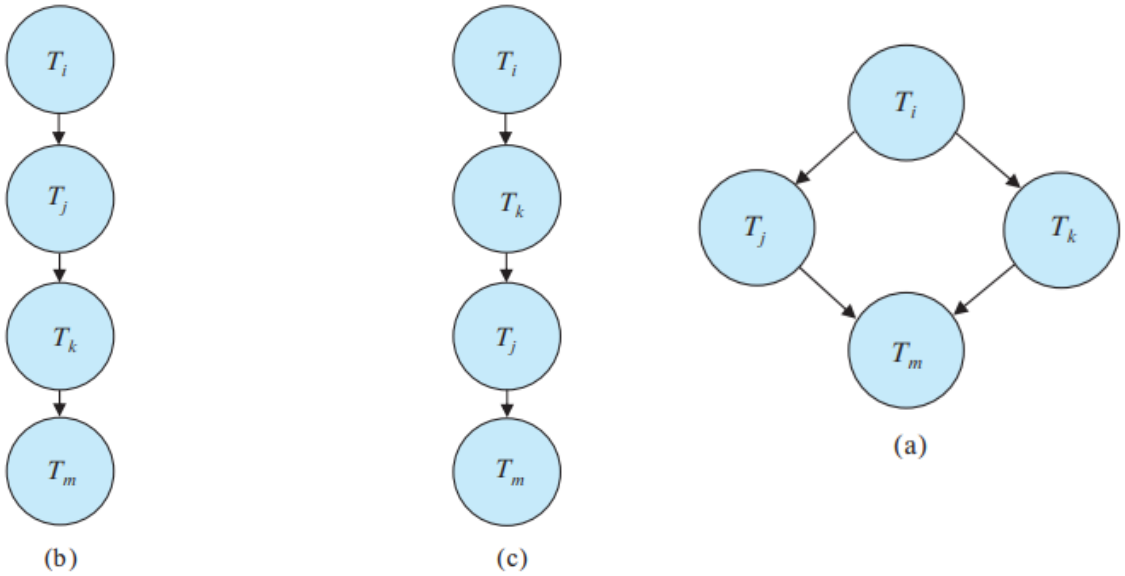


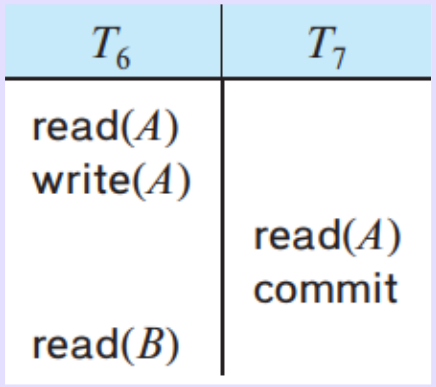
Figure 17.12 Illustration of topological sorting.

Transaction isolation and atomicity

So far, we have studied schedules while assuming implicitly that there are no transaction failures. We now address the effect of transaction failures during concurrent execution.

In a system that allows concurrent execution, the atomicity property requires that any transaction T_j that is dependent on T_i is also aborted. To achieve this, we need to place restrictions on the types of schedules permitted in the system.

❀ **Recoverable schedules**

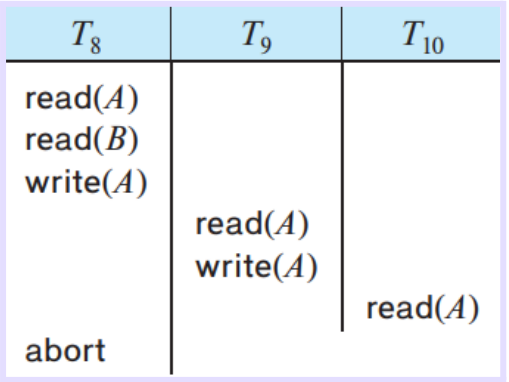


Consider this schedule, in which T_7 is a transaction that performs only one instruction: read(A). We call this a partial schedule because we have not included a commit or abort operation for T_6.

Now suppose that T_6 fails before it commits. T_7 has read the value of data item A written by T_6. Therefore, we say that T_7 is dependent on T_6. Because of this, we must abort T_7 to ensure atomicity

The schedule from the example is said to be a nonrecoverable schedule. A recoverable schedule is one where, for each pair of transactions T_i and T_j such that T_j reads a data item previously written by T_i , the commit operation of T_i appears before the commit operation of T_j .

❀ **Cascadeless schedules:** Even if a schedule is recoverable, to recover correctly from the failure of a transaction Ti , we may have to roll back several transactions.



Transaction T_8 writes a value of A that is read by transaction T_9. Transaction T_9 writes a value of A that is read by transaction T_{10}. Suppose that, at this point, T_8 fails. T_8 must be rolled back. Therefore, T_9 and T_{10} must be rolled back too.

Cascading rollback is undesirable, since it leads to the undoing of a significant amount of work.

A cascadeless schedule is one where… *definición de recoverable schedule*

Transaction isolation levels

If every transaction has the property that it maintains database consistency if executed alone, then serializability ensures that concurrent executions maintain consistency. However,

the protocols required to ensure serializability may allow too little concurrency for certain applications. In these cases, weaker levels of consistency are used.

The isolation levels specified by the SQL standard are as follows:

- ➔ Serializable: Usually ensures serializable execution.
- ➔ Repeatable read: Allows only committed data to be read and further requires that, between two reads of a data item by a transaction, no other transaction is allowed to update it.
- ➔ Read committed: Allows only committed data to be read, but does not require repeatable reads. Between two reads of a data item by the transaction, another transaction may have updated the data item and committed.
- ➔ Read uncommitted: Allows uncommitted data to be read.

All the isolation levels above additionally disallow dirty writes, that is, they disallow writes to a data item that has already been written by another transaction that has not yet committed or aborted.

Many database systems run, by default, at the read-committed isolation level.

set transaction isolation level serializable

By default, most databases commit individual statements as soon as they are executed. Such automatic commit of individual statements must be turned off to allow multiple statements to run as a single transaction.

Implementation of isolation levels

There are various concurrency-control policies that we can use to ensure that, even when multiple transactions are executed concurrently, only acceptable schedules are generated.

As a trivial example of a concurrency-control policy, consider this: A transaction acquires a lock on the entire database before it starts and releases the lock after it has committed. While a transaction holds a lock, no other transaction is allowed to acquire the lock, and all must therefore wait for the lock to be released. As a result of the locking policy, only one transaction can execute at a time (serializable).

Here we provide an overview of how some of the most important concurrency-control mechanisms work.

❀ **Locking:** Instead of locking the entire database, a transaction could instead lock only those data items that it accesses. Under such a policy, the transaction must hold locks long enough to ensure serializability, but for a period short enough not to harm performance excessively. Further improvements to locking result if we have two kinds of locks: shared (for data that T reads) and exclusive (for data that T writes).

❀ **Timestamps:** For each data item, the system keeps two timestamps.

- ➔ The read timestamp of a data item holds the largest timestamp of those transactions that read the data item.
- ➔ The write timestamp of a data item holds the timestamp of the transaction that wrote the current value of the data item.

Timestamps are used to ensure that transactions access each data item in order of the transactions' timestamps if their accesses conflict.

❀ **Multiple versions and snapshot isolation:** By maintaining more than one version of a data item, it is possible to allow a transaction to read an old version of a data item rather than a newer version written by an uncommitted transaction or by a transaction that should come later in the serialization order.

There are a variety of multiversion concurrency-control techniques. One in particular, called snapshot isolation: Where each transaction is given its own version, or snapshot, of the database when it begins (isolated from the updates made by other transactions).If the transaction updates the database, that update appears only in its own version; this updates are applied to the "real" database if the transaction commits.

Snapshot isolation ensures that attempts to read data never need to wait. Read-only transactions cannot be aborted; only those that modify data run a slight risk of aborting. It provides too much isolation.

Transactions as SQL statements

In a concurrent execution of these transactions (Al mismo tiempo que hacemos un select, otra query inserta un dato que cumple con los criterios), it is intuitively clear that they conflict, but this is a conflict that may not be captured by our simple model. This situation is referred to as the phantom phenomenon because a conflict may exist on "phantom" data.

So the same transaction, if run more than once, might reference different data items each time it is run if the values in the database change between runs.

It is not sufficient for concurrency control to consider only the tuples that are accessed by a transaction; the information used to find the tuples that are accessed by the transaction must also be considered for the purpose of concurrency control.

Capitulo 18

Review Terms

- Concurrency control
- Lock types
  - Shared-mode (S) lock
  - Exclusive-mode (X) lock
- Lock
  - Compatibility
  - Request
  - Wait
  - Grant
- Deadlock
- Starvation
- Locking protocol
- Legal schedule
- Two-phase locking protocol
  - Growing phase
  - Shrinking phase
  - Lock point
  - Strict two-phase locking
  - Rigorous two-phase locking
- Lock conversion
  - Upgrade
  - Downgrade
- Graph-based protocols
  - Tree protocol
  - Commit dependency
- Deadlock handling
  - Prevention
  - Detection
  - Recovery
- Deadlock prevention
  - Ordered locking
  - Preemption of locks
  - Wait–die scheme
  - Timeout-based schemes
- Deadlock detection
  - Wait-for graph
- Deadlock recovery
  - Total rollback
  - Partial rollback
- Multiple granularity
  - Explicit locks
  - Implicit locks
  - Intention locks
- Intention lock modes
  - Intention-shared (IS)
  - Intention-exclusive (IX)
  - Shared and intention-exclusive (SIX)
- Multiple-granularity locking protocol
- Timestamp
  - System clock
  - Logical counter
  - W-timestamp(Q)
  - R-timestamp(Q)
- Timestamp-ordering protocol
  - Thomas' write rule
- Validation-based protocols
  - Read phase
  - Validation phase
  - Validation test
- Multiversion timestamp ordering
- Multiversion two-phase locking
  - Read-only transactions
  - Update transactions
- Snapshot isolation
  - Lost update
  - First committer wins
  - First updater wins
  - Write skew
  - Select for update
- Insert and delete operations
- Phantom phenomenon
- Index-locking protocol
- Predicate locking
- Weak levels of consistency
  - Degree-two consistency
  - Cursor stability
- Optimistic concurrency control without read validation
- Conversations
- Concurrency in indices
  - Crabbing protocol
  - B-link trees
  - B-link-tree locking protocol
  - Next-key locking
- Latch-free data structures
- Compare-and-swap (CAS) instruction

Concurrency control

When several transactions execute concurrently in the database, however, the isolation property may no longer be preserved. To ensure that it is, the system must control the interaction among the concurrent transactions; this control is achieved through one of a variety of mechanisms called concurrency-control schemes.

Lock-based protocols

"While one transaction is accessing a data item, no other transaction can modify that data item." The most common method used to implement this requirement is to allow a transaction to access a data item only if it is currently holding a lock on that item.

❀ **Locks**

We introduce two modes in which a data item may be locked:

1. Shared: If a transaction Ti has obtained a shared-mode lock (denoted by S) on item Q, then Ti can read, but cannot write, Q.
2. Exclusive: If a transaction Ti has obtained an exclusive-mode lock (denoted by X) on item Q, then Ti can both read and write Q.

We require that every transaction request a lock in an appropriate mode on data item Q, depending on the types of operations that it will perform on Q. The transaction makes the request to the concurrency-control manager. The transaction can proceed with the operation only after the concurrency-control manager grants the lock to the transaction.

To state this more generally, given a set of lock modes, we can define a compatibility function on them as follows:

1. Let A and B represent arbitrary lock modes.
2. Ahora habla de que una transacción 1 pide un lock de modo A en Q, en donde la transacción 2 tiene un lock de modo B.
3. Si 1 puede acceder a Q, decimos que el modo A es compatible con el modo B

Note that shared mode is compatible with shared mode.

A transaction requests a shared lock on data item Q by executing the lock-S(Q) instruction. Similarly, a transaction requests an exclusive lock through the lock-X(Q) instruction. A transaction can unlock a data item Q by the unlock(Q) instruction.

Unfortunately, locking can lead to an undesirable situation. For example:

T_3	T_4
lock-X(B)	
read(B)	
$B := B - 50$	
write(B)	
	lock-S(A)
	read(A)
	lock-S(B)
lock-X(A)	

Since T_3 is holding an exclusive-mode lock on B and T_4 is requesting a shared-mode lock on B, T_4 is waiting for T_3 to unlock B. Similarly, since T_4 is holding a shared-mode lock on A and T_3 is requesting an exclusive mode lock on A, T_3 is waiting for T_4 to unlock A.

We have arrived at a state where neither of these transactions can ever proceed with its normal execution. This situation is called deadlock.

When deadlock occurs, the system must roll back one of the two transactions.

If we do not use locking, or if we unlock data items too soon after reading or writing them, we may get inconsistent states. On the other hand, if we do not unlock a data item before requesting a lock on another data item, deadlocks may occur.

Deadlocks are definitely preferable to inconsistent states, since they can be handled by rolling back transactions, whereas inconsistent states may lead to real-world problems that cannot be handled by the database system.

We shall require that each transaction in the system follow a set of rules, called a locking protocol. Locking protocols restrict the number of possible schedules. We shall present several locking protocols that allow only conflict-serializable schedules, and thereby ensure isolation. Before doing so, we introduce some terminology:

1. We say that T_i precedes T_j in S, written $T_i \rightarrow T_j$, if there exists a data item Q such that T_i has held lock mode A on Q, and T_j has held lock mode B on Q later, and comp(A,B) = false.
2. We say that a schedule S is legal under a given locking protocol if S is a possible schedule for a set of transactions that follows the rules of the locking protocol.
3. We say that a locking protocol ensures conflict serializability if and only if all legal schedules are conflict serializable

❀ **Granting of locks**

When a transaction requests a lock on a data item in a particular mode, and no other transaction has a lock on the same data item in a conflicting mode, the lock can be granted.

Acá entra un ejemplo en el que una transacción T hace un request de lock exclusivo, pero varios nodos hacen request de lock compartido, por lo que T tiene mucho tiempo de espera.The transaction T may never make progress, and is said to be starved.
We can avoid starvation of transactions by granting locks in the following manner:

When a transaction Ti requests a lock on a data item Q in a particular mode M, the concurrency-control manager grants the lock provided that:

- There is no other transaction holding a lock on Q in a mode that conflicts with M.
- There is no other transaction that is waiting for a lock on Q and that made its lock request before Ti .

❀ The two-phase locking protocol

The two-phase locking protocol ensures serializability and requires that each transaction issue lock and unlock requests in two phases:

1. Growing phase: A transaction may obtain locks, but may not release any lock.
2. Shrinking phase: A transaction may release locks, but may not obtain any new locks.

Consider any transaction. The point in the schedule where the transaction has obtained its final lock (the end of its growing phase) is called the lock point of the transaction. Now, transactions can be ordered according to their lock points.

Cascading rollbacks can be avoided by a modification of two-phase locking called the strict two-phase locking protocol. This protocol requires that all exclusive-mode locks taken by a transaction be held until that transaction commits.

Another variant of two-phase locking is the rigorous two-phase locking protocol, which requires that all locks be held until the transaction commits.

T_8	T_9
lock-S(a_1)	
	lock-S(a_1)
lock-S(a_2)	
	lock-S(a_2)
lock-S(a_3)	
lock-S(a_4)	
	unlock(a_1)
	unlock(a_2)
lock-S(a_n)	
upgrade(a_1)	

Figure 18.9 Incomplete schedule with a lock conversion.

This observation leads us to a refinement of the basic two-phase locking protocol, in which lock conversions are allowed. We shall provide a mechanism for upgrading a shared lock to an exclusive lock, and downgrading an exclusive lock to a shared lock.

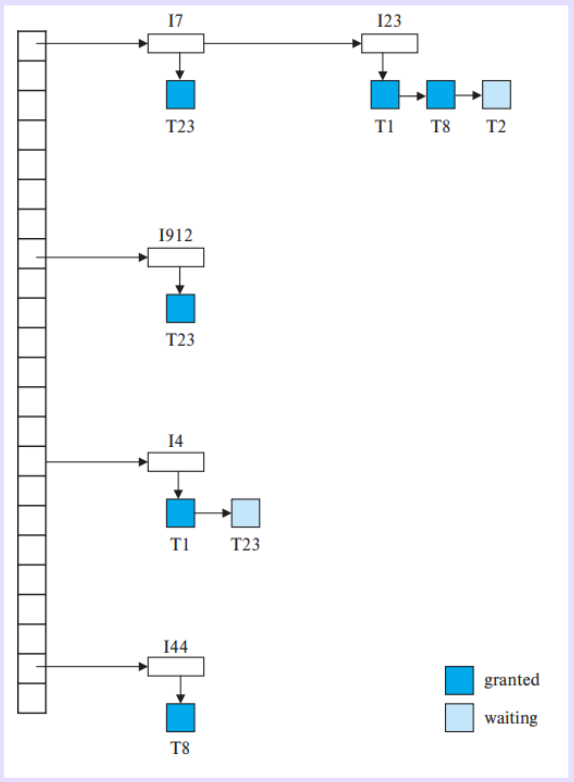
We denote conversion from shared to exclusive modes by upgrade, and from exclusive to shared by downgrade. Lock conversion cannot be allowed arbitrarily. Rather, upgrading can take place in only the growing phase, whereas downgrading can take place in only the shrinking phase.

Note that a transaction attempting to upgrade a lock on an item Q may be forced to wait. This enforced wait occurs if Q is currently locked by another transaction in shared mode.

❀ Implementation of locking

A lock manager can be implemented as a process that receives messages from transactions and sends messages in reply. The lock-manager process replies to lock-request messages with lock-grant messages, or with messages requesting rollback of the transaction. Unlock messages require only an acknowledgment in response, but may result in a grant message to another waiting transaction.

The lock manager uses this data structure: For each data item that is currently locked, it maintains a linked list of records, one for each request, in the order in which the requests arrived. It uses a hash table, indexed on the name of a data item, to find the linked list for a data item; this table is called the lock table. There is also a list of transactions that have been granted locks, or are waiting for locks, for each of the data items.



The table contains locks for five different data items, I4, I7, I23, I44, and I912.

Granted locks are the rectangles filled in a darker shade, while waiting requests are the rectangles filled in a lighter shade.

The lock manager processes requests this way:

1. When a lock request message arrives, it adds a record to the end of the linked list for the data item, if the linked list is present. Otherwise it creates a new linked list, containing only the record for the request.
2. When the lock manager receives an unlock message from a transaction, it deletes the record for that data item in the linked list corresponding to that transaction.
3. If a transaction aborts, the lock manager deletes any waiting request made by the transaction.

This algorithm guarantees freedom from starvation for lock requests, since a request can never be granted while a request received earlier is waiting to be granted.

❀ **Graph-based protocols**

If we wish to develop protocols that are not two phase, we need additional information on how each transaction will access the database.

The simplest model requires that we have prior knowledge about the order in which the database items will be accessed. Given such information, it is possible to construct locking protocols that are not two phase, but that, nevertheless, ensure conflict serializability.

To acquire such prior knowledge, we impose a partial ordering → on the set D = {d1, d2, …, dh} of all data items. If di → dj , then any transaction accessing both di and dj must access di before accessing dj .

The partial ordering implies that the set D may now be viewed as a directed acyclic graph, called a database graph.

In the tree protocol, the only lock instruction allowed is lock-X. Each transaction Ti can lock a data item at most once, and must observe the following rules:

1. The first lock by Ti may be on any data item.
2. Subsequently, a data item Q can be locked by Ti only if the parent of Q is currently locked by Ti .
3. Data items may be unlocked at any time.
4. A data item that has been locked and unlocked by Ti cannot subsequently be relocked by Ti .

All schedules that are legal under the tree protocol are conflict serializable.

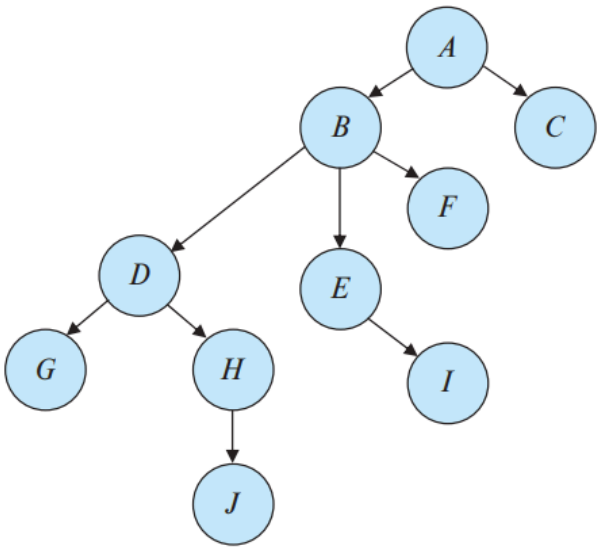


Figure 18.11 Tree-structured database graph.

T_{10}: lock-X(B); lock-X(E); lock-X(D); unlock(B); unlock(E); lock-X(G); unlock(D); unlock(G).
T_{11}: lock-X(D); lock-X(H); unlock(D); unlock(H).
T_{12}: lock-X(B); lock-X(E); unlock(E); unlock(B).
T_{13}: lock-X(D); lock-X(H); unlock(D); unlock(H).

It can be shown not only that the tree protocol ensures conflict serializability, but also that this protocol ensures freedom from deadlock.

The tree protocol does not ensure recoverability and cascadelessness. To ensure recoverability and cascadelessness, the protocol can be modified to not permit release of exclusive locks until the end of the transaction. Holding exclusive locks until the end of the transaction reduces concurrency.

Here is an alternative that improves concurrency, but ensures only recoverability:

1. For each data item with an uncommitted write, we record which transaction performed the last write to the data item.
2. Whenever a transaction T_i performs a read of an uncommitted data item, we record a commit dependency of T_i on the transaction that performed the last write to the data item.
3. Transaction T_i is then not permitted to commit until the commit of all transactions on which it has a commit dependency. If any of these transactions aborts, T_i must also be aborted.

On the tree-locking protocol, no rollbacks are required (because it is deadlock free). Earlier unlocking may lead to shorter waiting times and to an increase in concurrency.

In some cases, a transaction may have to lock data items that it does not access. For example, a transaction that needs to access data items A and J in the database graph of must lock not only A and J, but also data items B, D, and H. This additional locking results in:

- ➔ Increased locking overhead
- ➔ Possibility of additional waiting time
- ➔ Potential decrease in concurrency.

Further, without prior knowledge of what data items will need to be locked, transactions will have to lock the root of the tree, and that can reduce concurrency greatly.

Deadlock handling

A system is in a deadlock state if there exists a set of transactions such that every transaction in the set is waiting for another transaction in the set.

The only remedy to this undesirable situation is for the system to invoke some drastic action, such as rolling back some of the transactions involved in the deadlock. Rollback of a transaction may be partial: That is, a transaction may be rolled back to the point where it obtained a lock whose release resolves the deadlock.

There are two principal methods for dealing with the deadlock problem:
deadlock prevention protocol: This would ensure that the system will never enter a deadlock state.

deadlock detection and deadlock recovery scheme: we allow the system to enter a deadlock state, and then try to recover.

Both methods may result in transaction rollback. Prevention is commonly used if the probability that the system would enter a deadlock state is relatively high.

❀ **Deadlock prevention**

There are two approaches:

1. One approach ensures that no cyclic waits can occur by ordering the requests for locks, or requiring all locks to be acquired together.
2. The other approach is closer to deadlock recovery, and it performs transaction rollback instead of waiting for a lock whenever the wait could potentially result in a deadlock.

The simplest scheme under the first approach requires that each transaction locks all its data items before it begins execution.
Disadvantages (1): Often hard to predict what data items need to be locked before the transaction begins and data-item utilization may be very low.

Another approach is to impose an ordering of all data items and to require that a transaction lock data items only in a sequence consistent with the ordering (we have seen one such scheme in the tree protocol, which uses a partial ordering of data items). A variation of this approach is to use a total order of data items, in conjunction with two-phase locking.

The second approach for preventing deadlocks is to use preemption and transaction rollbacks. In preemption, when a transaction T_j requests a lock that transaction T_i holds, the lock granted to T_i may be preempted (adelantada) by rolling back of T_i , and granting of the lock to T_j .

To control the preemption, we assign a unique timestamp, based on a counter or on the system clock, to each transaction when it begins. The system uses these timestamps only to decide whether a transaction should wait or roll back. If a transaction is rolled back, it retains its old timestamp when restarted. Two different deadlock-prevention schemes using timestamps have been proposed:

1. The wait–die scheme is a nonpreemptive technique. When transaction T_i requests a data item currently held by T_j , T_i is allowed to wait only if it has a smaller timestamp than that of T_j. Otherwise, T_i is rolled back (dies).
2. The wound–wait scheme is a preemptive technique. It is a counterpart to the wait–die scheme. When transaction T_i requests a data item currently held by T_j , Ti is allowed to wait only if it has a larger timestamp than that of T_j (i.e., Ti is younger than Tj). Otherwise, Tj is rolled back

The major problem with both of these schemes is that unnecessary rollbacks may occur.

Another simple approach to deadlock prevention is based on lock timeouts.

In this approach, a transaction that has requested a lock waits for at most a specified amount of time. If the lock has not been granted within that time, the transaction is said to time out, and it rolls itself back and restarts. If there was in fact a deadlock, one or more transactions involved in the deadlock will time out and roll back. This scheme falls somewhere between deadlock prevention and deadlock detection and recovery.

The timeout scheme is particularly easy to implement, and it works well if transactions are short and if long waits are likely to be due to deadlocks. However, in general it is hard to decide how long a transaction must wait before timing out.

Starvation is also a possibility with this scheme. Hence, the timeout-based scheme has limited applicability.

❀ **Deadlock detection and recovery**

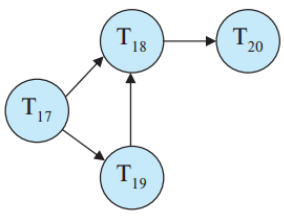
An algorithm that examines the state of the system is invoked periodically to determine whether a deadlock has occurred. If one has, then the system must attempt to recover from the deadlock. To do so, the system must:

- ➔ Maintain information about the current allocation of data items to transactions, as well as any outstanding data item requests.
- ➔ Provide an algorithm that uses this information to determine whether the system has entered a deadlock state.
- ➔ Recover from the deadlock when the detection algorithm determines that a deadlock exists.

❀ Deadlock detection

Deadlocks can be described precisely in terms of a directed graph called a wait-for graph. Graph G = (V,E) where transactions are vertices and the ordered pairs $T_i \rightarrow T_j$ are edges.

When transaction T_i requests a data item currently being held by transaction T_j, then the edge $T_i \rightarrow T_j$ is inserted in the wait-for graph. This edge is removed only when transaction T_j is no longer holding a data item needed by transaction T_i.



A deadlock exists in the system if and only if the wait-for graph contains a cycle.

Each transaction involved in the cycle is said to be deadlocked. To detect deadlocks, the system needs to maintain the wait-for graph, and periodically invoke an algorithm that searches for a cycle in the graph.

Since the graph has no cycle, the system is not in a deadlock state.

When should we invoke the detection algorithm? The answer depends on two factors:

1. How often does a deadlock occur?

If deadlocks occur frequently, then the detection algorithm should be invoked more frequently. Data items allocated to deadlocked transactions will be unavailable to other transactions until the deadlock can be broken. In the worst case, we would invoke the detection algorithm every time a request for allocation could not be granted immediately.

2. How many transactions will be affected by the deadlock?

❀ Recovery from deadlock

The most common solution is to roll back one or more transactions to break the deadlock. Three actions need to be taken:

1. Selection of a victim: Given a set of deadlocked transactions, we must determine which transaction/s to roll back to break the deadlock. We should roll back those transactions that will incur the minimum cost. But many factors may determine the cost of a roll back:
    a. How long the transaction has computed
    b. How much longer the transaction will compute
    c. How many data items the transaction has used.
    d. How many more data items the transaction needs for it to complete.
    e. How many transactions will be involved in the rollback.

2. Rollback: We must determine how far the selected transaction should be rolled back. The simplest solution is a total rollback: Abort the transaction and then restart it. Partial rollback requires the system to maintain additional information about the state of all the running transactions. The sequence of lock requests/grants and updates performed by the transaction needs to be recorded. The transactions must be capable of resuming execution after a partial rollback.
3. Starvation: It may happen that the same transaction is always picked as a victim. As a result, this transaction never completes its designated task, thus there is starvation. We must ensure that a transaction can be picked as a victim only a (small) finite number of times.

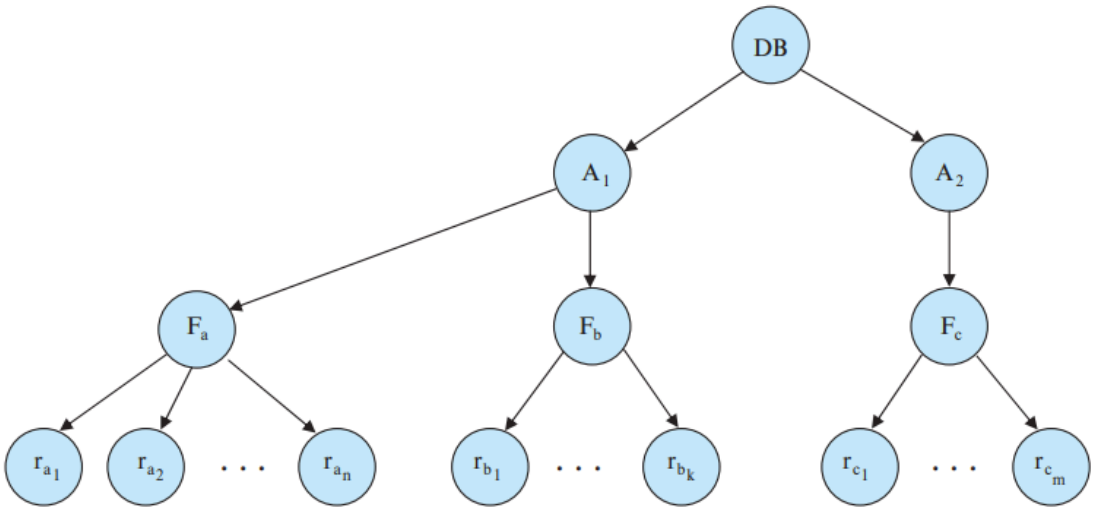
Multiple granularity

We have used each individual data item as the unit on which synchronization is performed, however, there are circumstances where it would be advantageous to group several data items, and to treat them as one individual synchronization unit.

What is needed is a mechanism to allow the system to define multiple levels of granularity. This is done by allowing data items to be of various sizes and defining a hierarchy of data

granularities, where the small granularities are nested within larger ones. Such a hierarchy can be represented graphically as a tree.

A nonleaf node of the multiple-granularity tree represents the data associated with its descendants. In the tree protocol, each node is an independent data item.



The highest level represents the entire database. Below it are nodes of type area; the database consists of exactly these areas. Each area in turn has nodes of type file as its children. No file is in more than one area. Finally, each file has nodes of type record. As before, the file consists of exactly those records that are its child nodes, and no record can be present in more than one file.

Each node in the tree can be locked individually. When a transaction locks a node, in either shared or exclusive mode, the transaction also has implicitly locked all the descendants of that node in the same lock mode.

How does the system determine if the root node can be locked?
One possibility is for it to search the entire tree. This solution, however, defeats the whole purpose of the multiple-granularity locking scheme.
A more efficient way to gain this knowledge is to introduce a new class of lock modes, called intention lock modes. Intention locks are put on all the ancestors of a node before that node is locked explicitly.

There is an intention mode associated with shared mode, and there is one with exclusive mode.

- ➔ If a node is locked in intention-shared (IS) mode, explicit locking is being done at a lower level of the tree, but with only shared-mode locks.
- ➔ If a node is locked in intention-exclusive (IX) mode, then explicit locking is being done at a lower level, with exclusive-mode or shared-mode locks.

➔ If a node is locked in shared and intention-exclusive (SIX) mode, the subtree rooted by that node is locked explicitly in shared mode, and that explicit locking is being done at a lower level with exclusive-mode locks.

	IS	IX	S	SIX	X
IS	true	true	true	true	false
IX	true	true	false	false	false
S	true	false	true	false	false
SIX	true	false	false	false	false
X	false	false	false	false	false

Figure 18.16 Compatibility matrix.

The multiple-granularity locking protocol uses these lock modes to ensure serializability. It requires that a transaction T_i that attempts to lock a node Q must follow these rules:

- Transaction T_i must observe the lock-compatibility function of Figure 18.16.
- Transaction T_i must lock the root of the tree first and can lock it in any mode.
- Transaction T_i can lock a node Q in S or IS mode only if T_i currently has the parent of Q locked in either IX or IS mode.
- Transaction T_i can lock a node Q in X, SIX, or IX mode only if T_i currently has the parent of Q locked in either IX or SIX mode.
- Transaction T_i can lock a node only if T_i has not previously unlocked any node (i.e., T_i is two phase).
- Transaction T_i can unlock a node Q only if T_i currently has none of the children of Q locked.

The multiple-granularity protocol requires that locks be acquired in topdown (root-to-leaf) order, whereas locks must be released in bottom-up (leaf-to-root) order. Deadlock is possible in the multiple-granularity protocol.

It is particularly useful in applications that include a mix of:

➔ Short transactions that access only a few data items
➔ Long transactions that produce reports from an entire file or set of files.

The number of locks that an SQL query may need to acquire can usually be estimated based on the relation scan operations performed by a query.

In case a transaction acquires a large number of tuple locks, the lock table may become overfull. To deal with this situation, the lock manager may perform lock escalation, replacing many lower level locks by a single higher level lock

Insert and delete operations and predicate reads

To examine how such transactions affect concurrency control, we introduce these additional operations: insert(Q), delete(Q)

❀ **Deletion**

We must decide when a delete instruction conflicts with another instruction. . Let I_i=delete(Q). We consider several instructions I_j .

- I_j = read(Q). I_i and I_j conflict. If I_i comes before I_j, T_j will have a logical error. If I_j comes before I_i, T_j can execute the read operation successfully.
- I_j = write(Q). I_i and I_j conflict. If I_i comes before I_j, T_j will have a logical error. If I_j comes before I_i, T_j can execute the write operation successfully.
- I_j = **delete**(Q). I_i and I_j conflict. If I_i comes before I_j, T_j will have a logical error. If I_j comes before I_i, T_i will have a logical error.
- I_j = **insert**(Q). I_i and I_j conflict. Suppose that data item Q did not exist prior to the execution of I_i and I_j. Then, if I_i comes before I_j, a logical error results for T_i.

Under the two-phase locking protocol, an exclusive lock is required on a data item before that item can be deleted.
Under the timestamp-ordering protocol, a test similar to that for a write must be performed.

❀ **Insertion**

Since an insert(Q) assigns a value to data item Q, an insert is treated similarly to a write for concurrency-control purposes:

- ➔ Under the two-phase locking protocol: if T_i performs an insert(Q) operation, T_i is given an exclusive lock on the newly created data item Q.
- ➔ Under the timestamp-ordering protocol: if T_i performs an insert(Q) operation, the values R-timestamp(Q) and W-timestamp(Q) are set to TS(T_i).

??

❀ **Predicate reads and the phantom phenomenon**

Supongamos que tenemos 2 transacciones: T_{30} que quiere hacer select count(*) en la relación de instructores y T_{31} que inserta un valor que coincide con esa búsqueda.
Si un schedule adopta esas 2 transacciones, nos podemos esperar que ocurra alguna de las siguientes cosas:

- ➔ T_{30} cuenta la tupla que acaba de insertar T_{31}, pero en un serial schedule, T_{31} debe venir antes que el T_{30}.
- ➔ Si T_{30} no tiene encuentra la tupla que se acaba de insertar, entonces en un serial schedule, T_{30} tiene que venir antes que T_{31}.

The second of these two cases is curious. T_{30} and T_{31} do not access any tuple in common, yet they conflict with each other. If concurrency control is performed at the tuple granularity, this conflict would go undetected. As a result, the system could fail to prevent a nonserializable schedule. This problem is an instance of the phantom phenomenon.

Phantom phenomena can occur not just with inserts, but also with updates.

T_{30} is an example of a transaction that reads information about what tuples are in a relation, and T_{31} is an example of a transaction that updates that information Clearly, it is not sufficient merely to lock the tuples that are accessed; the information used to find the tuples that are accessed by the transaction must also be locked.

Do not confuse the locking of an entire relation, as in multiple-granularity locking, with the locking of the data item corresponding to the relation. By locking the data item, a transaction only prevents other transactions from updating information about what tuples are in the relation.

The major disadvantage of locking a data item corresponding to the relation, or locking an entire index, is the low degree of concurrency .

A better solution is an index-locking technique that avoids locking the whole index. Any transaction that inserts a tuple into a relation must insert information into every index maintained on the relation. We eliminate the phantom phenomenon by imposing a locking protocol for indices.

The index-locking protocol takes advantage of the availability of indices on a relation, by turning instances of the phantom phenomenon into conflicts on locks on index leaf nodes. The protocol operates as follows:

1. Every relation must have at least one index.
2. A transaction Ti can access tuples of a relation only after first finding them through one or more of the indices on the relation. A relation scan is treated as a scan through all the leaves of one of the indices.
3. A transaction Ti that performs a lookup must acquire a shared lock on all the index leaf nodes that it accesses.
4. A transaction Ti may not insert, delete, or update a tuple t_i in a relation r without updating all indices on r.
5. Lock are obtained on tuples as usual
6. The rules of the two-phase locking protocol must be observed.
7. Ver libro en pag 889 pdf reader

Locking an index leaf node prevents any update to the node, even if the update did not actually conflict with the predicate.

Resumen: Bloqueo de Predicados (Predicate Locking)

El texto explica un enfoque alternativo para gestionar conflictos entre transacciones en bases de datos:

1. **El Concepto**: En lugar de bloquear registros individuales, el sistema adquiere bloqueos compartidos sobre **predicados** (condiciones lógicas) de una consulta. Por ejemplo, se bloquea la condición `"salario > 90000"`.
2. **Funcionamiento**:
    - Cualquier intento de **insertar o eliminar** datos debe verificarse contra ese predicado. Si los datos cumplen la condición, se genera un conflicto y la operación debe esperar.
    - En las **actualizaciones**, se deben revisar tanto el valor original como el nuevo para asegurar que no interfieran con el conjunto de tuplas seleccionado por la consulta original.
3. **Objetivo**: Evitar que transacciones concurrentes modifiquen el conjunto de resultados de una consulta activa que posee el bloqueo del predicado.
4. **Uso en la práctica**: Este protocolo no se utiliza habitualmente debido a que su implementación es mucho más costosa en cuanto a recursos que el bloqueo de índices (index-locking) y no ofrece beneficios adicionales significativos.

Timestamp-based protocols

Another method for determining the serializability order is to select an ordering among transactions in advance. The most common method for doing so is to use a timestamp-ordering scheme.

❀ **Timestamps**

With each transaction Ti in the system, we associate a unique fixed timestamp, denoted by $TS(T_i)$. There are two simple methods for implementing this scheme:

1. Use the value of the system clock as the timestamp; that is, a transaction's timestamp is equal to the value of the clock when the transaction enters the system.
2. Use a logical counter that is incremented after a new timestamp has been assigned; that is, a transaction's timestamp is equal to the value of the counter when the transaction enters the system.

To implement this scheme, we associate with each data item Q two timestamp values:

1. W-timestamp(Q) denotes the largest timestamp of any transaction that executed write(Q) successfully.
2. R-timestamp(Q) denotes the largest timestamp of any transaction that executed read(Q) successfully

❀ **The timestamp ordering protocol**

The timestamp-ordering protocol ensures that any conflicting read and write operations are executed in timestamp order. This protocol operates as follows:

- Suppose that transaction T_i issues read(Q).
  - If TS(T_i) < W-timestamp(Q), then T_i needs to read a value of Q that was already overwritten. Hence, the read operation is rejected, and T_i is rolled back.
  - If TS(T_i) ≥ W-timestamp(Q), then the read operation is executed, and R-timestamp(Q) is set to the maximum of R-timestamp(Q) and TS(T_i).
- Suppose that transaction T_i issues write(Q).
  - If TS(T_i) < R-timestamp(Q), then the value of Q that T_i is producing was needed previously, and the system assumed that that value would never be produced. Hence, the system rejects the write operation and rolls T_i back.
  - If TS(T_i) < W-timestamp(Q), then T_i is attempting to write an obsolete value of Q. Hence, the system rejects this write operation and rolls T_i back.
  - Otherwise, the system executes the write operation and sets W-timestamp(Q) to TS(T_i).

If a transaction T_i is rolled back by the concurrency-control scheme as result of issuance of either a read or write operation, the system assigns it a new timestamp and restarts it.

The timestamp-ordering protocol ensures conflict serializability. This is because conflicting operations are processed in timestamp order. The protocol ensures freedom from deadlock, since no transaction ever waits. However, there is a possibility of starvation of long transactions if a sequence of conflicting short transactions causes repeated restarting of the long transaction.

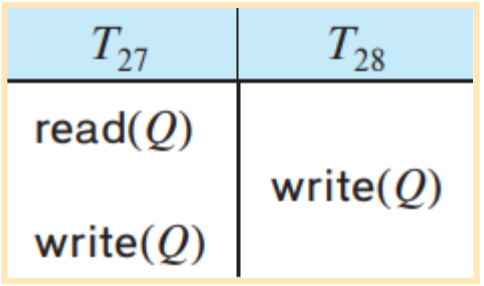
The protocol can generate schedules that are not recoverable. However, it can be extended to make the schedules recoverable, in one of several ways:

- ➔ Recoverability and cascadelessness can be ensured by performing all writes together at the end of the transaction.
- ➔ Recoverability and cascadelessness can also be guaranteed by using a limited form of locking, whereby reads of uncommitted items are postponed until the transaction that updated the item commits
- ➔ Recoverability alone can be ensured by tracking uncommitted writes and allowing a transaction T_i to commit only after the commit of any transaction that wrote a value that T_i read.

If the timestamp-ordering protocol is applied only to tuples, the protocol would be vulnerable to the phantom problems that we saw previously. To avoid this problem, the timestamp-ordering protocol could be applied to all data that is read by a transaction, including relation metadata and index data.

❀ **Thomas' write rule**

This modification to the TSOP allows greater potential concurrency.



Since T_{27} starts before T_{28}, we shall assume that TS(T_{27}) < TS(T_{28}). The read(Q) operation of T_{27} succeeds, as does the write(Q) operation of T_{28}. When T_{27} attempts its write(Q) operation, we find that TS(T_{27}) < W-timestamp(Q), since W-timestamp(Q) = TS(T_{28}). Thus, the write(Q) by T_{27} is rejected and transaction T_{27} must be rolled back.
Any transaction T_i with TS(T_i) < TS(T_{28}) that attempts a read(Q) will be rolled back, since TS(T_i) < W-timestamp(Q). Any transaction T_j with TS(T_j) > TS(T_{28}) must read the value of Q written by T_{28}, rather than the value that T_{27} is attempting to write.

This observation leads to a modified version of the timestamp-ordering protocol in which obsolete write operations can be ignored under certain circumstances (we change some of the rules for write operations):

2. If TS(T_i) < W-timestamp(Q), then Ti is attempting to write an obsolete value of Q. Hence, this write operation can be ignored.

Antes decía que había que hacer un rollback, ahora solo ignoramos el write. By ignoring the write, Thomas' write rule allows schedules that are not conflict serializable but are nevertheless correct.

Validation based protocols

The validation protocol requires that each transaction Ti executes in two or three different phases in its lifetime, depending on whether it is a read-only or an update transaction. The phases are, in order:

1. Read phase: During this phase, the system executes transaction Ti . It reads the values of the various data items and stores them in variables local to Ti . It performs all write operations on temporary local variables, without updates of the actual database.
2. Validation phase: The validation test is applied to transaction Ti . This determines whether Ti is allowed to proceed to the write phase without causing a violation of serializability. If a transaction fails the validation test, the system aborts the transaction.
3. Write phase: If the validation test succeeds for transaction Ti , the temporary local variables that hold the results of any write operations performed by Ti are copied to the database. Read-only transactions omit this phase.

Each transaction must go through the phases in the order shown. To perform the validation test, we need to know when the various phases of transactions took place (now we have startTS, validationTS, finishTS).

We determine the serializability order by the timestamp-ordering technique, using the value of the timestamp ValidationTS(Ti).

The validation test for transaction T_i requires that, for all transactions T_k with $TS(T_k) < TS(T_i)$, one of the following two conditions must hold:

1. $FinishTS(T_k) < StartTS(T_i)$. Since T_k completes its execution before T_i started, the serializability order is indeed maintained.
2. The set of data items written by T_k does not intersect with the set of data items read by T_i, and T_k completes its write phase before T_i starts its validation phase ($StartTS(T_i) < FinishTS(T_k) < ValidationTS(T_i)$).

The validation scheme automatically guards against cascading rollbacks, since the actual writes take place only after the transaction issuing the write has committed. However, there is a possibility of starvation of long transactions, due to a sequence of conflicting short transactions that cause repeated restarts of the long transaction.

This validation scheme is called the optimistic concurrency-control scheme since transactions execute optimistically, assuming they will be able to finish execution and validate at the end. In contrast, locking and timestamp ordering are pessimistic in that they force a wait or a rollback whenever a conflict is detected, even though there is a chance that the schedule may be conflict serializable.

Multiversion schemes

In multiversion concurrency-control schemes, each write(Q) operation creates a new version of Q. When a transaction issues a read(Q) operation, the concurrency control manager selects one of the versions of Q to be read. The concurrency-control scheme must ensure that the version to be read is selected in a manner that ensures serializability

❀ Multiversion timestamp ordering

The database system assigns this timestamp before the transaction starts execution, as described in timestamp-based protocols. With each data item Q, a sequence of versions <Q1, Q2, …, Qm> is associated. Each version Qk contains three data fields:

1. Content: Value of the version Qk
2. W-timestamp(Qk): timestamp of the transaction that created the version Qk.
3. R-timestamp(Qk): largest timestamp of any transaction that successfully read version Qk.

The multiversion timestamp-ordering scheme presented next ensures serializability. The scheme operates as follows (Let Qk denote the version of Q whose write timestamp is the largest write timestamp less than or equal to $TS(T_i)$):

1. If transaction Ti issues a read(Q), then the value returned is the content of version Qk.
2. If transaction Ti issues write(Q), and if TS(Ti) < R-timestamp(Qk), then the system rolls back transaction Ti . On the other hand, if TS(Ti) = W-timestamp(Qk), the system overwrites the contents of Qk; otherwise (if TS(Ti) > R-timestamp(Qk)), it creates a new version of Q.

The justification for rule 1 is clear. A transaction reads the most recent version that comes before it in time.

The valid interval of a version Qi of Q with W-timestamp t is defined as follows: if Qi is the latest version of Q, the interval is [t, ∞]; otherwise let the next version of Q have timestamp s; then the valid interval is [t,s).

Versions that are no longer needed are removed (se borra la versión mas vieja). The multiversion timestamp-ordering scheme has the desirable property that a read request never fails and is never made to wait.

The scheme, however, suffers from two undesirable properties. First, the reading of a data item also requires the updating of the R-timestamp field, resulting in two potential disk accesses, rather than one. Second, the conflicts between transactions are resolved through rollbacks, rather than through waits. This alternative may be expensive.

❀ **Multiversion two-phase locking**

Attempts to combine the advantages of multiversion concurrency control with the advantages of two-phase locking. This protocol differentiates between read-only transactions and update transactions.

Update transactions perform rigorous two-phase locking; that is, they hold all locks up to the end of the transaction. The timestamp in this case is not a real clock-based timestamp, but rather is a counter.

When an update transaction wants to write an item, it first gets an exclusive lock on the item and then creates a new version of the data item. The write is performed on the new version, and the timestamp of the new version is initially set to a value ∞, a value greater than that of any possible timestamp.

Read-only transactions see the old value of ts-counter until Ti has successfully committed. As a result, read-only transactions that start after Ti commits will see the values updated by Ti , whereas those that start before Ti commits will see the value before the updates by Ti . In either case, read-only transactions never need to wait for locks.

Snapshot isolation

Snapshot isolation involves giving a transaction a “snapshot” of the database at the time when it begins its execution. It then operates on that snapshot in complete isolation from concurrent transactions. This isolation is ideal for read-only transactions since they never wait and are never aborted by the concurrency manager.

Updates performed by a transaction must be validated before the transaction is allowed to commit.

When a transaction T is allowed to commit, the transition of T to the committed state and the writing of all of the updates made by T to the database must be conceptually done as an atomic action so that any snapshot created for another transaction either includes all updates by transaction T or none of them.

❀ **Multiversioning in snapshot isolation**

Transactions are given two timestamps: StartTS and CommitTS (when the T requested validation).

Snapshot isolation is based on multiversioning, and each transaction that updates a data item creates a version of the data item. Versions have only one timestamp, which is the write timestamp, indicating when the version was created. The timestamp of a version created by transaction Ti is set to CommitTS(Ti).

When a transaction Ti reads a data item, Ti does see the updates of all transactions that commit before it started (snapshot).

❀ **Validation steps for update transactions**

Deciding whether or not to allow an update transaction to commit requires some care. Potentially, two transactions running concurrently might both update the same data item.

There are two variants of snapshot isolation, both of which prevent lost updates. They are called first committer wins and first updater wins. Both approaches are based on testing the transaction against concurrent transactions.

Formally, Tj is concurrent with Ti if either
StartTS(Tj) ≤ StartTS(Ti) ≤ CommitTS(Tj), or
StartTS(Ti) ≤ StartTS(Tj) ≤ CommitTS(Ti).

First committer wins

When a transaction T_i starts validation:

- ➔ A test is made to see if any transaction that was concurrent with T_i has already written an update to the database for some data item that T_i intends to write.
- ➔ If any such data item is found, then T_i aborts.

➔ If no such data item is found, then T_i commits and its updates are written to the database.

First updater wins

(locking mechanism) When a transaction Ti attempts to update a data item, it requests a write lock on that data item. If the lock is not held by a concurrent transaction, the following steps are taken after the lock is acquired:

➔ If the item has been updated by any concurrent transaction, then Ti aborts.
➔ Otherwise Ti may proceed with its execution, including possibly committing.

If, however, some other concurrent transaction Tj already holds a write lock on that data item, then Ti cannot proceed, and the following rules are followed:

➔ Ti waits until Tj aborts or commits; if Tj commits, Ti must abort and vice versa

Locks are released when the transaction commits or aborts.

❀ **Serializability issues and solutions**

It is worth noting that integrity constraints that are enforced by the database, such as primary-key and foreign-key constraints, cannot be checked on a snapshot; this problem is handled by checking these constraints on the current state of the database, rather than on the snapshot.

Even with the above fix, there is still a serious problem with the snapshot isolation scheme as we have presented it and as it is implemented in practice: snapshot isolation does not ensure serializability!

T_i	T_j
read(A)	
read(B)	
	read(A)
	read(B)
A=B	
	B=A
write(A)	
	write(B)

Figure 18.20 Nonserializable schedule under snapshot isolation.

However, the execution is not serializable, since it results in swapping of the values of A and B, whereas any serializable schedule would set both A and B to the same value: either the initial value of A or the initial value of B, depending on the order of Ti and Tj .
??????????????????

This situation, where each of a pair of transactions has read a data item that is written by the other, but the set of data items written by the two transactions do not have any data item in common, is referred to as write skew.

Ver ejemplo en pag 905 pdf reader

The problems listed above seem to indicate that the snapshot isolation technique is vulnerable to many serializability problems and should never be used. However, serializability problems are relatively rare for two reasons:

1. The fact that the database must check integrity constraints at the time of commit, and not on a snapshot, helps avoid inconsistencies in many situations.
2. In many applications that are vulnerable to serializability problems, such as skew writes on some data items, the transactions conflict on other data items, ensuring such transactions cannot execute concurrently; ??????

Nonserializable may nevertheless occur with snapshot isolation. Several solutions:

➔ Using serializable snapshot isolation

Suppose we track all conflicts between transactions and construct a graph with that information; one way to ensure serializability is to look for cycles in the transaction precedence graph and roll back transactions if a cycle is found.

➔ Allowing different transactions to run under different isolation levels

Running long read-only transactions under the snapshot isolation level while running update transactions under the serializable isolation level ensures that the read-only transaction does not block updaters, while also ensuring that some anomalies cannot occur.

➔ Provide a way for SQL programmers to create artificial conflicts

Adding the for update clause causes the system to treat data that are read as if they had been updated for purposes of concurrency control.

Weak levels of consistency in practice

We first briefly outline some older terminology relating to consistency levels weaker than serializability and relate it to the SQL standard levels. We then discuss the issue of concurrency control for transactions that involve user interaction.

❀ **Degree-two consistency** (ver ejemplo 910 pdf reader / 881 libro)

The purpose of degree-two consistency is to avoid cascading aborts without necessarily ensuring serializability. The locking protocol for degree-two consistency uses the same two lock modes that we used for the two-phase locking protocol, but two-phase behavior is not required.

In contrast to the situation in two-phase locking, S-locks may be released at any time, and locks may be acquired at any time. Exclusive locks, however, cannot be released until the transaction either commits or aborts.

➔ Serializability is not ensured by this protocol.
➔ A transaction may read the same data item twice and obtain different results.
➔ Reads are not repeatable, but since exclusive locks are held until transaction commit, no transaction can read an uncommitted value.
➔ A transaction that is scanning an index may potentially see two versions of a record that was updated while the scan was in progress and may also potentially see neither version

❀ **Cursor stability**

Form of degree-two consistency designed for programs that iterate over tuples of a relation by using cursors. Instead of locking the entire relation, cursor stability ensures that:

➔ The tuple that is currently being processed by the iteration is locked in shared mode. Once the tuple is processed, the lock on the tuple can be released.

➔ Any modified tuples are locked in exclusive mode until the transaction commits.

These rules ensure that degree-two consistency is obtained. But locking is not done in a two-phase manner, and serializability is not guaranteed. Cursor stability is used in practice on heavily accessed relations as a means of increasing concurrency and improving system performance.

❀ **Concurrency control across user interactions**

Concurrency-control protocols usually consider transactions that do not involve user interaction.

If two-phase locking is used, the entire set of seats on a flight would be locked in shared mode until the user has completed the seat selection, and no other transaction would be able to update the seat allocation information in this period.

Timestamp protocols or validation could be used instead, which avoid the problem of locking, but both these protocols would abort the transaction for a user A if any other user B has updated the seat allocation information.

Snapshot isolation is a good option in this situation, since it would not abort the transaction of user A as long as B did not select the same seat as A. But this method requires the database to remember information about updates performed by a transaction even after it has committed, which can be problematic for long-duration transactions.

Another option is to split a transaction that involves user interaction into two or more transactions, such that no transaction spans a user interaction. The above idea has been generalized in an alternative concurrency control scheme, which uses version numbers stored in tuples to avoid lost updates.

When a transaction reads (for the first time) a tuple that it intends to update, it remembers the version number of that tuple. Updates are done locally and copied to the database as part of commit processing. El update se comitea si la versión del dato es la misma que la guardada, caso contrario se hace un rollback.

Unlike snapshot isolation, the reads performed by a transaction may not correspond to a snapshot of the database; and unlike the validation-based protocol, reads performed by the transaction are not validated.

We refer to the above scheme as optimistic concurrency control without read validation. This provides a weak level of serializability, and it does not ensure serializability.

The validation and update steps performed as part of commit processing are then executed as a single transaction in the database, using the concurrency-control scheme of the database to ensure atomicity for commit processing.

PEDIR EXPLICACION A LA IA

Advanced topics in concurrency control

Instead of using two-phase locking, special-purpose concurrency control techniques can be used for index structures, resulting in improved concurrency.

Concurrency control actions often become bottlenecks in main-memory databases. It is possible to consider operations, such as increment of a counter, as basic operations, and perform concurrency control on the basis of conflicts between operations.

❀ Online index creation

When we are dealing with large volumes of data (ranging in the terabytes), operations such as creating an index can take a long time.

The index contents must be consistent with the contents of the relation, and all further updates to the relation must maintain the index. One way to ensure this is to block all updates to the relation while the index is created. After the index is created, and the relation metadata are updated, locks can be released.

The above approach would make the system unavailable for updates to the relation for a very long time. Instead, most database systems support online index creation, which allows relation updates to occur even as the index is being created.

1. Index creation gets a snapshot of the relation and uses it to create the index; meanwhile, the system logs all updates to the relation that happen after the snapshot is created.
2. When the index on the snapshot data is complete, the log of updates to the relation is used to update the index. But while the index update is being carried out, further updates may be happening on the relation.
3. The index update then obtains a shared lock on the relation to prevent further updates and applies all remaining updates to the index. The relation metadata are then updated to indicate the existence of the new index.

Creation of materialized views that are maintained immediately, can also benefit from online construction techniques that are similar to online index construction. (Se ejecuta el query con un snapshot, se agregan los logs, se crea un shared lock y se actualiza la vista)

Schema changes such as adding or deleting attributes or constraints can also have a significant impact if relations are locked while the schema change is implemented on all tuples. (Podríamos usar un número de versión para cada tupla)(Para agregar restricciones, tenemos que verificar que los datos existentes las cumplan)

❀ Concurrency in index structures

It is possible to treat access to index structures like any other database structure and to apply the concurrency-control techniques discussed earlier. However, since indices are

accessed frequently, they would become a point of great lock contention, leading to a low degree of concurrency.

Luckily, indices do not have to be treated like other database structures; it is desirable to release index locks early, in a non-two-phase manner, to maximize concurrency.

Operation serializability for index operations is defined as follows: A concurrent execution of index operations on an index is said to be serializable if there is a serialization order of the operations that is consistent with the results that each index operation in the concurrent execution sees.

We outline two techniques for managing concurrent access to B+-trees as well as an index-concurrency control technique to prevent the phantom phenomenon.

1. Crabbing protocol: When searching for a key value, the crabbing protocol first locks the root node in shared mode. Then it acquires a shared lock on the child and releases the lock on the parent node. It repeats this process until it reaches a leaf. When inserting or deleting: Para hacer cambios requiere usar locks exclusivos (si hay que hacer un split en el parent, vuelve hacia arriba y luego sigue su camino)
   The system can easily handle such deadlocks by restarting the search operation from the root, after releasing the locks held by the operation.

Locks that are held for a short duration, instead of being held in a two-phase manner, are often referred to as latches. Latches are used internally in databases to achieve mutual exclusion on shared data structures

2. B-link-tree locking protocol: (B-link trees require that every node maintain a pointer to its right sibling) Holds locks on only one internal node at a time. The protocol releases the lock on the current internal node before requesting a lock on a child or parent node.
   The protocol ensures operation serializability while avoiding deadlocks between operations and increasing concurrency compared to the crabbing protocol. }

Instead of locking an entire index leaf node, some index concurrency-control schemes use key-value locking on individual key values, allowing other key values to be inserted or deleted from the same leaf.

To prevent the phantom phenomenon, the next-key locking technique is used: every index lookup must lock not only the keys found within the range (or the single key, in case of a point lookup) but also the next-key value.

❀ **Concurrency control in main memory databases**

When disk I/O is the bottleneck cost in a system, there is little benefit from optimizing other smaller costs, such as the cost of concurrency control.
Thus, it may be acceptable to perform locking at a coarse granularity: for example, the entire index could be locked using a single latch, the operation performed, and the latch released.

The reduced overhead of locking has been found to make up for the slightly reduced concurrency, and to improve overall performance.

There is another way to improve performance with in-memory indices, using atomic instructions to carry out index updates without acquiring any latches at all. Data structures implementations that support concurrent operations without requiring latches are called latch-free data structure implementations.

It is best to use latch-free data structure implementations (more often referred to as lock-free data structure implementations) that are provided by standard libraries.

❀ Long-Duration Transactions

Serious problems arise when this concept is applied to database systems that involve human interaction. Such transactions have these key properties:

Long duration: Once a human interacts with an active transaction, that transaction becomes a long-duration transaction from the perspective of the computer, since human response time is slow relative to computer speed.
Exposure of uncommitted data: Data generated and displayed to a user by a long duration transaction are uncommitted, since the transaction may abort.
Subtasks: An interactive transaction may consist of a set of subtasks initiated by the user.
Recoverability: It is unacceptable to abort a long-duration interactive transaction because of a system crash. The active transaction must be recovered to a state that existed shortly before the crash so that relatively little human work is lost.
Performance: Good performance in an interactive transaction system is defined as fast response time. Systems with high throughput make efficient use of system resources.

❀ Concurrency control with operations

> Two important points about the concept of correctness without serializability:
>
> 1. Correctness depends on the specific consistency constraints for the database.
> 2. Correctness depends on the properties of operations performed by each transaction.

Concurrency can be increased by treating some operations besides read and write as fundamental low-level operations and to extend concurrency control to deal with them.

The increment operation does not lock the variable in a two-phase manner; however, individual operations should be executed serially on the variable. Thus, if two increment operations are initiated concurrently on the same variable, one must finish before the other is allowed to start.
Compensating operation: If one of the transactions rolls back, the increment(v, n) operation must be rolled back by executing an operation increment(v,−n), which adds a negative of the original value

We can define a locking protocol to handle the preceding situation by defining an increment lock. The increment lock is compatible with itself but is not compatible with shared and exclusive locks.

Una banda de ejemplos que no quiero leer…

❀ **Real time transaction systems**

In certain applications, the constraints include deadlines by which a task must be completed. Deadlines are characterized as follows:

Hard deadline: Serious problems may occur if a task is not completed by its deadline.
Firm deadline: The task has zero value if it is completed after the deadline.
Soft deadlines: The task has diminishing value if it is completed after the deadline, with the value approaching zero as the degree of lateness increases.

Systems with deadlines are called real-time systems.

Due to the unpredictable nature of delays when reading data from disk, main memory databases are often used if real-time constraints have to be met. However, even if data are resident in main memory, variances in execution time arise from lock waits, transaction aborts, and so on.

Capitulo 19

Recovery System

A computer system, like any other device, is subject to failure from a variety of causes. The database system must take actions in advance to ensure that the atomicity and durability properties of transactions are preserved. An integral part of a database system is a recovery scheme that can restore the database to the consistent state that existed before the failure.

The recovery scheme must also support high availability (the database should be usable for a very high percentage of time). To support high availability in the face of machine failure, the recovery scheme must support the ability to keep a backup copy of the database synchronized with the current contents of the primary copy of the database.

Failure classification

We'll consider only the following types of failure:

- Transaction failure: There are two types of errors that may cause a transaction to fail:
  - Logical error: The transaction can no longer continue with its normal execution because of some internal condition, such as bad input, data not found, overflow, or resource limit exceeded.
  - System error: The system has entered an undesirable state (e.g., deadlock), as a result of which a transaction cannot continue with its normal execution. The transaction, however, can be re-executed at a later time.
- System crash: There is a hardware malfunction, or a bug in the database software or the operating system, that causes the loss of the content of volatile storage and brings transaction processing to a halt. The content of non-volatile storage remains intact and is not corrupted. The assumption that hardware errors and bugs in the software bring the system to a halt, but do not corrupt the non-volatile storage contents, is known as the fail-stop assumption. Well-designed systems have numerous internal checks, at the hardware and the software level, that bring the system to a halt when there is an error. Hence, the fail-stop assumption is a reasonable one.
- Disk failure: A disk block loses its content as a result of either a head crash or failure during a data-transfer operation. Copies of the data on other disks, or archival backups on tertiary media, such as DVD or tapes, are used to recover from the failure.

To determine how the system should recover from failures, we need to:

1. Identify modes of those devices used for storing data.
2. Consider how these failure modes affect the contents of the database.
3. Propose algorithms to ensure database consistency and transaction atomicity despite failures.

The algorithms consist of the actions taken during normal transaction processing (to ensure that enough information exists to allow recovery from failures) and actions taken after a failure to recover the db.

Storage

We identified three categories of storage:

1. Volatile storage
2. Non-Volatile storage
3. Stable storage

❀ **Stable storage implementation**

To implement stable storage, we need to replicate the needed information in several non-volatile storage media (usually disk) with independent failure modes and to update the information in a controlled manner to ensure that failure during data transfer does not damage the needed information.

RAID systems guarantee that the failure of a single disk will not result in loss of data. The simplest and fastest form of RAID is the mirrored disk, which keeps two copies of each block on separate disks. Other forms of RAID offer lower costs, but at the expense of lower performance.

Block transfer between memory and disk storage can result in:

- ➔ Successful completion: The transferred information arrived safely at its destination
- ➔ Partial failure: A failure occurred in the midst of transfer, and the destination block has incorrect information.
- ➔ Total failure: The failure occurred sufficiently early during the transfer that the destination block remains intact.

We require that, if a data-transfer failure occurs, the system detects it and invokes a recovery procedure to restore the block to a consistent state. To do so, the system must maintain two physical blocks for each logical database block. An output operation is executed as follows:

1. Write the information onto the first physical block.
2. When the first write completes successfully, write the same information onto the second physical block.
3. The output is completed only after the second write completes successfully.

If the system fails while blocks are being written, it is possible that the two copies of a block could be inconsistent with each other. During recovery, for each block, the system would need to examine two copies of the blocks.

The requirement of comparing every corresponding pair of blocks during recovery is expensive to meet. We can reduce the cost greatly by keeping track of block writes that are

in progress, using a small amount of non-volatile RAM. On recovery, only blocks for which writes were in progress need to be compared.

❀ **Data access**

The database is partitioned into fixed-length storage units called blocks. Blocks are the units of data transfer to and from disk and may contain several data items. We shall assume that no data item spans two or more blocks. This assumption is realistic for most data-processing applications, such as a bank or a university

The blocks residing on the disk are referred to as physical blocks; the blocks residing temporarily in main memory are referred to as buffer blocks. The area of memory where blocks reside temporarily is called the disk buffer.

Block movements between disk and main memory are initiated through input (physical block to memory) and output operations (buffer block to the disk and replaces physical block)

Each transaction has a private work area; the system creates this work area when the transaction is initiated, and the system removes the area when the transaction either commits or aborts. Transaction Ti interacts with the database system by transferring data to and from its work area to the system buffer.

1. read(X) assigns the value of data item X to the local variable x_i. It executes this operation as follows:
   a. If block B_X on which X resides is not in main memory, it issues input(B_X).
   b. It assigns to x_i the value of X from the buffer block.
2. write(X) assigns the value of local variable x_i to data item X in the buffer block. It executes this operation as follows:
   a. If block B_X on which X resides is not in main memory, it issues input(B_X).
   b. It assigns the value of x_i to X in buffer B_X.

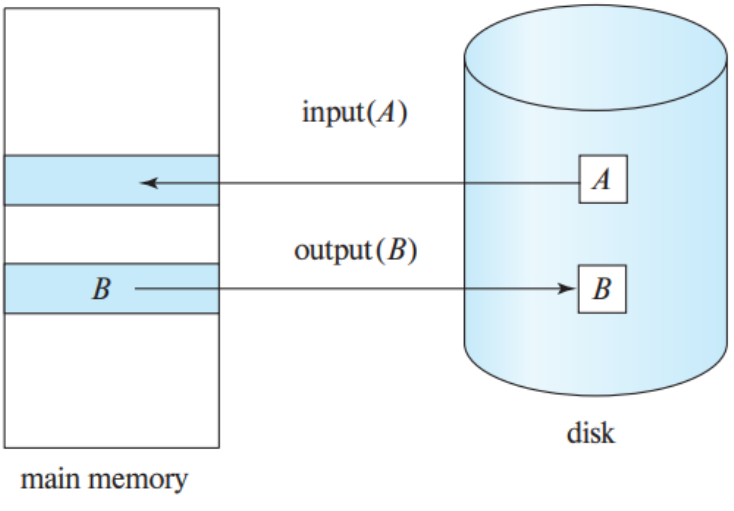


Figure 19.1 Block storage operations.

If the system crashes after the write(X) operation was executed but before output(BX) was executed, the new value of X is never written to disk and, thus, is lost.

Recovery and atomicity

Our goal is to perform either all or no database modifications made by Ti. However, if Ti performed multiple database modifications, several output operations may be required, and a failure may occur after some of these modifications have been made, but before all of them are made.

To achieve our goal of atomicity, we must first output to stable storage information describing the modifications, without modifying the database itself. As we shall see, this information can help us ensure that all modifications performed by committed transactions are reflected in the database. We also need to store information about the old value of any item updated by a modification in case the transaction performing the modification fails (aborts).

The most commonly used technique for recovery is based on log records, and we study log-based recovery in detail in this chapter.

❀ Log records

The log is a sequence of log records, recording all the update activities in the database. There are several types of log records. An update log record describes a single database write. It has these fields:

- ➔ Transaction identifier: which is the unique identifier of the transaction that performed the write operation.
- ➔ Data-item identifier: which is the unique identifier of the data item written. Typically, it is the location on disk of the data item, consisting of the block identifier of the block on which the data item resides and an offset within the block.
- ➔ Old value: which is the value of the data item prior to the write.
- ➔ New value: which is the value that the data item will have after the write.

Among the types of log records are:

- ➔ <Ti start> Transaction Ti has started.
- ➔ <Ti commit> Transaction Ti has committed.
- ➔ <Ti abort> Transaction Ti has aborted.

Once a log record exists, we can output the modification to the database if that is desirable. For log records to be useful for recovery from system and disk failures, the log must reside in stable storage.

❀ Database modification

A transaction creates a log record prior to modifying the database.
In order for us to understand the role of these log records in recovery, we need to consider the steps a transaction takes in modifying a data item:

1. The transaction performs some computations in its own private part of main memory.
2. The transaction modifies the data block in the disk buffer in main memory holding the data item.

3. The database system executes the output operation that writes the data block to disk.

We say a transaction modifies the database if it performs an update on a disk buffer, or on the disk itself. If a transaction does not modify the database until it has committed, it is said to use the deferred-modification technique. If database modifications occur while the transaction is still active, the transaction is said to use the immediate-modification technique.

The recovery algorithms we describe in this chapter support immediate modification. As described, they work correctly even with deferred modification, but they can be optimized to reduce overhead when used with deferred modification;

A recovery algorithm must take into account a variety of factors, including:

- ➔ The possibility that a transaction may have committed although some of its database modifications exist only in the disk buffer in main memory and not in the database on disk.
- ➔ The possibility that a transaction may have modified the database while in the active state and, as a result of a subsequent failure, may need to abort.

Because all database modifications must be preceded by the creation of a log record, the system has available both the old value prior to the modification of the data item and the new value that is to be written for the data item. This allows the system to perform undo and redo operations as appropriate.

❀ **Concurrency control and recovery**

Recovery algorithms usually require that if a data item has been modified by a transaction, no other transaction can modify the data item until the first transaction commits or aborts.

This requirement can be ensured by acquiring an exclusive lock on any updated data item and holding the lock until the transaction commits; in other words, by using strict two-phase locking. When either snapshot isolation or validation is used for concurrency control, database updates of a transaction are (conceptually) deferred until the transaction is partially committed.

❀ **Transaction commit**

We say that a transaction has committed when its commit log record has been output to stable storage; If a system crash occurs before a log record <Ti commit> is output to stable storage, transaction Ti will be rolled back.

❀ Using the log to redo and undo transactions

Consider our simplified banking system. Let T_0 be a transaction that transfers \$50 from account A to account B:

T_0: read(A);
$A := A - 50$;
write(A);
read(B);
$B := B + 50$;
write(B).

Let T_1 be a transaction that withdraws \$100 from account C:

T_1: read(C);
$C := C - 100$;
write(C).

Log	Database
$<T_0$ start$>$	
$<T_0, A, 1000, 950>$	
$<T_0, B, 2000, 2050>$	
	$A = 950$
	$B = 2050$
$<T_0$ commit$>$	
$<T_1$ start$>$	
$<T_1, C, 700, 600>$	
	$C = 600$
$<T_1$ commit$>$	

Using the log, the system can handle any failure that does not result in the loss of information in non-volatile storage. Two recovery procedures:

- ➔ **redo**(Ti): The procedure sets the value of all data items updated by transaction Ti to the new values.

- ➔ **undo**(Ti): The procedure restores the value of all data items updated by transaction Ti to the old values. The undo operation not only restores the data items to their old value, but also writes log records to record the updates performed as part of the undo process. These log records are special redo-only log records. When the undo operation for transaction Ti completes, it writes a log record <Ti abort>, indicating that the undo has completed.

After a system crash has occurred, the system consults the log to determine which transactions need to be redone and which need to be undone so as to ensure atomicity.

Undone if log contains the record Ti start but does not contain either Ti commit or abort
Redo if log contains Ti start and commit or abort.

(ejemplo del banco: pag 919 / 948 pdf reader)

❀ Checkpoints

When a system crash occurs, we must consult the log to determine those transactions that need to be redone and those that need to be undone. In principle, we need to search the entire log to determine this information.

To reduce these types of overhead (time-consuming search process and a lot of redones), we introduce checkpoints.

A checkpoint is performed as follows:

1. Output onto stable storage all log records currently residing in main memory.
2. Output to the disk all modified buffer blocks.
3. Output onto stable storage a log record of the form <checkpoint L>, where L is a list of transactions active at the time of the checkpoint.

The redo or undo operations need to be applied only to transactions in L, and to all transactions that started execution after the record was written to the log.

The requirement that transactions must not perform any updates to buffer blocks or to the log during checkpointing can be bothersome, since transaction processing has to halt while a checkpoint is in progress. A fuzzy checkpoint is a checkpoint where transactions are allowed to perform updates even while buffer blocks are being written out.

Recovery algorithm

We are now ready to present the full recovery algorithm using log records for recovery from transaction failure and a combination of the most recent checkpoint and log records to recover from a system crash.

The recovery algorithm described in this section requires that a data item that has been updated by an uncommitted transaction cannot be modified by any other transaction, until the first transaction has either committed or aborted. (strict two-phase locking)

❀ **Recovery after a system crash**

1. In the redo phase, the system replays updates of all transactions by scanning the log forward from the last checkpoint. The log records that are replayed include log records for transactions that were rolled back before system crash, and those that had not committed when the system crash occurred. This phase also determines all transactions that were incomplete at the time of the crash, and must therefore be rolled back.

At the end of the redo phase, undo-list contains the list of all transactions that are incomplete, that is, they neither committed nor completed rollback before the crash.

2. In the undo phase, the system rolls back all transactions in the undo-list. It performs rollback by scanning the log backward from the end.

After the undo phase of recovery terminates, normal transaction processing can resume.
REVISAR LAS 3 ETAPAS DENTRO DE CADA ETAPA (pag 923 / 953 pdf reader) Y EJEMPLOS EN LA SIG PAGINA

The actions are repeated in the same order in which they were originally carried out; hence, this process is called repeating history. Although it may appear wasteful, repeating history even for failed transactions simplifies recovery schemes.

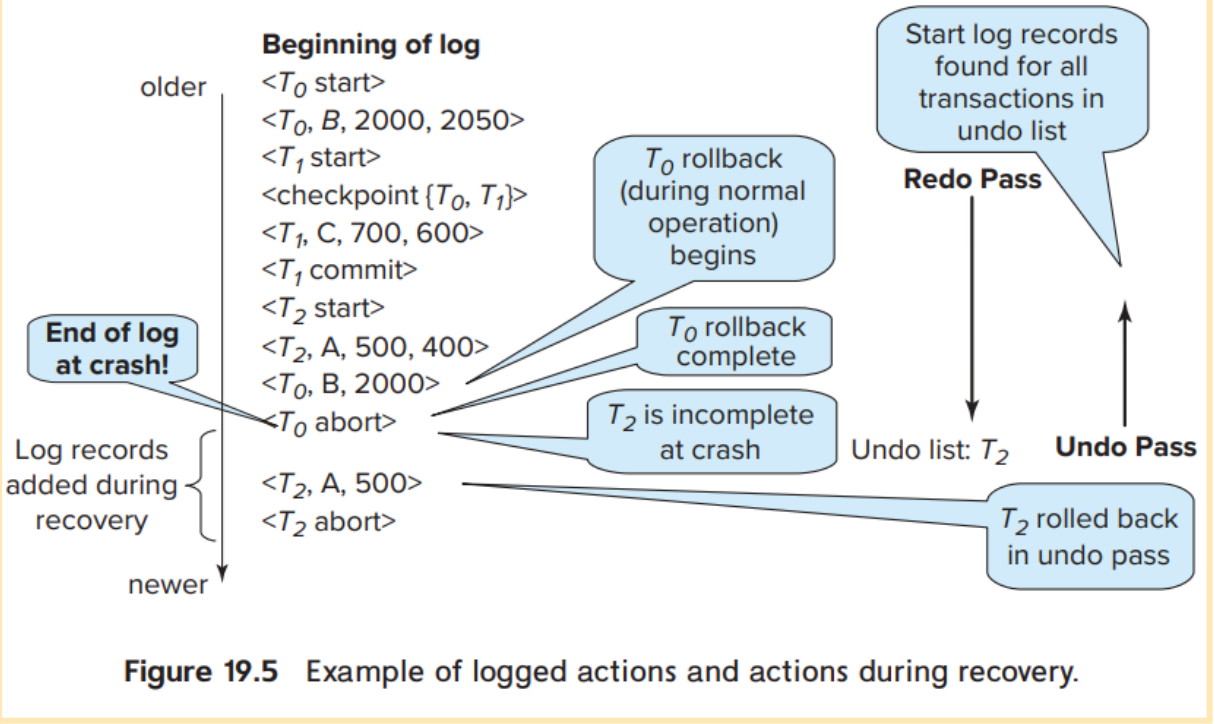


Figure 19.5 Example of logged actions and actions during recovery.

❀ **Optimizing commit processing**

Committing a transaction requires that its log records have been forced to disk. If a separate log flush is done for each transaction, each commit incurs a significant log write overhead.

The rate of transaction commit can be increased using the group-commit technique; Instead of attempting to force the log as soon as a transaction completes, the system waits until several transactions have completed, or a certain period of time has passed since a transaction completed execution. It then commits the group of transactions that are waiting, together. The system can ensure that blocks are full when they are written to stable storage without making transactions wait excessively.

Although group commit reduces the overhead imposed by logging, it results in a slight delay in commit of transactions that perform updates. When the rate of commits is low, the delay may not be worth the benefit, but with high rates of transaction commit, the overall delay in commit is actually reduced by using group commit.

Buffer management

❀ **Log-record buffering**

So far, we have assumed that every log record is output to stable storage at the time it is created. This assumption imposes a high overhead on system execution.

The cost of outputting a block to stable storage is sufficiently high that it is desirable to output multiple log records at once. To do so, we write log records to a log buffer in main memory, where they stay temporarily until they are output to stable storage. Multiple log records can be gathered in the log buffer and output to stable storage in a single output operation. The order of log records in the stable storage must be exactly the same as the order in which they were written to the log buffer.

Con esta táctica los logs solo permanecen en memoria hasta que son mandados a disco, por lo tanto, si el sistema falla, los logs se pierden.

Additional requirements on the recovery techniques to ensure transaction atomicity:

- Transaction T_i enters the commit state after the $<T_i$ commit> log record has been output to stable storage.
- Before the $<T_i$ commit> log record can be output to stable storage, all log records pertaining to transaction T_i must have been output to stable storage.
- Before a block of data in main memory can be output to the database (in non-volatile storage), all log records pertaining to data in that block must have been output to stable storage.

This rule is called the write-ahead logging (WAL) rule. There is no problem resulting from the output of log records earlier than necessary. Writing the buffered log to disk is sometimes referred to as a log force.

❀ **Database buffering**

Since main memory is typically much smaller than the entire database, it may be necessary to overwrite a block B1 in main memory when another block B2 needs to be brought into memory. If B1 has been modified, B1 must be output prior to the input of B2.

One might expect that transactions would force-output all modified blocks to disk when they commit. Such a policy is called the force policy. The alternative, the no-force policy, allows a transaction to commit even if it has modified some blocks that have not yet been written back to disk.

Similarly, one might expect that blocks modified by a transaction that is still active should not be written to disk. This policy is called the no-steal policy. The alternative, the steal policy, allows the system to write modified blocks to disk even if the transactions that made those modifications have not all committed.

When a block B1 is to be output to disk, all log records pertaining to data in B1 must be output to stable storage before B1 is output. It is important that no writes to the block B1 be in progress while the block is being output, since such a write could violate the write-ahead logging rule. We can ensure that there are no writes in progress by using a special means of locking:

- The following sequence of actions is taken when a block is to be output:
  - Obtain an exclusive lock on the block, to ensure that no transaction is performing a write on the block.
  - Output log records to stable storage until all log records pertaining to block B_1 have been output.
  - Output block B_1 to disk.
  - Release the lock once the block output has completed.

Database systems usually have a process that continually cycles through the buffer blocks, outputting modified buffer blocks back to disk. The above locking protocol must of course be followed when the blocks are output. As a result of continuous output of modified blocks, the number of dirty blocks in the buffer, that is, blocks that have been modified in the buffer but have not been subsequently output, is minimized.

❀ **Operating system role in buffer management**

We can manage the database buffer by using one of two approaches:

1. The database system reserves part of main memory to serve as a buffer that it, rather than the operating system, manages. The database system manages datablock transfer in accordance with the requirements in section 19.5.2. This approach has the drawback of limiting flexibility in the use of main memory..

2. The database system implements its buffer within the virtual memory provided by the operating system. Since the operating system knows about the memory requirements of all processes in the system, ideally it should be in charge of deciding what buffer blocks must be force-output to disk, and when.

Unfortunately, almost all current-generation operating systems retain complete control of virtual memory. The operating system reserves space on disk for storing virtual-memory pages that are not currently in main memory; this space is called swap space.

❀ **Fuzzy checkpointing**

The checkpointing technique described in Section 19.3.6 requires that all updates to the database be temporarily suspended while the checkpoint is in progress. If the number of pages in the buffer is large, a checkpoint may take a long time to finish, which can result in an unacceptable interruption in processing of transactions.

To avoid such interruptions, the checkpointing technique can be modified to permit updates to start once the checkpoint record has been written, but before the modified buffer blocks are written to disk. The checkpoint thus generated is a fuzzy checkpoint.

Since pages are output to disk only after the checkpoint record has been written, it is possible that the system could crash before all pages are written.

El problema: Los checkpoints (puntos de control) en el disco pueden quedar incompletos si el sistema falla durante su escritura.
La solución: La ubicación del **último checkpoint completo** se guarda en una posición fija del disco llamada `last-checkpoint`.
El proceso:

1. Antes de escribir el checkpoint, el sistema crea una lista de todos los bloques de memoria (buffers) modificados.
2. El registro del checkpoint en el disco **no** se actualiza inmediatamente.
3. La información de `last-checkpoint` se actualiza **solo después** de que todos los bloques modificados de la lista se hayan guardado por completo en el disco.

Failure with loss of non-volatile storage

The basic scheme is to dump the entire contents of the database to stable storage periodically—say, once per day. If a failure occurs that results in the loss of physical database blocks, the system uses the most recent dump in restoring the database to a previous consistent state.

One approach to database dumping requires that no transaction may be active during the dump procedure, and it uses a procedure similar to checkpointing:

1. Output all log records currently residing in main memory onto stable storage.
2. Output all buffer blocks onto the disk.
3. Copy the contents of the database to stable storage.
4. Output a log record onto the stable storage.

A dump of the database contents is also referred to as an archival dump, since we can archive the dumps and use them later to examine old states of the database.

Such dumps are useful when migrating data to a different instance of the database, or to a different version of the database software, since the physical locations and layout may be different in the other database instance or database software version.

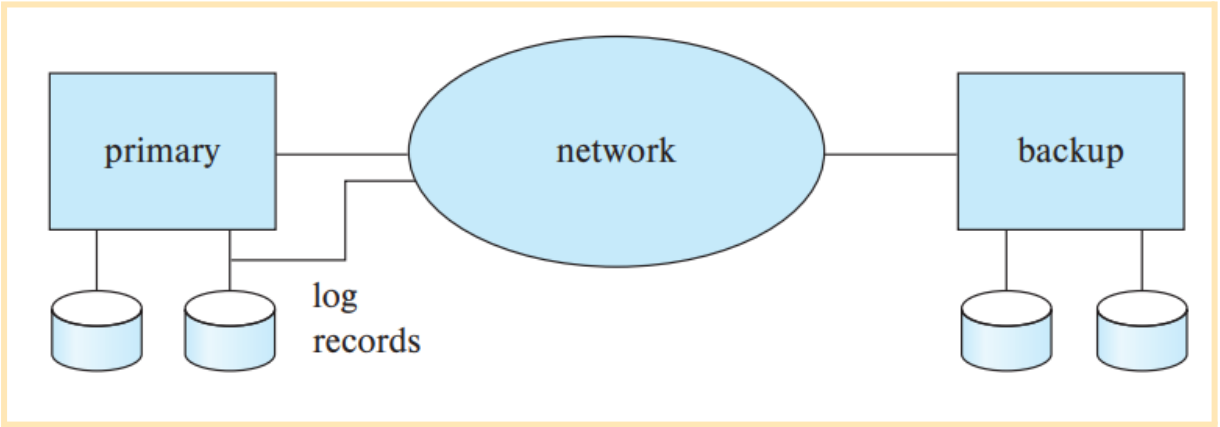
Fuzzy dump schemes have been developed that allow transactions to be active while the dump is in progress.

High availability using remote backup systems

Today's applications need transaction-processing systems that can function in spite of system failures or environmental disasters. Such systems must provide high availability; that is, the time for which the system is unusable must be extremely short.

We can achieve high availability by performing transaction processing at one site, called the primary site, and having a remote backup site where all the data from the primary site are replicated. The remote backup site is sometimes also called the secondary site. The remote

site must be kept synchronized with the primary site as updates are performed at the primary. We achieve synchronization by sending all log records from the primary site to the remote backup site.



El backup debe estar en un lugar físico diferente que la unidad de procesamiento para tener datos de respaldo frente a inundaciones u otros factores.

When the primary site fails, the remote backup site takes over processing. First, however, it performs recovery, using its (perhaps outdated) copy of the data from the primary and the log records received from the primary. In effect, the remote backup site is performing recovery actions that would have been performed at the primary site when the latter recovered.

Standard recovery algorithms, with minor modifications, can be used for recovery at the remote backup site. Several issues must be addressed in designing a remote backup system:

- ➔ Detection of failure: It is important for the remote backup system to detect when the primary has failed. Failure of communication lines can fool the remote backup into believing that the primary has failed. To avoid this problem, we maintain several communication links with independent modes of failure between the primary and the remote backup.

- ➔ Transfer of control: When the primary fails, the backup site takes over processing and becomes the new primary. The decision to transfer control can be done manually or can be automated using software provided by database system vendors. Queries must now be sent to the new primary. When the original primary site recovers, it can either play the role of remote backup or it can take over the role of primary site again. In either case, the old primary must receive a log of updates carried out by the backup site while the old primary was down.

- ➔ Time to recover: If the log at the remote backup grows large, recovery will take a long time. A hot-spare configuration can make takeover by the backup site almost instantaneous. In this configuration, the remote backup site continually processes redo log records as they arrive, applying the updates locally.

➔ Time to commit: To ensure that the updates of a committed transaction are durable, a transaction must not be declared committed until its log records have reached the backup site. This delay can result in a longer wait to commit a transaction, and some systems therefore permit lower degrees of durability. The degrees of durability can be classified as follows:

- One-safe: A transaction commits as soon as its commit log record is written to stable storage at the primary site.
- Two-very-safe: A transaction commits as soon as its commit log record is written to stable storage at the primary and the backup site
- Two-safe: This scheme is the same as two-very-safe if both primary and backup sites are active.

Some systems instead use a high availability proxy machine. Application clients do not connect to the database directly, but connect through the proxy. The proxy transparently routes application requests to the current primary.

Remote backups are kept synchronized with the primary by ensuring that all block writes performed at the primary are also replicated at the backup

If the database forces a block to disk and then performs some other update actions at the primary, the block must also be forced to disk at the backup, before subsequent updates are performed at the backup system.